THE CELLULAR AUTOMATA THAT COMMUTE WITH RULE 54, AND RIGIDITY OF THE SECOND-ORDER AUTOMATON 94R

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ABSTRACT. For an elementary rule f, the second-order automaton F_f on $(\{0,1\}^2)^{\mathbb{Z}}$, given by $y_{t+1} = y_{t-1} \oplus f(y_t)$, is reversible, and the automorphism group of its spacetime subshift is the group of the invertible cellular automata that commute with F_f. We determine this group for the rules 94 and 95. The automaton 95R consists of two interleaved copies of the reversible automaton known as Rule 54, and we show that the group of the invertible cellular automata that commute with Rule 54 is $\mathcal{N} \times \mathcal{N}$, where $\mathcal{N}$ is the group of automorphisms of the golden mean shift that preserve the number of 1s of every finite point. The proof uses the known description of Rule 54 by particles that move at light speed and are delayed at their crossings, which we state and prove in the form that is needed, for all configurations. For 95R the group is $(\mathcal{N} \times \mathcal{N} \times \mathbb{Z})/\langle(\sigma^{-1}, \sigma, -2)\rangle$. It contains every finite group, and $\langle \sigma, F_{95} \rangle$ is central of infinite index in it; yet F_{95} has no blocking word and no eigenfield of any speed. Thus a large automorphism group needs neither a blocking word nor linearity. The automorphisms outside $\langle \sigma, F_{95} \rangle$ have radius at least 7 (computer-assisted). For 94 the group is $\langle \sigma, F_{94} \rangle \cong \mathbb{Z}^2$; the proof uses the scattering of kinks. For these rules and for rule 91 the spacetime relation is a discrete wave equation with a source, and we prove that a block map of any radius that commutes with F_f is determined by its values on the fixed points of $\sigma^{\pm 1} F_f$. For 91 the group is not determined; we give partial results, and show where the method fails for rule 155.

1. INTRODUCTION

Let $\mathcal{A}$ be a finite alphabet and σ the left shift on $\mathcal{A}^{\mathbb{Z}}$. A reversible cellular automaton F on $\mathcal{A}^{\mathbb{Z}}$ determines a two-dimensional subshift, its *spacetime* $X_F \subset \mathcal{A}^{\mathbb{Z}^2}$, whose rows are the configurations $F^n(c)$, $n \in \mathbb{Z}$. The automorphism group of X_F is the centraliser of F in the group $\mathrm{Aut}(\mathcal{A}^{\mathbb{Z}})$ of invertible block maps (Lemma 2.2): its elements are the invertible cellular automata that commute with F, that is, the invertible maps given by a translation-invariant local rule that commute with the dynamics. It always contains $\langle \sigma, F \rangle$. We determine it for two nonlinear automata, with opposite outcomes, and obtain partial results for a third. The second of the two, 95R, consists of two interleaved copies of the reversible automaton of Bobenko, Bordemann, Gunn and Pinkall [4], known as Rule 54, and the group of the invertible cellular automata that commute with Rule 54 is determined as well (Theorem B(0)).

The automata are second-order elementary automata in the sense of Fredkin and Wolfram [45, 46]. For an elementary rule $f\colon \{0,1\}^3 \to \{0,1\}$ define, on $\mathcal{A} = \{0,1\}^2$ with cells written (x, y),

$$F_f(c)_i = \big(y_i,\ x_i \oplus f(y_{i-1}, y_i, y_{i+1})\big). \tag{1}$$

Reading y as the current row and x as the previous one, this is the recursion $y_{t+1} = y_{t-1} \oplus f(y_t)$, reversible for every f. Following Takesue [41], one writes fR for F_f. As models of statistical mechanics these automata were introduced by Takesue [41, 42], who reduced them, by reflection and the cellwise complement, to 88 classes [42, Table I]. We consider the rules 91, 94 and

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95, and their conjugates 218, 122 and 250 under the cellwise complement. For these rules $f(a,b,c) = a \oplus c \oplus N(a,b,c)$ with a function N that vanishes on most triples, so that the rows of a spacetime diagram satisfy a discrete wave equation with a source, and the fixed points of σF_f and of $\sigma^{-1}F_f$, the configurations that move at speed 1 to the right and to the left, form two copies of a shift of finite type Φ_f of positive entropy (Section 2.3).

1.1. **Results.**

Theorem A. $\mathrm{Aut}(X_{F_{94}}) = \langle \sigma, F_{94} \rangle \cong \mathbb{Z}^2$, *and the same holds for rule* 122.

This is Theorem 5.9. The automaton 122R is the example with which irreversible behaviour of a reversible rule is illustrated in [46, p. 442]. For the second theorem let $\mathsf{G} \subset \{0,1\}^{\mathbb{Z}}$ be the golden mean shift, the set of sequences with no two consecutive 1s, and let $\mathcal{N}$ be the group of those automorphisms of (G, σ) that preserve the number of 1s of every point with finitely many 1s. Let $\mathcal{W}$ be the set of all $u\colon \mathbb{Z}^2 \to \{0,1\}$ with $u(j,k) \oplus u(j-1,k-1) = u(j,k-1) \vee u(j-1,k)$ for all (j,k); it is the spacetime of the Rule 54 automaton in light-cone coordinates (Section 1.3), and $\mathrm{Aut}(\mathcal{W})$ is the group of the homeomorphisms of $\mathcal{W}$ that commute with the translations of $\mathbb{Z}^2$.

Theorem B. *There is an isomorphism*

$$\mathrm{Aut}(X_{F_{95}}) \;\cong\; \big(\mathcal{N} \times \mathcal{N} \times \mathbb{Z}\big) \big/ \big\langle (\sigma^{-1}, \sigma, -2) \big\rangle,$$

under which σ *is the class of* $(\mathrm{id}, \mathrm{id}, 1)$ *and* F_{95} *the class of* $(\mathrm{id}, \sigma, -1)$. *Moreover:*

(0) *For the spacetime* $\mathcal{W}$ *of the Rule 54 automaton,* $\mathrm{Aut}(\mathcal{W}) \cong \mathcal{N} \times \mathcal{N}$, *the two light-cone translations corresponding to* (σ, id) *and* (id, σ) *(Theorem 6.7); that is, the group of the invertible cellular automata that commute with Rule 54 is* $\mathcal{N} \times \mathcal{N}$.

(1) $\bigoplus_{m \geq 1} S_m$ *embeds in* $\mathcal{N}$; *so* $\mathrm{Aut}(X_{F_{95}})$ *contains every finite group and is not virtually* $\mathbb{Z}^n$ *for any* n.

(2) $\langle \sigma, F_{95} \rangle \cong \mathbb{Z}^2$ *is central in* $\mathrm{Aut}(X_{F_{95}})$, *with quotient* $(\mathcal{N}/\langle \sigma \rangle)^2$; *in particular it has infinite index.*

(3) F_{95} *has no blocking word, and for all* $(k,p) \in \mathbb{Z}^2$ *with* $p \geq 1$ *and all* $M \geq 0$ *the window graph* $W_M^{(k,p)}$ *of* $\sigma^k F_{95}^p$ *is strongly connected; equivalently,* F_{95} *has no eigenfield of any speed.*

(4) (computer-assisted) *Every automorphism of* $X_{F_{95}}$ *of radius at most* 6 *lies in* $\langle \sigma, F_{95} \rangle$, *and exactly* 36 *automorphisms of radius* 7 *do not.*

The same holds for rule 250.

The isomorphism is Theorem 6.9; (0) is Theorem 6.7, in which $\mathcal{W}$ is defined by (10); (1) is Lemma 6.8 and Corollary 6.10, (2) is Corollary 6.11, (3) is Propositions 6.17 and 6.18, and (4) is Proposition 6.15; blocking words, window graphs and eigenfields are defined in Section 2.4, and the statements for 122 and 250 follow by Lemma 2.3 (κ is a cellwise conjugacy commuting with σ, so blocking words and the graphs $W_M^{(k,p)}$ are carried over as well). The automaton 95R, in the form 250R, is the reversible automaton known as Rule 54, taken on both sublattices of spacetime (Section 1.3); so Theorem B(0) gives the automorphism group of the spacetime of that automaton, and the displayed isomorphism that of two interleaved copies.

In the constructions of large automorphism groups for second-order automata in [13], the automorphisms outside $\langle \sigma, F \rangle$ of rules that are not affine come from a blocking word [34, 3] with an autonomous degree of freedom next to it (a blocking word cuts the line into parts that do not influence each other, and a local rule can then act on a degree of freedom that the word confines), and the affine rules 90 and 165, which have no blocking word, show that linearity can take its place. One may ask whether, for a rule that is not affine, the absence of blocking

words forces $\mathrm{Aut}(X_{F_f})$ to be virtually $\mathbb{Z}^2$. By (1) and (3) the answer is negative (Corollary 6.19): $f_{95}(a,b,c) = 1 \oplus ac$ is not affine. Whether the class of 95 is the only exception among the elementary rules is Question 1.6 of [13]. Part (4) shows that an exhaustive search at small radius does not detect such a group: the block maps of radius at most 4 that commute with F_{95} and are invertible are the maps $\sigma^a F^b$ (Lemma 6.20 and Proposition 6.21, computer-assisted).

Both theorems rest on the following statement, which we call Lemma L. Let Φ^+ and Φ^- be the sets of fixed points of σF_f and of $\sigma^{-1} F_f$.

Theorem C (Lemma L)**.** *Let $f \in \{91, 94, 95\}$. A block map of any radius that commutes with F_f and fixes every point of $\Phi^+ \cup \Phi^-$ is the identity. Consequently two block maps that commute with F_f and agree on $\Phi^+ \cup \Phi^-$ are equal, and the restriction $\mathrm{Aut}(X_{F_f}) \to \mathrm{Aut}(\Phi^+, \sigma) \times \mathrm{Aut}(\Phi^-, \sigma)$ is injective.*

This is Theorem 3.1 with Corollaries 3.2 and 3.9. For rule 91 it is the main result: $\mathrm{Aut}(X_{F_{91}})$ is not determined. We show that an automorphism maps the points of $\Phi^\pm$ with a single 0 to points that satisfy the constraints of Proposition 7.4, and why the arguments used for 94 and 95 stop there (Remark 7.5); Section 7.1 contains two further statements on the scattering for this rule. For rule 155, which is of the same form with a source N that is not symmetric under reflection, the method of proof of Lemma L fails: a front carrying the pair 11 makes the source fire at every step (Proposition 8.1), while to the right of a perturbation the medium is transparent (Proposition 8.3 and Corollary 8.4). Whether Lemma L holds for 155 is open.

1.2. **The method.** *The wave equation.* For $f \in \{91, 94, 95\}$ write $f(a,b,c) = a \oplus c \oplus N(a,b,c)$; then N is the indicator of 000 for 91, of 010 for 94, and of $\{000, 010\}$ for 95. The y-bits $Y(t,i)$ of the rows $F^t(c)$ satisfy

$$Y(t+1,i) \oplus Y(t-1,i) \oplus Y(t,i-1) \oplus Y(t,i+1) = N\big(Y(t,i-1), Y(t,i), Y(t,i+1)\big),$$

and we say that Y *fires* where the right side is 1. In the light-cone coordinates $q = t - i$, $p = t + i$ the points of Φ^+ are the solutions $\pi(q)$ and those of Φ^- the solutions $\xi(p)$, with $\pi, \xi \in \Phi_f$, the sequences in which no triple is in the support of N. On a quadrant bounded by two light rays a solution is determined by its values on the two rays, as in a Goursat problem, and it is the sum of the two boundary values and the parity of the number of firings in a rectangle (the d'Alembert formula, Lemma 3.3).

Lemma L. Let c differ in finitely many cells from a spatially periodic configuration without firings. Along the light rays of each family, the first position at which the source can fire moves outwards by at least 3 in every two steps (Lemmas 3.4 and 3.8), and the product of the two rates exceeds 1; so Y_c fires only finitely often in positive time (Lemma 3.5), and every window of a late row $F^t(c)$ is a window of a point of $\Phi^+ \cup \Phi^-$ (Proposition 3.6). A block map h that commutes with F_f and fixes $\Phi^+ \cup \Phi^-$ therefore fixes $F^t(c)$ for large t, hence c, and such configurations are dense.

Scattering. Let $P \in \Phi^+$ be placed to the left of $X \in \Phi^-$, both with eventually 2-periodic diagrams. The diagram of the resulting configuration fires finitely often and then consists of a point $\Gamma \in \Phi^-$ on the left and a point $\Omega \in \Phi^+$ on the right, which move apart (Lemma 4.1); we write $\mathcal{S}(P, X) = (\Gamma, \Omega)$. Time reversal inverts $\mathcal{S}$ (Proposition 4.2), and every block map h that commutes with F_f intertwines it: $\mathcal{S}(hP, hX) = (h\Gamma, h\Omega)$ (Lemma 4.5).

Kink rigidity. For 94 the sequences of Φ_{94} are those without 010, and the simplest non-periodic points of $\Phi^\pm$ are the *bare kinks* $\cdots 000111 \cdots$ and $\cdots 111000 \cdots$. Two kinks that cross with the value 1 between them and at equal parity make the source fire twice, and both are delayed by one step; otherwise they cross freely (Lemma 4.4). From these crossings and the intertwining property we show that every automorphism translates the bare kinks of Φ^+ by one amount and those of Φ^- by another, of the same parity, so that after composition with an element of $\langle \sigma, F \rangle$

it fixes all of them (Theorem 4.9). The same holds for 95 with the *bare defects* $\cdots 111\,0\,111 \cdots$ in place of the kinks (Theorem 4.12).

The window property and prepending, for 94. Let g be an automorphism of radius R that fixes the bare kinks. Its set of fixed points in $\Phi^+ \cup \Phi^-$ is closed under $\mathcal{S}$ and $\mathcal{S}^{-1}$, and it contains every point of Φ^+ all of whose words of length $2R+2$ are words of points of Φ^+ that g fixes, and likewise for Φ^-: the *window property*. The *scattering closure* C_L is the smallest set that contains the two constant configurations and the bare kinks, is invariant under the shift, the reflection and the time reversal, and has these properties for windows of length L (Definition 5.1 and Lemma 5.2). A kink is prepended to a point U of C_L as follows (Lemma 5.3): bare kinks are scattered off the point $[b;U]$ with the additional kink in such a way that the new kink is never delayed, while the first kink of the rest is delayed at every second scattering; after $2L$ scatterings the two are so far apart that the window property applies, and the closure under $\mathcal{S}^{-1}$ leads back to $[b;U]$. With a similar lemma for isolated zeros this shows that C_L contains every point with finitely many kinks (Theorem 5.8), hence, by the window property, all of $\Phi^+ \cup \Phi^-$, and Lemma L gives $g = \mathrm{id}$.

The particle description, for 95. For 95 the scattering closure of the bare defects is a proper subset of $\Phi^+ \cup \Phi^-$, because $\mathcal{S}$ only translates the zeros of a point (Corollary 6.5), and a normalised automorphism need not be the identity. Since f_{95} does not read its middle argument, the two sublattices of a diagram evolve independently, each as a copy of a $\mathbb{Z}^2$-subshift $\mathcal{W}$ (Lemma 6.1), the spacetime of Rule 54. We use the known particle picture of Rule 54 [4] in the following form (Theorem 6.3; see Section 1.3): $\mathcal{W}$ is the image of $\mathsf{G} \times \mathsf{G}$ under a continuous map Ψ, at most two-to-one, with $\Psi(\sigma^a x, \sigma^b y) = \tau^{\theta_{x,y}(a,b)}\Psi(x,y)$ for a cocycle θ. The 1s of x and of y are right- and left-moving particles, and the delays at their crossings are removed by this change of coordinates. An element φ of $\mathcal{N}$ has a flux λ_φ (Lemma 6.6), and for $\varphi, \psi \in \mathcal{N}$ the map $\Psi(x,y) \mapsto \Psi(\varphi x, \psi y)$, corrected by the translation by $(\lambda_\psi(y), \lambda_\varphi(x))$, is a well-defined automorphism of $\mathcal{W}$; by Lemma L these are all, so $\mathrm{Aut}(\mathcal{W}) \cong \mathcal{N} \times \mathcal{N}$ (Theorem 6.7). The group $\mathrm{Aut}(X_{F_{95}})$ is generated by the diagonal action of $\mathrm{Aut}(\mathcal{W})$ on the two sublattices and by σ, which exchanges them (Theorem 6.9); the proof uses Theorem 4.12 and Lemma L.

1.3. **Relation to the literature.** *Rule 54.* On one sublattice the relation of 95R is, after complementation, the update rule $s_x^{t+1} = s_x^{t-1} \oplus (s_{x-1}^t \vee s_{x+1}^t)$ on a checkerboard lattice: the system $\mathcal{W}$ is the spacetime of the reversible automaton introduced by Bobenko, Bordemann, Gunn and Pinkall [4], known as Rule 54 or RCA54 and much studied as an exactly solvable model [7]. In [4] the rule is written on a diagonally oriented lattice, as it is in the light-cone coordinates used here. That the spacetime pattern of a rule that does not read its middle argument separates into two independent parts, like the two colours of a checkerboard, is noted in [42, p. 386], where the classes of these rules are listed, among them that of 95R, and again for 90R, 95R and 165R in [43, Section 4]; and that 250R is decomposed into two independent copies of RCA54 in [27, Section 3.1]. (RCA54 is not the second-order automaton 54R, whose automorphism group also contains every finite group, for a different reason [13].)

The particle picture of Rule 54 is known, and we do not claim it: left- and right-moving particles at light speed whose only interaction is a delay by one step at each crossing. The conservation of the particles and their pairwise interaction are proved in [4, Theorem 1, Corollary 1] for arbitrary configurations, by an inspection of cases; the delays are derived there for finitely many particles on the background 0 [4, Corollary 2]. Local rules that read off the particles of both families from an arbitrary configuration are given in [23, 1, 18, 31]. The description of a state by the asymptotic spacings of its particles, with a hard-core constraint, is given in [19, 18], where it is inferred from simulations: in the asymptotic coordinates no two particles of one family are adjacent, while on the lattice two particles of one family are separated by at least two sites or

have a particle of the other family between them [7, Section 6]. None of these sources states or proves the global parametrisation: that every point of $\mathcal{W}$ is $u_{x,y}$ for some $(x, y) \in \mathsf{G} \times \mathsf{G}$, with the fibres of Theorem 6.3(d) and the action of the translations of (b). The proofs of Theorems 6.7 and 6.9 need exactly this, and since no proof in this form is available in the sources, Theorem 6.3 is proved here. A rigorous description of another soliton automaton in coordinates in which the solitons move freely is the soliton decomposition of the box-ball system [17].

Local conserved quantities of Rule 54 whose number grows with the size of the support like the Fibonacci numbers, identified with arrangements of particles that move in one direction, are reported from numerical searches in [31, Section 2.2.3] and [7, Section 3.3], with the remark in [31] that a systematic construction of these quantities is missing; because of them the model is called superintegrable in [21, 22, 37]. Operators that commute with the dynamics are known in the quantum setting: local Hamiltonians and transfer matrices [18, 22] and charges of quantum deformations of the model [37]; Rule 54 is not among the Yang–Baxter integrable models with three-site interactions [20]. These are quantum operators and not cellular automata. Theorems 6.3 and 6.7 are a rigorous counterpart of these observations within cellular automata: the arrangements of the particles of one family are the points of the golden mean shift G, and every automorphism of G that conserves the number of particles, applied to one family, is an invertible cellular automaton that commutes with the dynamics. We have not found the group of the invertible cellular automata that commute with the dynamics (Theorems 6.7 and 6.9), or the group $\mathcal{N}$, in the literature.

Conserved quantities. The additive conserved quantities of the second-order elementary automata are determined for densities on two adjacent cells in [42, Table II] and on at most three adjacent cells in [24, Tables 3–5], and counted for densities on at most four adjacent cells in [44, Table 1]; see also [43]. The two that we use for 91 are recalled in Remark 7.5; for one of them no verification is written out in the sources, and one is included (Lemma 7.6). That a cellular automaton on a full shift conserves an additive quantity if and only if the change of its density is the difference of a flux and its translate is known [24, Theorems 2.1, 2.2, 2.6 and 2.7], [5, 38, 36]; the framework on shifts of finite type is in [29, Section 4.1], where the implication from conserved to locally conserved is stated for full shifts. Lemma 6.6 is this criterion on the golden mean shift, with the cocycle identities that we need.

Automorphism groups. The automorphism group of a mixing shift of finite type such as G contains every finite group and free groups [6]; the subgroup $\mathcal{N}$ of Theorem B still contains $\bigoplus_m S_m$, and we do not know its structure (Section 9). The general results on automorphism groups of $\mathbb{Z}^2$-subshifts do not apply to the spacetime X_F: the positive-entropy theorem of Hochman [26], because X_F has zero $\mathbb{Z}^2$-entropy; the methods for subshifts of low complexity [10, 15], because the rows of X_F are all of $\mathcal{A}^{\mathbb{Z}}$; and the algebraic rigidity of Kitchens and Schmidt [30], because the rules of this paper are not affine. The spacetime of a general endomorphism is studied in [9], and the problem of finding the block maps that commute with a given one goes back to [8].

Gliders. The scheme of the proof of Theorem A has a precedent in the main lemma of Kopra on glider automorphisms [32, Lemma 1], [33, Lemma 5.7]. There X is a subshift with a group $\mathcal{G}$ of automorphisms that has left- and right-going fleets of gliders, defined as the finite points on which an element of $\mathcal{G}$ acts as a power of the shift, such that every finite point is mapped by an element of $\mathcal{G}$ to a point all of whose windows of a given length are windows of fleets, and such that an element of $\mathcal{G}$ delays two fleets that meet at a prescribed distance; the conclusion is that the automorphisms that commute with $\mathcal{G}$ and preserve the orbit of $0^{\mathbb{Z}}$ are the powers of σ. The three steps are those of Lemma L, kink rigidity and the closure: the fleets correspond to $\Phi^{\pm}$, the diffusion to Proposition 3.6, and the delays to Lemma 4.4. In [32, 33] the group $\mathcal{G}$ is constructed for the purpose, on mixing shifts of finite type and on conjugates of transitive sofic shifts, and the

result is a finitary form of the theorem of Ryan [39] that the centre of the automorphism group of a full shift consists of the powers of the shift: two automorphisms whose common centraliser is $\langle\sigma\rangle$ (for the full 4-shift a finite set with this property is given in [40, Theorem 2.2]). Here the automaton F is given and not constructed, the sets $\Phi^{\pm}$ are shifts of finite type of positive entropy with several backgrounds, the delays occur only at equal parity, and the conclusion concerns the centraliser of the single map F.

Companion papers. The group $\mathrm{Aut}(X_{F_f})$ is $\langle\sigma, F_f\rangle$ for the non-affine rules that are permutive in an outer variable [12] and for the classes of 110, 14, 46 and 79, and it is $\langle\sigma, F_f\rangle \times \mathbb{Z}/2$ for the classes of 24 and 189 [11]; in the cases with group $\langle\sigma, F_f\rangle$ an automorphism is first normalised by an element of $\langle\sigma, F\rangle$ and then shown to be the identity, as in the proof of Theorem A, but the structures used are different: for 110 the fixed points of $\sigma^{\pm 1}F$ are two constants and one orbit of kinks, and here they form a shift of finite type of positive entropy. Large groups from walls and from linearity are constructed in [13], and [14] records what is known for each of the 88 classes. No result of the present paper depends on these papers.

1.4. **Computer-assisted statements.** The following statements rest on computations, and each is marked as such where it is stated: the enumeration of the particle-conserving block maps of G that depend on at most 8 cells (Lemma 6.14) and Proposition 6.15, that is, Theorem B(4); the census of the block maps of radius at most 4 that commute with F_{95} (Lemma 6.20) and their identification (Proposition 6.21); the search for 175 (Remark 6.22); the computations reported in Remarks 7.5 and 7.7 for 91 and 94; and those reported in Remark 8.2 and Appendix B for 155. All of them are exact and exhaustive over a finite set. In Proposition 6.15(b), that the explicit block map h_* of radius 7 is an automorphism of $X_{F_{95}}$ outside $\langle\sigma, F_{95}\rangle$ is verified on all windows from its tables, without the theory of Section 6. Everything else is proved by hand, in particular Theorem A, Theorem B with (1)–(3), and Theorem C. Appendix A describes the computations, and in addition computations that confirm results proved by hand; these are not used in any proof.

Organisation. Section 2 fixes notation, describes the fixed points of $\sigma^{\pm 1}F_f$, and defines blocking words, window graphs and eigenfields. Section 3 proves Lemma L. Section 4 introduces the scattering map and proves the kink rigidity theorems for 94 and 95, and Section 5 proves Theorem A. Section 6 treats rule 95 and proves Theorem B. Sections 7 and 8 contain the results on 91 and 155, and Section 9 lists questions. Appendix A describes the computations, and Appendix B the front strips of 155R.

2. Preliminaries

Throughout, $\mathcal{A} = \{0,1\}^2$, and a cell $(x,y) \in \mathcal{A}$ is also written as the integer $2x+y \in \{0,1,2,3\}$; thus $3 = (1,1)$ and $0 = (0,0)$. The left shift is $\sigma(c)_i = c_{i+1}$. A *block map of radius* R is a map $g\colon \mathcal{A}^{\mathbb{Z}} \to \mathcal{A}^{\mathbb{Z}}$ of the form $g(c)_i = \gamma(c_{i-R}, \dots, c_{i+R})$ for a local rule $\gamma\colon \mathcal{A}^{2R+1} \to \mathcal{A}$. A block map of radius R is also a block map of every larger radius, so "radius at most R" and "radius R" describe the same set of maps. By the Curtis–Hedlund–Lyndon theorem [25], the continuous maps of $\mathcal{A}^{\mathbb{Z}}$ into itself that commute with σ are the block maps, and $\mathrm{Aut}(\mathcal{A}^{\mathbb{Z}})$ is the group of invertible ones.

For an elementary rule f, identified with its Wolfram number $\sum_{a,b,c} f(a,b,c)2^{4a+2b+c}$, the automaton F_f is defined by (1). It has radius 1, and F_f^b has radius at most $|b|$, so $\sigma^a F_f^b$ is a block map of radius at most $|a|+|b|$ that commutes with F_f.

Definition 2.1. The *spacetime subshift* of a reversible cellular automaton F on $\mathcal{A}^{\mathbb{Z}}$ is

$$X_F = \left\{(F^n(c))_{n\in\mathbb{Z}} :\ c \in \mathcal{A}^{\mathbb{Z}}\right\} \subset \mathcal{A}^{\mathbb{Z}^2},$$

with the $\mathbb{Z}^2$-action generated by the horizontal shift σ and the vertical shift τ, $(\tau z)_n = z_{n+1}$. An *automorphism* of X_F is a homeomorphism of X_F commuting with both.

Lemma 2.2. *For reversible F the map $c \mapsto (F^n(c))_{n\in\mathbb{Z}}$ is a homeomorphism $\mathcal{A}^{\mathbb{Z}} \to X_F$ conjugating (σ, F) to (σ, τ), and it induces an isomorphism*

$$\mathrm{Aut}(X_F) \;\cong\; \{g \in \mathrm{Aut}(\mathcal{A}^{\mathbb{Z}}) :\; gF = Fg\}.$$

Under this isomorphism $\langle\sigma, \tau\rangle$ corresponds to $\langle\sigma, F\rangle$.

Proof. The map is continuous, injective (the 0-th row recovers c), and surjective onto X_F by definition, hence a homeomorphism of compact spaces; it carries σ to σ and F to τ since $F^n(F(c)) = F^{n+1}(c)$. A homeomorphism of $\mathcal{A}^{\mathbb{Z}}$ commuting with σ is a block map by Curtis–Hedlund–Lyndon, and commuting with τ means commuting with F. □

Thus determining $\mathrm{Aut}(X_F)$ means determining which invertible block maps commute with F, and we regard the elements of $\mathrm{Aut}(X_F)$ as block maps. The block maps that commute with F and are not invertible correspond in the same way to the endomorphisms of X_F that are not automorphisms; if F fixes a constant configuration, the constant map onto it is one. A constant configuration fixed by F_f is of the form $(a, a)^{\mathbb{Z}}$ with $f(a, a, a) = 0$; for $f \in \{91, 95\}$ the only one is $(1, 1)^{\mathbb{Z}}$, and for $f = 94$ there are two, $(0, 0)^{\mathbb{Z}}$ and $(1, 1)^{\mathbb{Z}}$.

2.1. **Symmetries.** Three involutions of $\mathcal{A}^{\mathbb{Z}}$ act on second-order automata: the cellwise complement $\kappa(x, y) = (1 - x, 1 - y)$, the reflection $\mathrm{rev}(c)_i = c_{-i}$, and the time reversal $\rho(x, y) = (y, x)$, applied in every cell. The first two are the Boolean conjugation and the reflection of [42, Section 2], and that exchanging the two rows reverses time is noted in [41, p. 2500].

Lemma 2.3. *For every elementary rule f, $\kappa F_f \kappa = F_{\kappa f}$ with $\kappa f(a, b, c) = f(1 - a, 1 - b, 1 - c)$; $\mathrm{rev}\, F_f\, \mathrm{rev} = F_{f^r}$ with $f^r(a, b, c) = f(c, b, a)$, and $\mathrm{rev}\,\sigma\,\mathrm{rev} = \sigma^{-1}$; and $\rho F_f \rho = F_f^{-1}$. For $\theta \in \{\kappa, \mathrm{rev}\}$ the map $g \mapsto \theta g \theta^{-1}$ preserves the radius of block maps, maps the block maps that commute with F_f onto those that commute with $F_{f'}$ and $\mathrm{Aut}(X_{F_f})$ onto $\mathrm{Aut}(X_{F_{f'}})$, where $f' = \kappa f$ or f^r, and maps $\langle\sigma, F_f\rangle$ onto $\langle\sigma, F_{f'}\rangle$.*

Proof. By (1), $F_f(\kappa c)_i = (1 - y_i,\, 1 - x_i \oplus f(1 - y_{i-1}, 1 - y_i, 1 - y_{i+1}))$, and applying κ gives $(y_i,\, x_i \oplus \kappa f(y_{i-1}, y_i, y_{i+1}))$. Similarly $\mathrm{rev}\, F_f\, \mathrm{rev}(c)_i = F_f(\mathrm{rev}\, c)_{-i} = (y_i,\, x_i \oplus f(y_{i+1}, y_i, y_{i-1}))$. And $F_f(\rho c)_i = (x_i,\, y_i \oplus f(x_{i-1}, x_i, x_{i+1}))$, whose image under ρ is $F_f^{-1}(c)_i$. Conjugation by a cellwise map or by rev preserves the radius, and the remaining statements follow, since κ commutes with σ and rev conjugates σ to σ^{-1}. □

So $\mathrm{Aut}(X_{F_f})$, and every statement below about block maps of bounded radius, depends only on the class of f under κ and reflection; there are 88 classes [41], [42, Table I]. The rules 91, 94 and 95 are invariant under reflection, and their classes are $\{91, 218\}$, $\{94, 122\}$ and $\{95, 250\}$; the class of 155 is $\{155, 203, 211, 217\}$.

2.2. **Diagrams.** For $c \in \mathcal{A}^{\mathbb{Z}}$ the *diagram* of c is $Y_c\colon \mathbb{Z}^2 \to \{0, 1\}$, where $Y_c(t, i)$ is the y-bit of $F^t(c)_i$. The x-row of $F(d)$ is the y-row of d, so the x-bit of $F^t(c)_i$ is $Y_c(t - 1, i)$, and $c_i = (Y_c(-1, i), Y_c(0, i))$. By (1) every diagram satisfies the *spacetime relation*

$$Y(t + 1, i) \oplus Y(t - 1, i) = f\big(Y(t, i - 1), Y(t, i), Y(t, i + 1)\big) \qquad \text{for all } (t, i) \in \mathbb{Z}^2, \tag{2}$$

so two consecutive rows determine all rows, and $c \mapsto Y_c$ is a bijection from $\mathcal{A}^{\mathbb{Z}}$ onto the solutions of (2). For $t \geq 0$, $Y_c(t, i)$ depends only on $c_{[i-t, i+t]}$. Moreover $Y_{\sigma c}(t, i) = Y_c(t, i + 1)$, $Y_{F(c)}(t, i) = Y_c(t + 1, i)$ and $Y_{\rho c}(t, i) = Y_c(-1 - t, i)$, the last because $F^t(\rho c) = \rho F^{-t}(c)$.

2.3. **The fixed points of $\sigma^{\pm 1}F_f$.** The fixed points of σF_f and of $\sigma^{-1}F_f$ are the configurations whose diagrams are constant along the light rays of one of the two families. They are described by a shift of finite type on $\{0,1\}$.

Lemma 2.4. *For an elementary rule f let $\Phi_f \subseteq \{0,1\}^{\mathbb{Z}}$ be the set of sequences φ with*

$$\varphi(k-1) \oplus \varphi(k+1) = f\big(\varphi(k-1), \varphi(k), \varphi(k+1)\big) \qquad \text{for all } k \in \mathbb{Z}.$$

The maps $\varphi \mapsto c$, $c_i = (\varphi(i+1), \varphi(i))$, and $\varphi \mapsto d$, $d_i = (\varphi(i-1), \varphi(i))$, are bijections from Φ_f onto the fixed points of σF_f and of $\sigma^{-1}F_f$ respectively. If Φ_f contains a sequence that is not periodic, then $(a,b) \mapsto \sigma^a F_f^b$ is injective, so $\langle \sigma, F_f\rangle \cong \mathbb{Z}^2$ and F_f has infinite order.

Proof. If $c_i = (\varphi(i+1), \varphi(i))$, then by (1), $F_f(c)_i = \big(\varphi(i),\ \varphi(i+1) \oplus f(\varphi(i-1), \varphi(i), \varphi(i+1))\big)$, which is $(\varphi(i), \varphi(i-1)) = c_{i-1}$ exactly when the displayed equation holds at $k = i$. So $F_f c = \sigma^{-1}c$, that is $\sigma F_f c = c$. Conversely, if $\sigma F_f c = c$, then $\varphi(i) = y(c_i)$ satisfies $x(c_i) = y(F_f^{-1}c)_i = y(\sigma c)_i = \varphi(i+1)$, and the equation holds by the same computation. The case of $\sigma^{-1}F_f$ is identical with $F_f d = \sigma d$. Let $\varphi \in \Phi_f$ be non-periodic, so that the corresponding c and d are not periodic under σ. If $\sigma^a F_f^b = \mathrm{id}$, then $c = \sigma^a F_f^b c = \sigma^{a-b}c$ and $d = \sigma^{a+b}d$, so $a - b = a + b = 0$. □

The set Φ_f is the shift of finite type on $\{0,1\}$ given by the allowed triples (u,v,w) with $u \oplus w = f(u,v,w)$. For the four rules of this paper it is the set of sequences that contain no 000 for $f = 91$; no 010 for $f = 94$; neither 000 nor 010, that is, no two zeros at distance 2, for $f = 95$; and none of 000, 110, 111 for $f = 155$ (Sections 3 and 8). In each case Φ_f has positive entropy and contains sequences that are not periodic, for instance $\cdots 111\,0\,111 \cdots$ for 91 and 95, $\cdots 000\,111 \cdots$ for 94 and $\cdots 0101\,001\,0101 \cdots$ for 155. So by Lemma 2.4,

$$\langle \sigma, F_f\rangle \cong \mathbb{Z}^2 \qquad \text{for } f \in \{91, 94, 95, 155\}.$$

For rule 110, by contrast, Φ_{110} consists of $0^{\mathbb{Z}}$, $1^{\mathbb{Z}}$ and one σ-orbit of non-periodic points, the *kinks* on which the determination of $\mathrm{Aut}(X_{F_{110}})$ in [11] is built.

2.4. **Blocking words, window graphs and eigenfields.** Let F be a cellular automaton of radius 1 on $\mathcal{A}^{\mathbb{Z}}$. A word w on an interval I of cells is *blocking* for F (cf. the proof of [34, Theorem 3] and, for the term, [3, Section 2.2]) if there is a cell $i_0 \in I$ such that $F^n(c)_{i_0} = F^n(c')_{i_0}$ for all $n \geq 0$ and all configurations c, c' that carry w on I. A word that is fixed by F in every context is blocking.

Let F be a bijective cellular automaton on $\mathcal{A}^{\mathbb{Z}}$. For $(k,p) \in \mathbb{Z}^2$ with $p \geq 1$ and $M \geq 0$, let $W_M^{(k,p)}$ be the graph on $\mathcal{A}^{M+1}$ with an edge from each word to each of its possible images under $\sigma^k F^p$, the cells outside the window ranging over all values: there is an edge from $c_{[0,M]}$ to $\sigma^k F^p(c)_{[0,M]}$ for every $c \in \mathcal{A}^{\mathbb{Z}}$. The graph $W_M^{(1,1)}$ is the light-speed window graph used for the permutive rules in [12], and $W_M^{(0,1)}$ is a zero-speed graph. A nonconstant block map $\vartheta\colon \mathcal{A}^{\mathbb{Z}} \to B^{\mathbb{Z}}$ to sequences over a finite alphabet B with $\vartheta \circ F^p = \sigma^{-k} \circ \vartheta$ is an *eigenfield for* (k,p), of speed k/p; composing with the indicator of a letter shows that one exists if and only if one exists with $B = \{0,1\}$.

Lemma 2.5. *Let F be a bijective cellular automaton on $\mathcal{A}^{\mathbb{Z}}$ and $(k,p) \in \mathbb{Z}^2$ with $p \geq 1$. The following are equivalent.*

1. *The graph $W_M^{(k,p)}$ is strongly connected for every $M \geq 0$.*
2. *There is no nonconstant block map $\vartheta\colon \mathcal{A}^{\mathbb{Z}} \to \{0,1\}^{\mathbb{Z}}$ with $\vartheta \circ F^p = \sigma^{-k} \circ \vartheta$.*
3. *Every continuous function $v\colon \mathcal{A}^{\mathbb{Z}} \to \{0,1\}$ with $v \circ \sigma^k F^p = v$ is constant.*

Proof. Write $\varsigma = \sigma^k F^p$. If ϑ is as in (2), then $v(c) = \vartheta(c)_0$ is continuous and $v \circ \varsigma = v$, because ϑ commutes with σ; conversely, if v is as in (3), then $\vartheta(c)_i = v(\sigma^i c)$ is a block map with $\vartheta \circ F^p = \sigma^{-k} \circ \vartheta$, and ϑ is constant if and only if v is. So (2) and (3) are equivalent.

Suppose that (3) fails, and let v be a nonconstant continuous function with $v \circ \varsigma = v$. It depends on the cells of some interval, which after replacing v by $v \circ \sigma^a$ is $[0, M]$; let T be the set of words $w \in \mathcal{A}^{M+1}$ with $v = 1$ on the configurations that carry w on $[0, M]$. Every edge of $W_M^{(k,p)}$ leads from $c_{[0,M]}$ to $\varsigma(c)_{[0,M]}$ for some c, and $v(\varsigma(c)) = v(c)$; so no edge joins T and its complement, both of which are nonempty, and (1) fails.

Suppose that (1) fails for some M. Every vertex of $W_M^{(k,p)}$ has an outgoing edge, so the graph has a strongly connected component T that no edge leaves, and T is not the whole vertex set. Let U be the set of configurations c with $c_{[0,M]} \in T$; it is open and closed, nonempty and not all of $\mathcal{A}^{\mathbb{Z}}$, and $\varsigma(U) \subseteq U$. The uniform Bernoulli measure on $\mathcal{A}^{\mathbb{Z}}$ is invariant under σ and under the surjective cellular automaton F [25], so $\varsigma(U)$ and U have the same measure; since ς is a homeomorphism, $U \setminus \varsigma(U)$ is an open set of measure 0, hence empty. So $\varsigma(U) = U$, the indicator function of U is a nonconstant continuous function invariant under ς, and (3) fails. □

Condition (1) says that $\sigma^k F^p$ is chain transitive. By the lemma, the graphs $W_M^{(\pm 1,1)}$ are strongly connected for every M if and only if F has no eigenfield for $(k, p) = (\pm 1, 1)$, and the graphs $W_M^{(0,1)}$ if and only if it has none for $(0, 1)$. An eigenfield for $(0, 1)$ is a local conservation law in the sense of Takesue [41], [42, Section 4], and $W_M^{(0,1)}$ is the graph of the transition matrix on words of $M + 1$ cells considered there; for $(k, p) = (0, 1)$ the equivalence of (1) and (2) is stated in [42, Section 4] for the second-order elementary automata (see also [24, Proposition 4.1]). The proof above is included because the lemma is needed for all (k, p) and no proof is written out there.

3. Lemma L at every radius

For $f \in \{91, 94, 95\}$ a block map that commutes with F_f is determined by what it does on the fixed points of $\sigma^{\pm 1} F_f$. We give the argument for $f \in \{91, 94\}$ and extend it to 95 in Lemma 3.8 and Corollary 3.9. Except where 95 is named, $f \in \{91, 94\}$, $F = F_f$, and

$$\Phi^+ = \{c : \sigma F(c) = c\}, \qquad \Phi^- = \{c : \sigma^{-1} F(c) = c\}.$$

Write $f(a, b, c) = a \oplus c \oplus N(a, b, c)$. Then $N = N_{91}$ is the indicator of $abc = 000$ and $N = N_{94}$ that of $abc = 010$; both are invariant under $(a, b, c) \mapsto (c, b, a)$. The spacetime relation becomes

$$Y(t+1, i) \oplus Y(t-1, i) \oplus Y(t, i-1) \oplus Y(t, i+1) = N\big(Y(t, i-1), Y(t, i), Y(t, i+1)\big), \tag{3}$$

a discrete wave equation with a source, and Φ_f is the set of sequences with no triple in the support of N: no 000 for 91, no 010 for 94. By Lemma 2.4 and its proof, the diagrams of the points of Φ^+ are the $Y(t, i) = \varphi(i - t)$ and those of Φ^- the $Y(t, i) = \varphi(i + t)$, $\varphi \in \Phi_f$; both sets are invariant under σ and F.

Theorem 3.1 (Lemma L)**.** *Let $f \in \{91, 94\}$. A block map h of any radius that commutes with F_f and fixes every point of $\Phi^+ \cup \Phi^-$ is the identity.*

Corollary 3.2. *Let $f \in \{91, 94\}$. Two block maps commuting with F_f that agree on $\Phi^+ \cup \Phi^-$ are equal, and the restriction homomorphism*

$$\mathrm{Aut}(X_{F_f}) \to \mathrm{Aut}(\Phi^+, \sigma) \times \mathrm{Aut}(\Phi^-, \sigma), \qquad h \mapsto (h|_{\Phi^+}, h|_{\Phi^-}),$$

is injective.

Invertibility of h is not assumed. The restriction is defined because an automorphism commutes with $\sigma^{\pm1}F$ and therefore maps $\Phi^{\pm}$ onto itself. The proof occupies the rest of this section. Its idea is that every finite perturbation of a valid periodic background splits, after finitely many steps, into a left-moving and a right-moving light-speed wave, so that every window of a late row is a window of $\Phi^+ \cup \Phi^-$.

Light-cone coordinates. Put $q = t - i$ and $p = t + i$, and write $Y[q,p]$ for $Y(t,i)$; the lattice is $\{[q,p] : q \equiv p \pmod 2\}$, the lines $q = \text{const}$ are right-moving light rays and the lines $p = \text{const}$ left-moving ones. A point $C = [q,p]$ has the neighbours $\mathsf{N} = [q+1, p+1] = (t+1, i)$, $\mathsf{S} = [q-1, p-1] = (t-1, i)$, $\mathsf{W} = [q+1, p-1] = (t, i-1)$ and $\mathsf{E} = [q-1, p+1] = (t, i+1)$, and (3) reads $Y(\mathsf{N}) \oplus Y(\mathsf{S}) \oplus Y(\mathsf{W}) \oplus Y(\mathsf{E}) = N(Y(\mathsf{W}), Y(C), Y(\mathsf{E}))$. We say that Y *fires* at C if the right side is 1.

Fix $q_0 \equiv p_0 \pmod 2$ and the quadrant $G = \{[q,p] : q \geq q_0,\ p \geq p_0\}$. For $q \geq q_0$ let $A(q) = Y[q,p]$ with $p \in \{p_0, p_0+1\}$, $p \equiv q$, and for $p \geq p_0$ let $B(p) = Y[q,p]$ with $q \in \{q_0, q_0+1\}$, $q \equiv p$; these are the values on two thick boundary rays of G, which meet in the two corners $[q_0, p_0]$ and $[q_0+1, p_0+1]$. For $P = [q,p] \in G$ let $\nu(P)$ be the value of Y at the corner $[q_0 + e, p_0 + e]$ with $e \equiv q - q_0 \pmod 2$. Every point $[q,p]$ of G with $q \geq q_0 + 2$ and $p \geq p_0 + 2$ is the point N of the centre $[q-1, p-1]$, whose S, W, E lie in G, so

$$Y[q,p] = Y[q,p-2] \oplus Y[q-2,p-2] \oplus Y[q-2,p] \oplus N\big(Y[q,p-2], Y[q-1,p-1], Y[q-2,p]\big), \tag{4}$$

and Y on G is determined by A and B, by induction on $q + p$.

Lemma 3.3 (d'Alembert formula)**.** *For $P = [q,p] \in G$,*

$$Y(P) = A(q) \oplus B(p) \oplus \nu(P) \\ \oplus \#\{\textit{firings at centres } [q',p'] \textit{ with } q_0 < q' < q,\ p_0 < p' < p,\ q' \not\equiv q\} \pmod 2.$$

Proof. Let $Y'(P) = Y(P) \oplus A(q) \oplus B(p) \oplus \nu(P)$. On the two boundary rays $Y' = 0$: for instance, if $p \in \{p_0, p_0+1\}$ then $Y(P) = A(q)$, and $B(p)$ is the value at the corner $[q_0 + e, p_0 + e]$ with $p_0 + e = p$, which is $\nu(P)$ because $q \equiv p$. For a centre $C = [q,p]$ with $q > q_0$, $p > p_0$, the A-terms of $\mathsf{N}, \mathsf{S}, \mathsf{W}, \mathsf{E}$ cancel in pairs (N, W have first coordinate $q+1$, and S, E have $q-1$), so do the B-terms, and ν is the same at all four points. So $Y'(\mathsf{N}) \oplus Y'(\mathsf{S}) \oplus Y'(\mathsf{W}) \oplus Y'(\mathsf{E})$ is 1 exactly if Y fires at C. Let $\chi(P)$ be the count in the statement. For a centre $C = [q,p]$ the four sets counted by $\chi(\mathsf{N})$, $\chi(\mathsf{W})$, $\chi(\mathsf{E})$, $\chi(\mathsf{S})$ consist of the centres $[q',p']$ with $q' \equiv q$ and $q' \leq q$, $p' \leq p$; $q' \leq q$, $p' \leq p-2$; $q' \leq q-2$, $p' \leq p$; and $q' \leq q-2$, $p' \leq p-2$ (and $q' > q_0$, $p' > p_0$), so modulo 2 they add up to the single centre C. Hence χ satisfies the same relation as Y', and it vanishes on the boundary rays. By induction on $q + p$, $Y' = \chi$ on G. □

Finite perturbations. Call $\beta = (\beta_0, \beta_1) \in \{0,1\}^2$ a *valid background* if $Y_\beta(t,i) = \beta_{(t+i) \bmod 2}$ solves (3). The left side of (3) vanishes for Y_β, so β is valid if and only if neither $(\beta_0, \beta_1, \beta_0)$ nor $(\beta_1, \beta_0, \beta_1)$ is in the support of N: the valid backgrounds are 11, 01, 10 for 91 and 00, 11 for 94. Let b_β be the configuration with diagram Y_β. A *finite perturbation* of β is a configuration that differs from b_β in finitely many cells. Finite perturbations of a fixed valid background are dense in $\mathcal{A}^{\mathbb{Z}}$. Let c be one; since $\sigma b_\beta = b_{(\beta_1, \beta_0)}$ and (β_1, β_0) is again valid, we may shift c and assume that it differs from b_β only in cells of $[0, w)$, for some $w \geq 1$. Let $Y = Y_c$. For $t \geq 0$, $Y(t,i)$ depends only on $c_{[i-t, i+t]}$, so $Y = Y_\beta$ at every point with $t \geq 0$ and $p < 0$ or $q \leq -w$. We take

$$p_0 = -2, \qquad q_0 \in \{-w-1, -w-2\} \text{ even}.$$

Then $q_0 \equiv p_0$, the ray carrying A lies in $\{p \leq -1\}$, and the ray carrying B lies in $\{q \leq -w\}$. Hence $A(q) = \beta_{q \bmod 2}$ for $q \geq 2$ and $B(p) = \beta_{p \bmod 2}$ for $p \geq w + 2$ (these are the points of the rays with $t \geq 0$), and every point with $t \geq 0$ at which $Y \neq Y_\beta$ lies in G.

Column and row steps. For $k \geq 0$ let $Z_k(p)$, $p \geq p_0$, be $Y[q,p]$ for the $q \in \{q_0+k, q_0+k+1\}$ with $q \equiv p$; so $Z_0 = B$. The words Z_k and Z_{k+1} differ only on the class $U_k = \{p \geq p_0 : p \equiv q_0+k\}$, where $Z_k(p) = Y[q_0+k,p]$ and $Z_{k+1}(p) = Y[q_0+k+2,p]$. Let p_b be the least element of U_k and $D(p) = Z_{k+1}(p) \oplus Z_k(p)$ for $p \in U_k$. Then $D(p_b) = A(q_0+k+2) \oplus A(q_0+k)$, and (4) at $[q_0+k+2,p]$ gives, for $p \in U_k$, $p \geq p_b+2$,

$$D(p) = D(p-2) \oplus N\big(Z_{k+1}(p-2), Z_k(p-1), Z_k(p)\big), \qquad Z_{k+1}(p) = Z_k(p) \oplus D(p), \tag{5}$$

where the N-term is the firing at the centre $[q_0+k+1, p-1]$. Since $A(q) = \beta_{q \bmod 2}$ for $q \geq 2$, $D(p_b) = 0$ for all $k \geq k_0 = 2-q_0$. Exchanging the roles of q and p, and using that N is invariant under reversal of the triple, gives the row step from the words $Z'_\ell(q)$, $q \geq q_0$, built from the rays $p \in \{p_0+\ell, p_0+\ell+1\}$, with $Z'_0 = A$; it has the same form (5), and its initial value $B(p_0+\ell+2) \oplus B(p_0+\ell)$ vanishes for $\ell \geq \ell_0 = w+2-p_0$.

For a word Z on $\{p \geq p_0\}$, a *defect* is a position $p \geq p_0+2$ with $(Z(p-2), Z(p-1), Z(p))$ in the support of N, and $\eta(Z)$ is the least defect (∞ if there is none).

Lemma 3.4 (defect climbing)**.** *Let $N \in \{N_{91}, N_{94}\}$, let Z be a word on $\{p \geq p_0\}$, let U be one of the two residue classes modulo 2 in $\{p \geq p_0\}$, with least element p_b, and let Z' be obtained from Z by the step (5) on U with $D(p_b) = 0$ (so $Z' = Z$ off U). Put $\eta = \eta(Z)$.*

(i) *$Z' = Z$ on $\{p < \eta\}$, no N-term of (5) with $p < \eta$ is 1, and $\eta(Z') \geq \eta$.*
(ii) *If $\eta \in U$ then $\eta(Z') \geq \eta+3$; if $\eta \notin U$ then $\eta(Z') = \eta$.*
(iii) *If Z'' is obtained from Z' by the step on the other class, then $\eta(Z'') \geq \eta+3$.*

Proof. (i) For $p \in U$, $p < \eta$, induction on p gives $D(p) = 0$: if $D(p-2) = 0$ then $Z'(p-2) = Z(p-2)$, so the N-term is $N(Z(p-2), Z(p-1), Z(p)) = 0$, as p is not a defect. A defect of Z' below η would use only positions where $Z' = Z$, so it would be a defect of Z.

(ii) Let $\eta \in U$. Then $D(\eta-2) = 0$ by (i), and the N-term at η is $N(Z(\eta-2), Z(\eta-1), Z(\eta)) = 1$, so $D(\eta) = 1$ and $Z'(\eta) = 1 \oplus Z(\eta) = 1$, because the last letter of 000 and of 010 is 0. A triple of Z' ending at η, $\eta+1$ or $\eta+2$ contains the position η, with value 1, in its third, second or first place. For N_{91} no such triple is 000. For N_{94}, a 1 in the first or third place is not compatible with 010, and the triple with 1 in the second place begins with $Z'(\eta-1) = Z(\eta-1)$, the middle letter of the defect 010 of Z at η, which is 1. With (i), $\eta(Z') \geq \eta+3$. Next let $\eta \notin U$. Then η and $\eta-2$ are not in U, and $\eta-1 < \eta$, so by (i) the three letters of the defect at η are unchanged, and $\eta(Z') = \eta$ by (i).

(iii) If $\eta \in U$, then $\eta(Z'') \geq \eta(Z') \geq \eta+3$ by (ii) and (i). If $\eta \notin U$, then $\eta(Z') = \eta$ lies in the other class, and (ii) for the second step gives $\eta(Z'') \geq \eta+3$. □

Lemma 3.5 (finitely many firings)**.** *In the setting above there are a^*, b^* such that every firing of Y at a centre $[q', p']$ with $q' > q_0$, $p' > p_0$ has $q' - q_0 \leq a^*$ and $p' - p_0 \leq b^*$.*

Proof. A firing at such a centre is the N-term of the column step $k = q'-q_0-1$ at $p = p'+1$, and of the row step $\ell = p'-p_0-1$ at $q = q'+1$. Put $x = q'-q_0$ and $y = p'-p_0$, and $\eta_k = \eta(Z_k)$. The steps with $k \geq k_0$ satisfy the hypothesis of Lemma 3.4, so η_k is nondecreasing for $k \geq k_0$, $\eta_{k+2} \geq \eta_k+3$, and $\eta_{k_0} \geq p_0+2$. A firing in step $k \geq k_0$ lies at height $p'+1 \geq \eta_k \geq p_0+2+3\lfloor (k-k_0)/2 \rfloor$ by Lemma 3.4(i). Hence there is a constant c_1 with $y \geq \frac{3}{2}x - c_1$ whenever $x \geq k_0+1$, and likewise, from the row steps, a constant c_2 with $x \geq \frac{3}{2}y - c_2$ whenever $y \geq \ell_0+1$. If both hypotheses hold, then $x \geq \frac{9}{4}x - \frac{3}{2}c_1 - c_2$, so x is bounded, and then $y \leq \frac{2}{3}(x+c_2)$ is bounded. If only one holds, the other coordinate is bounded by assumption and the inequality bounds the first; if neither holds, both are bounded. □

The proof needs only that the product of the two climbing rates, $\frac{3}{2} \cdot \frac{3}{2}$, exceeds 1.

Proposition 3.6 (separation)**.** *Let c be a finite perturbation of a valid background and $R \geq 0$. There is T such that for every $t \geq T$ and every i, the window $F^t(c)_{[i-R,i+R]}$ equals the window at the same position of some point of $\Phi^+ \cup \Phi^-$.*

Proof. Use the setting above, and put $Q^* = \max(q_0 + a^* + 2,\, 2)$ and $P^* = \max(p_0 + b^* + 2,\, w + 2)$; so Q^* and P^* exceed the firing box of Lemma 3.5 and the initial segments of A and B that are not background.

Let $[q, p] \in G$ with $q \geq Q^*$. In Lemma 3.3, $A(q) = \beta_{q \bmod 2} = \beta_{p \bmod 2}$; ν depends on $q \bmod 2 = p \bmod 2$; and since every firing has $q' < q$, the count is that of the firings with $p_0 < p' < p$ and $p' \not\equiv p$. So $Y[q, p] = \varphi^-(p)$ depends only on p. Put $\varphi^-(p) = \beta_{p \bmod 2}$ for $p < p_0$. Then $Y[q, p] = \varphi^-(p)$ at every point with $q \geq Q^*$ and $t \geq 0$, since such points with $p < p_0$ are background. Moreover $\varphi^- \in \Phi_f$: given p, take a centre $C = [q, p]$ with $q \geq Q^* + 1$ and $t \geq 1$; its row triple is $(\varphi^-(p-1), \varphi^-(p), \varphi^-(p+1))$, and Y does not fire at C, because for $p > p_0$ the centre lies in G with $q - q_0 > a^*$, and for $p \leq p_0$ the triple is a triple of the valid background. Let $d^- \in \Phi^-$ be the point with diagram $\varphi^-(t + i)$; it agrees with Y on $\{q \geq Q^*,\ t \geq 0\}$. In the same way, with B in place of A, $Y[q, p] = \varphi^+(q)$ on $\{p \geq P^*,\ t \geq 0\}$, where $\varphi^+(q) = \beta_{q \bmod 2}$ for $q < q_0$, the sequence $\varphi(k) = \varphi^+(-k)$ lies in Φ_f, and the point $d^+ \in \Phi^+$ with diagram $\varphi(i - t) = \varphi^+(t - i)$ agrees with Y on $\{p \geq P^*,\ t \geq 0\}$.

The window $F^t(c)_{[i-R,i+R]}$ consists of the values of Y at $(t-1, j)$ and (t, j), $|j - i| \leq R$. These points have $q \geq t - 1 - i - R$ and $p \geq t - 1 + i - R$, and the two lower bounds add up to $2t - 2 - 2R$. Let $T = R + 2 + \lceil (Q^* + P^*)/2 \rceil$. For $t \geq T$ all these points satisfy $q \geq Q^*$, or all satisfy $p \geq P^*$, and $t - 1 \geq 0$. In the first case the window is that of $F^t(d^-) \in \Phi^-$, in the second that of $F^t(d^+) \in \Phi^+$, at the same position. □

Proposition 3.7 (decay between two half-lines)**.** *Let $f \in \{91, 94, 95\}$, and let c agree with some $\psi \in \Phi^-$ on $(-\infty, -L)$ and with some $\varphi \in \Phi^+$ on (L, ∞). Then Y_c fires only finitely often in $t \geq 0$, and there are C, T such that for $t \geq T$, $F^t(c)$ agrees with a point of Φ^- on $(-\infty, t/2 - C]$ and with a point of Φ^+ on $[-t/2 + C, \infty)$.*

Proof. Let $g = 3$ for 91 and 94 and $g = 4$ for 95 (Lemmas 3.4(iii) and 3.8). Take the quadrant G with corner $[q_0, p_0] = [L + 2, -L - 2]$, the point $t = 0$, $i = -L - 2$, and the quadrant G' with corner $[-L - 2, L + 2]$. For $t \geq 0$, $Y(t, i)$ depends only on $c_{[i-t,i+t]}$, so the ray of G carrying A lies where $Y = Y_\psi$, which is a function $\psi(p)$ of p; with $p \in \{p_0, p_0 + 1\}$ determined by the parity of q, A is 2-periodic, and $D(p_b) = A(q_0 + k + 2) \oplus A(q_0 + k) = 0$ at every column step. Likewise the ray of G' carrying B lies where $Y = \varphi(-q)$, and the row steps of G' start with $D(p_b) = 0$. As in the proof of Lemma 3.5, every firing of G at $[q', p']$ has $p' - p_0 \geq \frac{g}{2}(q' - q_0) - g$; applying this to the reflection $i \mapsto -i$, which commutes with F because N is invariant under reversal and exchanges q and p, every firing of G' has $q' + L + 2 \geq \frac{g}{2}(p' + L + 2) - g$. A firing at a centre (t, i) with $t \geq 0$ reads Y at $(t, i-1)$, (t, i), $(t, i+1)$, which depend only on $c_{[i-1-t,i+1+t]}$, so firings in $t \geq 0$ lie in $|i| \leq L + t + 1$; outside $G \cup G'$ this region has $t < L + 2$. In $G \setminus G'$ the coordinate p' is bounded, hence by the inequality so is q', and symmetrically in $G' \setminus G$; in $G \cap G'$ the two inequalities together bound both coordinates because $g/2 > 1$. So there are finitely many firings.

For the second statement let $T_0 \geq 1$ exceed the time of every firing in $t \geq 0$, and let $H = \{[q, p] : q + p \geq 2T_0 - 2\}$, the rows $t \geq T_0 - 1$. At every centre of H with $t \geq T_0$ the relation (3) has right side 0, so $Y[q+1, p+1] \oplus Y[q+1, p-1] = Y[q-1, p+1] \oplus Y[q-1, p-1]$. Fix one of the two sublattices $q \equiv p \equiv e \pmod 2$. Its points in the rows $T_0 - 1$ and T_0 form a staircase on which each point adds one new value of q or of p, so there are functions u, v with $Y[q, p] = u(q) \oplus v(p)$ there; $u(q) \oplus v(p)$ satisfies the same relation, and the rows $T_0 - 1$, T_0 determine Y on all later rows, so $Y = u(q) \oplus v(p)$ on the sublattice in H. For $t \geq 0$, $Y(t, i)$ depends only on $c_{[i-t,i+t]}$, so $Y = Y_\psi$, a function of p, on $\{p < -L,\ t \geq 0\}$. Let $q_1 = 2T_0 + L + 2$.

Two points $[q,p]$, $[q',p]$ of the sublattice with $q,q' \geq q_1$ and p the element of $\{-L-1,-L-2\}$ congruent to e lie in H and in this region, so $u(q) = u(q')$: u is constant on $\{q \geq q_1\}$ in each sublattice. Hence $Y[q,p] = \psi'(p)$ on $\{q \geq q_1\} \cap H$ for a sequence ψ' defined on all of $\mathbb{Z}$. Given p, take a centre $[q,p]$ with $q \geq q_1 + 1$ and $t \geq T_0$; its points N, E share $p+1$ and S, W share $p-1$, so the left side of (3) vanishes and the triple $(\psi'(p-1), \psi'(p), \psi'(p+1))$ is not in the support of N; so $\psi' \in \Phi_f$, and the point $d^- \in \Phi^-$ with diagram $\psi'(t+i)$ agrees with Y on $\{q \geq q_1\} \cap H$. For $t \geq T_0$ and $i \leq t/2 - C$ with $C = q_1 + 1$, the points $(t-1, i)$ and (t, i) have $q \geq t/2 + C - 1 \geq q_1$ and lie in H, so $F^t(c)_i = F^t(d^-)_i$; thus $T = T_0$ and $C = q_1 + 1$ work. The statement on $[-t/2 + C, \infty)$ follows by applying this to the reflection $i \mapsto -i$, which exchanges q and p and Φ^+ and Φ^-. □

Proof of Theorem 3.1 and Corollary 3.2. Let h have radius R and local rule $\hat{h}$, let c be a finite perturbation of a valid background, and take T from Proposition 3.6. For each i the window of $F^T(c)$ at i is the window at i of some $e \in \Phi^+ \cup \Phi^-$, so $h(F^T(c))_i = \hat{h}(e_{[i-R,i+R]}) = h(e)_i = e_i = F^T(c)_i$. Hence $F^T(h(c)) = h(F^T(c)) = F^T(c)$, and $h(c) = c$ because F is bijective. Finite perturbations are dense and h is continuous, so $h = \mathrm{id}$. If g_1, g_2 commute with F and agree on $\Phi^+ \cup \Phi^-$, the same argument gives $g_1(F^T(c)) = g_2(F^T(c))$, hence $g_1(c) = g_2(c)$, for every finite perturbation c, and so $g_1 = g_2$. For automorphisms, apply the first statement to $h_1^{-1} h_2$. □

Localised patterns. On a valid background, 91R and 94R have localised patterns that move at speed ± 1 without changing shape: the points of $\Phi^\pm$ that are finite perturbations of a valid background, for instance those with $\varphi = \cdots 111\,0\,111 \cdots$ for 91 and $\varphi = \cdots 000\,11\,000 \cdots$ for 94. They have none that stays in place: by Proposition 3.7 with $\psi = \varphi = b_\beta$, a finite perturbation c of b_β such that $F^t(c)$ agrees with $F^t(b_\beta)$ outside a fixed bounded interval for every $t \geq 0$ is b_β itself.

Rule 95. For $f = 95$ write again $f(a,b,c) = a \oplus c \oplus N(a,b,c)$. Then $N = N_{95}$ is the indicator of $a = c = 0$, that is of the triples 000 and 010; it is invariant under reversal, and 11 is the only valid background. Lemmas 3.3 and 3.4(i) hold for every N with these proofs; (ii) fails for N_{95}, and we replace (iii) by the following.

Lemma 3.8 (climbing for 95)**.** *Let $N = N_{95}$, and let Z, U, p_b, Z', Z'' and η be as in Lemma 3.4. Then $\eta(Z') \geq \eta$ and $\eta(Z'') \geq \eta + 4$.*

Proof. Since $N_{95}(a,b,c) = [a = c = 0]$ does not read b, the N-term $N(Z'(p-2), Z(p-1), Z(p))$ of (5) involves only positions of U, and whether p is a defect depends only on $Z(p-2)$, $Z(p)$, which lie in the same class. So the step on U changes only $Z|_U$ and depends only on $Z|_U$, and the defects of Z in the other class are defects of Z'. Let η_U be the least defect of Z in U (∞ if none). As in Lemma 3.4(i), $D(p) = 0$ and $Z'(p) = Z(p)$ for $p \in U$, $p < \eta_U$. At η_U the N-term is $N(Z(\eta_U - 2), \cdot, Z(\eta_U)) = 1$, so $D(\eta_U) = 1$ and $Z'(\eta_U) = 1 \oplus Z(\eta_U) = 1$. The N-term at $\eta_U + 2$ has first letter $Z'(\eta_U) = 1$, so it vanishes. The positions η_U and $\eta_U + 2$ are therefore not defects of Z': the first has last letter $Z'(\eta_U) = 1$, the second first letter 1. Positions of U below η_U are not defects of Z', since $Z' = Z$ there. Hence $\eta_U(Z') \geq \eta_U + 4$, while the other class is unchanged; in particular $\eta(Z') \geq \eta$. The second step treats the other class U' in the same way, so $\eta_U(Z'') = \eta_U(Z') \geq \eta_U + 4$ and $\eta_{U'}(Z'') \geq \eta_{U'}(Z) + 4$, and $\eta(Z'') = \min(\eta_U(Z''), \eta_{U'}(Z'')) \geq \eta + 4$. □

Corollary 3.9. *Theorem 3.1 and Corollary 3.2 hold for $f = 95$.*

Proof. The proofs of Lemma 3.5, Proposition 3.6, Theorem 3.1 and Corollary 3.2 carry over verbatim, with the valid background 11, Lemma 3.4(i) and Lemma 3.8 in place of Lemma 3.4, and the rate 2 in place of $\frac{3}{2}$ (the product $2 \cdot 2$ exceeds 1). □

For 95 the two sublattices $t+i$ even and odd evolve independently, since $f_{95} = \overline{a \wedge c}$ does not read b [42, p. 386]; but the map that advances one sublattice by two steps commutes with σ^2 and $\sigma^{\pm 1}F$ and not with σ or F, so it is not an automorphism. The decoupling is used in the proof of Theorem 4.12 and in Section 6.

4. Scattering and kink rigidity for 94R and 95R

We now study how a point of Φ^+ placed to the left of a point of Φ^- scatters off it, and use the scattering to normalise the automorphisms of $X_{F_{94}}$ and $X_{F_{95}}$. For 94 the normalisation is the first step of the proof of $\mathrm{Aut}(X_{F_{94}}) = \langle \sigma, F \rangle$ in Section 5; for 95 it is used in Section 6 to determine the group (Theorem 6.9).

Placed pairs. Let $f \in \{91, 94, 95\}$. The diagram of a point of Φ^+ is a function $\pi(q)$ of q alone and that of a point of Φ^- a function $\xi(p)$ of p alone, and we describe these points by π and ξ. A *placed pair* is a pair $(P, X) \in \Phi^+ \times \Phi^-$ whose diagrams $\pi(q)$ and $\xi(p)$ are eventually 2-periodic in both directions, together with a *corner* $[q_0, p_0]$, $q_0 \equiv p_0 \pmod 2$, and a valid background β such that

$$\pi(q) = \beta_{q \bmod 2} \quad (q \le q_0 + 1), \qquad \xi(p) = \beta_{p \bmod 2} \quad (p \le p_0 + 1).$$

The *left tail* β^L of P is the valid background with $\pi(q) = \beta^L_{q \bmod 2}$ for all large q, and the *right tail* β^R of X the one with $\xi(p) = \beta^R_{p \bmod 2}$ for all large p; they need not be equal to β. (A 2-periodic tail of a sequence of Φ_f is a valid background, since its two triples are not in the support of N.) Let $Y_{P,X}$ be the function on the lattice that equals $\pi(q)$ on $\{p < p_0\}$, $\xi(p)$ on $\{q < q_0\}$ (on the intersection both are $\beta_{q \bmod 2}$), and on the quadrant $G = \{q \ge q_0,\ p \ge p_0\}$ is the solution of (4) with boundary values $A = \pi$ and $B = \xi$; these agree at the two corners. Then $Y_{P,X} = \pi(q)$ on $\{p \le p_0 + 1\}$ and $Y_{P,X} = \xi(p)$ on $\{q \le q_0 + 1\}$. The relation (3) holds at every centre $C = [q, p]$: if $p \le p_0$, the points $\mathsf{N}, \mathsf{S}, \mathsf{W}, \mathsf{E}, C$ lie in $\{p \le p_0 + 1\}$, where $Y_{P,X}$ is the diagram of P; if $q \le q_0$, likewise with X; otherwise $\mathsf{S}, \mathsf{W}, \mathsf{E} \in G$ and $Y_{P,X}(\mathsf{N})$ is given by (4). So $Y_{P,X}$ is a diagram. The sets $\{p \le p_0 + 1\}$ and $\{q \le q_0 + 1\}$ together contain the rows $t_0 - 1$ and t_0, $t_0 = (q_0 + p_0)/2$, so $Y_{P,X}$ is the unique diagram that equals $\pi(q)$ on $\{p \le p_0 + 1\}$ and $\xi(p)$ on $\{q \le q_0 + 1\}$. In particular it does not change if the corner is replaced by a smaller one, and in the rows $t_0 - 1$, t_0 it is the diagram of P on the columns $i < i_0 = (p_0 - q_0)/2$ and that of X on $i \ge i_0$. So it is the diagram of the configuration that is P on the left and X on the right, and a point of Φ^+ that is eventually 2-periodic on both sides, placed far to the left of a point of Φ^- with the same property whose left tail is the right tail of the former, forms a placed pair with it. We call $Y_{P,X}$ the diagram of the pair. By (4) and induction on $q + p$ we have *Goursat causality*: for $[q, p] \in G$, $Y_{P,X}[q, p]$ depends only on π on $[q_0, q]$ and ξ on $[p_0, p]$.

Lemma 4.1 (scattering)**.** *Let $f \in \{91, 94, 95\}$, let (P, X) be a placed pair with corner $[q_0, p_0]$, let β^L be the left tail of P and β^R the right tail of X, and let $Y = Y_{P,X}$.*

(i) *Y fires at finitely many centres, and each of them is a centre $[q', p']$ with $q' > q_0$ and $p' > p_0$.*

(ii) *There are $q_1 \ge q_0$, $p_1 \ge p_0$, sequences $\gamma, o \in \Phi_f$ and a valid background β' such that $Y[q, p] = \gamma(p)$ whenever $q \ge q_1$, and $Y[q, p] = o(q)$ whenever $p \ge p_1$. Moreover $\gamma(p) = \beta^L_{p \bmod 2}$ for $p \le p_0 + 1$, $o(q) = \beta^R_{q \bmod 2}$ for $q \le q_0 + 1$, $\gamma(p) = \beta'_{p \bmod 2}$ for $p \ge p_1$ and $o(q) = \beta'_{q \bmod 2}$ for $q \ge q_1$.*

(iii) *For $k \ge 0$ let $P_k \in \Phi^+$ have the diagram $\pi(q - 2k)$. Then (P_k, X) is a placed pair with corner $[q_0 + 2k, p_0]$, and $Y_{P_k,X}[q, p] = Y[q - 2k, p]$.*

(iv) *Let $\Gamma \in \Phi^-$ have the diagram $\gamma(p)$, let $q_2 \ge q_1$ with $q_2 \equiv p_0 \pmod 2$, and let $P' \in \Phi^+$ have a diagram $\pi'(q)$ that is eventually 2-periodic, with $\pi'(q) = \beta^L_{q \bmod 2}$ for $q \le q_2 + 1$. Let $P' * P \in \Phi^+$ have the diagram equal to $\pi(q)$ for $q \le q_2 + 1$ and to $\pi'(q)$ for $q \ge q_2$.*

Then $(P' * P, X)$ is a placed pair with corner $[q_0, p_0]$, (P', Γ) is a placed pair with corner $[q_2, p_0]$, and $Y_{P'*P,X}$ equals Y on $\{q \le q_2 + 1\}$ and $Y_{P',\Gamma}$ on $\{q \ge q_2\}$.

Proof. (i) Let c be the configuration with diagram Y. For some L, the rows $t_0 - 1$ and t_0 of Y agree with the diagram of b_{β^L} at the columns $i \le i_0 - L$ (there $p < p_0$ and q is large) and with that of b_{β^R} at the columns $i \ge i_0 + L$. The diagram $\beta_{(t+i) \bmod 2}$ of a background is of both forms $\varphi(i - t)$ and $\varphi(i + t)$, so $b_{\beta^L}, b_{\beta^R} \in \Phi^+ \cap \Phi^-$. Since the rows $t_0 - 1$ and t_0 of Y are the configuration $F^{t_0}(c)$, the configuration $\sigma^{i_0} F^{t_0}(c)$ agrees with $\sigma^{i_0} b_{\beta^L} \in \Phi^-$ on $(-\infty, -L]$ and with $\sigma^{i_0} b_{\beta^R} \in \Phi^+$ on $[L, \infty)$, which is the hypothesis of Proposition 3.7; so Y fires only finitely often in the rows $t \ge t_0$. A centre $[q', p']$ with $p' \le p_0$ does not fire: its points $\mathsf{W}, C, \mathsf{E}$ lie in $\{p \le p_0 + 1\}$, so its row triple is $(\pi(q' + 1), \pi(q'), \pi(q' - 1))$, a triple of the sequence $k \mapsto \pi(-k)$ of Φ_f. Likewise a centre with $q' \le q_0$ has the row triple $(\xi(p' - 1), \xi(p'), \xi(p' + 1))$ and does not fire. The remaining centres lie in the rows $t > t_0$.

(ii) By (i) there is $q_1 \ge q_0$ such that every firing has $q' \le q_1 - 2$ and $\pi(q) = \beta^L_{q \bmod 2}$ for $q \ge q_1$. Let $[q, p] \in G$ with $q \ge q_1$. In Lemma 3.3, whose proof does not use the form of N, $A(q) = \beta^L_{q \bmod 2} = \beta^L_{p \bmod 2}$, ν depends only on $q \bmod 2 = p \bmod 2$, and the firings counted are those with $p' < p$ and $p' \not\equiv p$; so $Y[q, p]$ depends only on p. Points with $q \ge q_1$ and $p < p_0$ carry $\pi(q) = \beta^L_{p \bmod 2}$. So $Y[q, p] = \gamma(p)$ on $\{q \ge q_1\}$ for a sequence γ on $\mathbb{Z}$, and $\gamma(p) = \pi(q) = \beta^L_{p \bmod 2}$ for $p \le p_0 + 1$. A centre $[q, p]$ with $q \ge q_1 + 1$ does not fire and has the row triple $(\gamma(p - 1), \gamma(p), \gamma(p + 1))$, so $\gamma \in \Phi_f$. In the same way, with $p_1 \ge p_0$ such that every firing has $p' \le p_1 - 2$ and $\xi(p) = \beta^R_{p \bmod 2}$ for $p \ge p_1$, we get $Y[q, p] = o(q)$ on $\{p \ge p_1\}$ with $o \in \Phi_f$ and $o(q) = \xi(p) = \beta^R_{q \bmod 2}$ for $q \le q_0 + 1$. On $\{q \ge q_1,\ p \ge p_1\}$ we have $\gamma(p) = o(q)$; fixing p and varying q in one residue class shows that o is constant on each residue class of $\{q \ge q_1\}$, so $o(q) = \beta'_{q \bmod 2}$ there, and then $\gamma(p) = \beta'_{p \bmod 2}$ for $p \ge p_1$. As a 2-periodic tail of γ, β' is valid.

(iii) The map $[q, p] \mapsto [q + 2k, p]$ is the translation $(t, i) \mapsto (t + k, i - k)$, so $Y[q - 2k, p]$ is a diagram. It equals $\pi(q - 2k)$ on $\{p \le p_0 + 1\}$ and $\xi(p)$ on $\{q \le q_0 + 2k + 1\}$, and $\pi(q - 2k) = \beta_{q \bmod 2}$ for $q \le q_0 + 2k + 1$. By the uniqueness above it is $Y_{P_k,X}$.

(iv) The diagram of $P' * P$ is well defined, since π and π' both equal $\beta^L_{q \bmod 2}$ at q_2 and $q_2 + 1$, and it lies in Φ_f, since every three consecutive integers lie in $(-\infty, q_2 + 1]$ or in $[q_2, \infty)$. It equals π on $(-\infty, q_0 + 1]$, so $(P' * P, X)$ is placed at $[q_0, p_0]$; and $\pi'(q) = \beta^L_{q \bmod 2}$ for $q \le q_2 + 1$ and $\gamma(p) = \beta^L_{p \bmod 2}$ for $p \le p_0 + 1$, so (P', Γ) is placed at $[q_2, p_0]$. Let $Y'' = Y_{P'*P,X}$. By Goursat causality, $Y'' = Y$ on $G \cap \{q \le q_2 + 1\}$, and outside G the two diagrams are given by the same values of π and ξ there. On the quadrant $G_2 = \{q \ge q_2,\ p \ge p_0\}$, Y'' and $Y_{P',\Gamma}$ both satisfy (4), both equal $\pi'(q)$ on the ray $p \in \{p_0, p_0 + 1\}$, and both equal $\gamma(p)$ on the ray $q \in \{q_2, q_2 + 1\}$, where $Y'' = Y = \gamma(p)$ because $q_2 \ge q_1$. So they agree on G_2, and on $\{q \ge q_2,\ p < p_0\}$ both are $\pi'(q)$. □

For a placed pair (P, X) let $\Gamma \in \Phi^-$ and $\Omega \in \Phi^+$ be the points with diagrams $\gamma(p)$ and $o(q)$, and put $\mathcal{S}(P, X) = (\Gamma, \Omega)$. A point of Φ^- is determined by its diagram on a half-plane $\{q \ge q_1\}$, so Γ is the unique point of Φ^- whose diagram agrees with $Y_{P,X}$ on such a half-plane, and likewise for Ω. The point of Φ^- given by Proposition 3.7 for the configuration with diagram $Y_{P,X}$ agrees with $Y_{P,X}$ on a region that meets every line $p = \text{const}$ inside such a half-plane, so Γ and Ω are the outgoing Φ^- and Φ^+ parts given by that proposition.

Links, objects and probes. Let $f \in \{91, 94, 95\}$. We identify a point of $\Phi^\pm$ with its sequence in Φ_f. A *link* is a point of Φ^+ or Φ^- that agrees with a 2-periodic point off a bounded set. A Φ^- *object* is a point $X \in \Phi^-$ that is eventually 2-periodic on both sides, with a left tail λ and a right tail λ' among the valid backgrounds (0^∞, 1^∞ for 94; 1^∞, $(01)^\infty$, $(10)^\infty$ for 91; 1^∞ for 95). Its *support* $\operatorname{supp} X$ is the length of its core, the shortest word w with $X = \cdots \lambda\, w\, \lambda' \cdots$. So

$\operatorname{supp} X = 0$ exactly when X is a background or, for 91 and 94, a *bare kink*, the two different tails abutting at a junction. The core and the support of a link are defined in the same way. A *probe* for X is a link $P \in \Phi^+$ both of whose tails are λ; then (P, X) is a placed pair at every sufficiently small corner, and we write $\mathcal{S}(P, X) = (\Gamma_P(X), \Omega_X(P))$ and call $\Omega_X(P)$ the *response* of X to P. By Lemma 4.1(iii), moving the probe by $2k$ along its rays leaves $\Gamma_P(X)$ unchanged and replaces $o(q)$ by $o(q-2k)$; so $\Gamma_P(X)$ does not depend on how far to the left P is placed, and $\Omega_X(P)$ only up to this translation. By Lemma 4.1(iv), a probe P' placed far enough to the left of P acts on X through $\Gamma_P(X)$: $\Gamma_{P'*P}(X) = \Gamma_{P'}(\Gamma_P(X))$.

Time reversal. We use the time reversal to invert $\mathcal{S}$. Let $\mu = \mathrm{rev} \circ \rho$, so that $(\mu c)_i = \rho(c_{-i})$. Since $f(a,b,c) = f(c,b,a)$ for $f \in \{91, 94, 95\}$, Lemma 2.3 gives $\mathrm{rev}\, F\, \mathrm{rev} = F$, hence $\mu F \mu = F^{-1}$; and $Y_{\mu c}(t,i) = Y_c(-1-t,-i)$, since $F^t(\mathrm{rev}\, c) = \mathrm{rev}\, F^t(c)$ and $Y_{\rho c}(t,i) = Y_c(-1-t,i)$ (Section 2). In light-cone coordinates μ is the point reflection $[q,p] \mapsto [-1-q,-1-p]$, which exchanges N with S and W with E. It maps Φ^+ and Φ^- to themselves, the diagram $\pi(q)$ to $\pi(-1-q)$ and $\xi(p)$ to $\xi(-1-p)$, and backgrounds to backgrounds. On pairs put $\mu(P, X) = (\mu X, \mu P)$ for $(P, X) \in \Phi^+ \times \Phi^-$ and $\mu(\Gamma, \Omega) = (\mu\Omega, \mu\Gamma)$ for $(\Gamma, \Omega) \in \Phi^- \times \Phi^+$.

Proposition 4.2 (time-reversal duality)**.** *Let $f \in \{91, 94, 95\}$, let (P, X) be a placed pair, and let $\mathcal{S}(P, X) = (\Gamma, \Omega)$. Then $(\mu\Omega, \mu\Gamma)$ is a placed pair, and*

$$\mathcal{S}(\mu\Omega, \mu\Gamma) = (\mu X, \mu P).$$

Consequently $\mathcal{S}\mu\mathcal{S} = \mu$ on placed pairs, $\mathcal{S}$ is a bijection from the set of placed pairs onto the set of pairs $(\Gamma, \Omega) \in \Phi^- \times \Phi^+$ for which $\mu(\Gamma, \Omega)$ is placed, and $\mathcal{S}^{-1} = \mu\mathcal{S}\mu$.

Proof. Let $Y = Y_{P,X}$, with corner $[q_0, p_0]$, and take $q_1, p_1, \gamma, o, \beta'$ from Lemma 4.1(ii). Put $Y'[q,p] = Y[-1-q,-1-p]$. The relation at a centre C for Y is the relation at the reflected centre for Y', with W and E exchanged in the argument of N; since N is invariant under reversal of the triple, Y' is a diagram (it is $Y_{\mu c}$, c the configuration of Y). Let $\pi'(q) = o(-1-q)$ and $\xi'(p) = \gamma(-1-p)$, the diagrams of $\mu\Omega$ and $\mu\Gamma$. Then $Y' = \pi'(q)$ on $\{p \le -1-p_1\}$ and $Y' = \xi'(p)$ on $\{q \le -1-q_1\}$. Choose $q_0' \le -2-q_1$ and $p_0' \le -2-p_1$ with $q_0' \equiv p_0'$. For $q \le q_0' + 1$ we have $-1-q \ge q_1$, so $\pi'(q) = \beta'_{(-1-q) \bmod 2}$, and likewise $\xi'(p) = \beta'_{(-1-p) \bmod 2}$ for $p \le p_0' + 1$. So $(\mu\Omega, \mu\Gamma)$ is a placed pair with corner $[q_0', p_0']$, and by uniqueness its diagram is Y'. Now $Y = \xi(p)$ on $\{q \le q_0 + 1\}$, so $Y' = \xi(-1-p)$ on $\{q \ge -2-q_0\}$, a half-plane on which Y' is the diagram of μX; and $Y = \pi(q)$ on $\{p \le p_0 + 1\}$, so $Y' = \pi(-1-q)$ on $\{p \ge -2-p_0\}$, the diagram of μP. By the characterisation of the outgoing parts, $\mathcal{S}(\mu\Omega, \mu\Gamma) = (\mu X, \mu P)$.

The first statement reads $\mathcal{S}(\mu(\mathcal{S}(x))) = \mu(x)$ for every placed pair x. Since μ is an involution, $\mu\mathcal{S}\mu$ is a left inverse of $\mathcal{S}$, so $\mathcal{S}$ is injective. If $y \in \Phi^- \times \Phi^+$ and $\mu(y)$ is placed, then $x = \mu(\mathcal{S}(\mu(y)))$ is placed by the first statement, and $\mathcal{S}(x) = \mu(\mu(y)) = y$. So the image of $\mathcal{S}$ is as stated, and $\mu\mathcal{S}\mu$ is the inverse. □

Corollary 4.3. *Let $f \in \{91, 94, 95\}$ and let (P, X) be a placed pair, with $\mathcal{S}(P, X) = (\Gamma, \Omega)$.*

1. *If X is a background, then Γ is the background b_{β^L} of the left tail of P, and $\Omega = P$.*
2. *If Γ is a background, then so is X. In particular, for a probe P and an object X, $\Gamma_P(X)$ is a background only if X is.*

Proof. (1) A background X in a placed pair is the background b_β of the corner. Then $\pi(q)$ is a diagram that equals $\pi(q)$ on $\{p \le p_0 + 1\}$ and $\beta_{p \bmod 2} = \pi(q)$ on $\{q \le q_0 + 1\}$, so $Y_{P,X} = \pi(q)$ everywhere; reading it on the half-planes of Lemma 4.1(ii) gives $\Gamma = b_{\beta^L}$ and $\Omega = P$. (2) If Γ is a background, so is $\mu\Gamma$, and by (1) the first component of $\mathcal{S}(\mu\Omega, \mu\Gamma)$ is a background. By Proposition 4.2 it is μX, so X is a background. □

Crossing kinks for 94. For $f = 94$ the valid backgrounds are the constants 0 and 1, and a point of $\Phi^{\pm}$ on these backgrounds is described by its *kinks*, the positions a with $\pi(a-1) \ne \pi(a)$,

or b with $\xi(b-1) \neq \xi(b)$. The elementary event in the interaction of a probe with an object is the crossing of one kink of each. For an integer a let $\theta_a(k) = 1$ if $k \geq a$ and $\theta_a(k) = 0$ if $k < a$.

Lemma 4.4 (crossing of two kinks)**.** *Let $f = 94$, let G be the quadrant with corner $[q_0, p_0]$, let $m \in \{0,1\}$, $a \geq q_0 + 2$, $b \geq p_0 + 2$, and let Y be the solution of (4) on G with $A(q) = m \oplus \theta_a(q)$ and $B(p) = m \oplus \theta_b(p)$.*

(i) *If $m = 0$, or if $a \not\equiv b \pmod 2$, then Y does not fire, and $Y[q,p] = m \oplus \theta_a(q) \oplus \theta_b(p)$.*
(ii) *If $m = 1$ and $a \equiv b \pmod 2$, then Y fires exactly at the centres $[a-1, b-1]$ and $[a+1, b+1]$, and $Y[q,p] = 1$ if $q < a$ and $p < b$ or if $q > a$ and $p > b$, and $Y[q,p] = 0$ otherwise.*

Here A carries a Φ^+ kink on the ray $q = a$ and B a Φ^- kink on the ray $p = b$, and m is the value of the region $\{q < a,\ p < b\}$ between them before they meet. In case (i), for $p \geq b$, $Y[q,p] = \bar{m} \oplus \theta_a(q)$, and for $q \geq a$, $Y[q,p] = \bar{m} \oplus \theta_b(p)$: the kinks cross without interacting. In case (ii), $Y[q,p] = \theta_{a+1}(q)$ for $p > b$ and $Y[q,p] = \theta_{b+1}(p)$ for $q > a$: after two firings both kinks continue on the next ray, that is, both are delayed by one step.

Proof. For a finite set Σ of centres $[q', p']$ with $q' > q_0$ and $p' > p_0$ let $\chi_\Sigma(P)$, for $P = [q,p] \in G$, be the parity of the number of elements $[q', p'] \in \Sigma$ with $q' < q$, $p' < p$ and $q' \not\equiv q$, and put $Y_\Sigma = A(q) \oplus B(p) \oplus m \oplus \chi_\Sigma$; here $m = \nu$ is the value at both corners. The proof of Lemma 3.3 shows that Y_Σ has the boundary values A and B and that $Y_\Sigma(\mathsf{N}) \oplus Y_\Sigma(\mathsf{S}) \oplus Y_\Sigma(\mathsf{W}) \oplus Y_\Sigma(\mathsf{E}) = 1$ exactly when $C \in \Sigma$, at every centre $C = [q,p]$ with $q > q_0$, $p > p_0$. So if the centres at which Y_Σ fires are exactly those of Σ, then Y_Σ satisfies (4), and $Y_\Sigma = Y$. For $N = N_{94}$, Y_Σ fires at C when $(Y_\Sigma(\mathsf{W}), Y_\Sigma(C), Y_\Sigma(\mathsf{E})) = (0,1,0)$, with $\mathsf{W} = [q+1, p-1]$ and $\mathsf{E} = [q-1, p+1]$. The eight cases given by m and the parities of a and b reduce to the two of the statement.

(i) Take $\Sigma = \emptyset$, so $Y_\emptyset \oplus m = \theta_a(q) \oplus \theta_b(p)$. At a centre $C = [q,p]$,

$$Y_\emptyset(\mathsf{W}) \oplus Y_\emptyset(C) = [q = a-1] \oplus [p = b], \qquad Y_\emptyset(C) \oplus Y_\emptyset(\mathsf{E}) = [q = a] \oplus [p = b-1],$$

where $[\cdot]$ is 1 for a true statement and 0 otherwise, because $\theta_a(q+1) \oplus \theta_a(q) = [q = a-1]$ and $\theta_b(p-1) \oplus \theta_b(p) = [p = b]$. A firing needs both sums to be 1. If $q = a-1$ this forces $p = b-1$; if $q \neq a-1$ it forces $p = b$ and then $q = a$. So the only candidates are $[a-1, b-1]$ and $[a,b]$, which are lattice points only if $a \equiv b$. At both of them the row triple is $m \oplus (1,0,1)$, since $Y_\emptyset[a-1, b-1] = Y_\emptyset[a,b] = m$, while at the points W and E of these centres exactly one of $\theta_a(q), \theta_b(p)$ is 1. This is $(0,1,0)$ only if $m = 1$. So under the hypothesis of (i), $Y_\emptyset$ does not fire, and $Y = Y_\emptyset$.

(ii) Take $\Sigma = \{[a-1, b-1], [a+1, b+1]\}$ and write $x = q-a$, $y = p-b$, so $x \equiv y$. The first centre contributes to χ_Σ at the points with $x \geq 0$, $y \geq 0$, x even, and the second at those with $x \geq 2$, $y \geq 2$, x even. With $Y_\emptyset = 1 \oplus \theta_0(x) \oplus \theta_0(y)$ this gives $Y_\Sigma = 1$ if $x, y < 0$ or $x, y > 0$, and $Y_\Sigma = 0$ if $\min(x,y) \leq 0 \leq \max(x,y)$; for even $x, y \geq 0$, for instance, $Y_\Sigma = Y_\emptyset \oplus 1 \oplus \theta_2(x)\theta_2(y) = \theta_2(x)\theta_2(y)$, and for odd $x, y > 0$, $Y_\Sigma = Y_\emptyset = 1$. A firing at $C = (x,y)$ needs $Y_\Sigma(C) = 1$ and $Y_\Sigma = 0$ at $\mathsf{W} = (x+1, y-1)$ and $\mathsf{E} = (x-1, y+1)$. If $x, y > 0$, then $Y_\Sigma(\mathsf{W}) = 0$ forces $y - 1 \leq 0$ and $Y_\Sigma(\mathsf{E}) = 0$ forces $x - 1 \leq 0$, so $C = (1,1)$; if $x, y < 0$, then $x + 1 \geq 0$ and $y + 1 \geq 0$, so $C = (-1,-1)$. At both centres the triple is indeed $(0,1,0)$, since $(2,0), (0,2), (0,-2), (-2,0)$ all have $\min \leq 0 \leq \max$. So Y_Σ fires exactly on Σ, and $Y = Y_\Sigma$. □

Kink rigidity. For 94 and 95 we now normalise an arbitrary automorphism by an element of $\langle \sigma, F \rangle$ so that it fixes the simplest non-periodic points of $\Phi^\pm$, and for 94 we then show, in Section 5, that the normalised automorphism is the identity; for 95 it need not be, and the normalisation is a step in the determination of the group (Theorem 6.9). Let $f \in \{91, 94, 95\}$ and let h be a block map of radius R that commutes with σ and F. It commutes with $\sigma^{\pm 1}F$, so it maps $\Phi^\pm$ into itself; for $Z \in \Phi^+ \cup \Phi^-$ we write hZ for $h(Z)$. Since $F^t(c)_j = (Y_c(t-1, j), Y_c(t,j))$

and $F^t(h(c)) = h(F^t(c))$, the value $Y_{h(c)}(t,i)$ depends only on the values of Y_c at the points (s,j) with $s \in \{t-1, t\}$ and $|j-i| \le R$, and these points have $q(t,i) - R - 1 \le q \le q(t,i) + R$ and $p(t,i) - R - 1 \le p \le p(t,i) + R$. Hence, for configurations c, d and $m \in \mathbb{Z}$:

$$\begin{aligned} Y_c = Y_d \text{ on } \{p \le m\} \;&\Rightarrow\; Y_{h(c)} = Y_{h(d)} \text{ on } \{p \le m - R\}, \\ Y_c = Y_d \text{ on } \{p \ge m\} \;&\Rightarrow\; Y_{h(c)} = Y_{h(d)} \text{ on } \{p \ge m + R + 1\}, \end{aligned} \tag{6}$$

and the same with q in place of p. In particular, if $Z \in \Phi^-$ has the diagram $\xi(p)$ and hZ the diagram $\xi'(p)$, then $\xi'(p)$ depends only on ξ on $[p-R-1, p+R]$, and likewise on Φ^+. Since $Y_{\sigma c}[q,p] = Y_c[q-1, p+1]$ and $Y_{F(c)}[q,p] = Y_c[q+1, p+1]$, the shift σ acts on the diagrams of points of Φ^+ by $\pi(q) \mapsto \pi(q-1)$ and on those of points of Φ^- by $\xi(p) \mapsto \xi(p+1)$, and F acts by $\pi(q) \mapsto \pi(q+1)$ and $\xi(p) \mapsto \xi(p+1)$; so h commutes with these translations of diagrams. Finally, for a valid background β the point $h(b_\beta)$ lies in $\Phi^+ \cap \Phi^-$, so its diagram is a function of q alone and of p alone; it therefore depends only on the parity of q, and $h(b_\beta) = b_{\beta''}$ for a valid background β''. A *kink* of a sequence ω is a position x with $\omega(x-1) \ne \omega(x)$; it is *descending* if $\omega(x-1) = 1$ and $\omega(x) = 0$, and *ascending* otherwise.

Lemma 4.5 (intertwining)**.** *Let $f \in \{91, 94, 95\}$, let h be a block map of radius R commuting with σ and F, let (P, X) be a placed pair with corner $[q_0, p_0]$ and background β, and let $\mathcal{S}(P, X) = (\Gamma, \Omega)$ and $h(b_\beta) = b_{\beta''}$. Then (hP, hX) is a placed pair with corner $[q_0 - R, p_0 - R]$ and background β'', its diagram is $Y_{h(c)}$, where c is the configuration with diagram $Y_{P,X}$, and*

$$\mathcal{S}(hP, hX) = (h\Gamma, h\Omega).$$

Proof. Write $Y = Y_{P,X} = Y_c$, $Y' = Y_{h(c)}$, and π', ξ' for the diagrams of hP, hX. The diagram of P agrees with that of b_β on $\{q \le q_0 + 1\}$, so by (6) the diagram of hP agrees with that of $b_{\beta''}$ on $\{q \le q_0 - R + 1\}$, that is, $\pi'(q) = \beta''_{q \bmod 2}$ for $q \le q_0 - R + 1$; likewise $\xi'(p) = \beta''_{p \bmod 2}$ for $p \le p_0 - R + 1$. If π agrees with the diagram of b_{β^L} on $\{q \ge m\}$, then π' agrees with the diagram of the background $h(b_{\beta^L})$ on $\{q \ge m + R + 1\}$; so π', and in the same way ξ', is eventually 2-periodic in both directions. Since $q_0 - R \equiv p_0 - R$, (hP, hX) is a placed pair with corner $[q_0 - R, p_0 - R]$ and background β''. Next, Y agrees with the diagram of P on $\{p \le p_0 + 1\}$ and with that of X on $\{q \le q_0 + 1\}$, so by (6) Y' equals $\pi'(q)$ on $\{p \le p_0 - R + 1\}$ and $\xi'(p)$ on $\{q \le q_0 - R + 1\}$. By the uniqueness of the diagram of a placed pair, $Y' = Y_{hP,hX}$. Finally, by Lemma 4.1(ii), Y agrees with the diagram of Γ on a half-plane $\{q \ge q_1\}$ and with that of Ω on a half-plane $\{p \ge p_1\}$, so Y' agrees with the diagram of $h\Gamma$ on $\{q \ge q_1 + R + 1\}$ and with that of $h\Omega$ on $\{p \ge p_1 + R + 1\}$. By the characterisation of the outgoing parts after Lemma 4.1, $\mathcal{S}(hP, hX) = (h\Gamma, h\Omega)$. □

The next lemma is the counterpart of Lemma 4.1(iv) on the side of the object: an object that consists of two parts far apart is crossed by the probe one part at a time.

Lemma 4.6 (gluing on the side of the object)**.** *Let $f \in \{91, 94, 95\}$, let (P, X) be a placed pair with corner $[q_0, p_0]$, diagram Y and $\mathcal{S}(P, X) = (\Gamma, \Omega)$, let $\gamma(p)$ and $o(q)$ be the diagrams of Γ and Ω, and let β^R be the right tail of X. Let $p_2 \ge p_0$ with $p_2 \equiv q_0 \pmod 2$ be such that*

$$\xi(p) = \beta^R_{p \bmod 2} \quad (p \ge p_2), \qquad Y[q,p] = o(q) \quad (q \ge q_0,\ p \in \{p_2, p_2 + 1\});$$

by Lemma 4.1(ii) this holds for every large $p_2 \equiv q_0$. Let $X' \in \Phi^-$ have an eventually 2-periodic diagram $\xi'(p)$ with $\xi'(p) = \beta^R_{p \bmod 2}$ for $p \le p_2 + 1$, and let $X * X' \in \Phi^-$ have the diagram equal to $\xi(p)$ for $p \le p_2 + 1$ and to $\xi'(p)$ for $p \ge p_2$. Then $(P, X * X')$ is a placed pair with corner $[q_0, p_0]$, (Ω, X') is a placed pair with corner $[q_0, p_2]$ and background β^R, and $Y_{P,X*X'}$ equals Y on $\{p \le p_2 + 1\}$ and $Y_{\Omega,X'}$ on $\{p \ge p_2\}$. Consequently, if $\mathcal{S}(\Omega, X') = (\Gamma', \Omega')$ and $\gamma'(p)$ is the diagram of Γ', then $\gamma(p) = \gamma'(p)$ for $p \in \{p_2, p_2 + 1\}$, and $\mathcal{S}(P, X * X') = (\Gamma * \Gamma', \Omega')$, where $\Gamma * \Gamma' \in \Phi^-$ has the diagram equal to $\gamma(p)$ for $p \le p_2 + 1$ and to $\gamma'(p)$ for $p \ge p_2$.

Proof. The diagram of $X * X'$ is well defined, since ξ and ξ' both equal $\beta^R_{p \bmod 2}$ at p_2 and p_2+1, and it lies in Φ_f, since any three consecutive integers lie in $(-\infty, p_2+1]$ or in $[p_2, \infty)$. It equals ξ on $(-\infty, p_0+1]$, so $(P, X * X')$ is placed at $[q_0, p_0]$. By Lemma 4.1(ii), $o(q) = \beta^R_{q \bmod 2}$ for $q \le q_0+1$, and o is eventually 2-periodic; with $\xi'(p) = \beta^R_{p \bmod 2}$ for $p \le p_2+1$ this shows that (Ω, X') is placed at $[q_0, p_2]$, with background β^R. Let $Y'' = Y_{P, X*X'}$. By Goursat causality $Y'' = Y$ on $G \cap \{p \le p_2+1\}$, and outside G the two diagrams are given by the same values of π and ξ; so $Y'' = Y$ on $\{p \le p_2+1\}$. On the quadrant $G_2 = \{q \ge q_0,\ p \ge p_2\}$, Y'' and $Y_{\Omega, X'}$ both satisfy (4); both equal $\xi'(p)$ on the ray $q \in \{q_0, q_0+1\}$, and both equal $o(q)$ on the ray $p \in \{p_2, p_2+1\}$, where $Y'' = Y = o(q)$ by hypothesis. So they agree on G_2, and on $\{q < q_0,\ p \ge p_2\}$ both are $\xi'(p)$. For the last statement, read the outgoing parts of $(P, X * X')$ on half-planes $\{q \ge q_1''\}$ and $\{p \ge p_1''\}$ with q_1'', p_1'' large: on the first, Y'' is $\gamma(p)$ for $p \le p_2+1$ and $\gamma'(p)$ for $p \ge p_2$; on the second, $Y'' = Y_{\Omega, X'}$ is the diagram of Ω'. On the overlap $p \in \{p_2, p_2+1\}$ both descriptions of Y'' on the first half-plane apply, so $\gamma = \gamma'$ there, $\Gamma * \Gamma'$ is well defined, and it lies in Φ_f because any three consecutive integers lie in $(-\infty, p_2+1]$ or in $[p_2, \infty)$. □

For the rest of this paragraph let $f = 94$, and write $b_0 = b_{00}$ and $b_1 = b_{11}$ for its two backgrounds. For $a, b \in \mathbb{Z}$ let $K^+_a, \bar{K}^+_a \in \Phi^+$ be the points with diagrams $\theta_a(q)$ and $1 \oplus \theta_a(q)$, and $K^-_b, \bar{K}^-_b \in \Phi^-$ the points with diagrams $\theta_b(p)$ and $1 \oplus \theta_b(p)$. These are the *bare kinks*; together with the backgrounds b_0, b_1 they are the points of $\Phi^+ \cup \Phi^-$ with at most one kink. The crossing lemma gives, at any corner $[q_0, p_0]$ with $q_0 \le a-2$, $p_0 \le b-2$ and $q_0 \equiv p_0$, and by reading Y on the half-planes of Lemma 4.1(ii),

$$\mathcal{S}(K^+_a, K^-_b) = (\bar{K}^-_b, \bar{K}^+_a), \qquad \mathcal{S}(\bar{K}^+_a, \bar{K}^-_b) = \begin{cases} (K^-_{b+1}, K^+_{a+1}), & a \equiv b \pmod 2, \\ (K^-_b, K^+_a), & a \not\equiv b \pmod 2, \end{cases} \tag{7}$$

the first on the background 00 by Lemma 4.4(i) with $m = 0$, the second on the background 11 by Lemma 4.4(ii) and (i) with $m = 1$.

Lemma 4.7 (one exact ray)**.** *Let $f = 94$, let G be the quadrant with corner $[q_0, p_0]$, let $\varepsilon \in \{0, 1\}$, and let Y be the solution of (4) on G with boundary values A, B such that*

$$A(q) = \varepsilon \ \ (q_0 \le q < a_1), \qquad A(a_1) = 1 - \varepsilon, \qquad a_1 \ge q_0 + 2, \qquad B(p_0) = B(p_0+1) = \varepsilon,$$

and $(B(p-2), B(p-1), B(p)) \ne (0, 1, 0)$ for all $p \ge p_0+2$. If B has no descending kink $x \ge p_0+1$ with $x \equiv a_1 \pmod 2$, then Y fires at no centre $[a_1 - 1, p']$ with $p' > p_0$. Otherwise it fires at exactly one such centre, $[a_1-1, x_1-1]$, where x_1 is the least such kink. The same holds with q and p exchanged: if $B(p) = \varepsilon$ for $p_0 \le p < b_1$, $B(b_1) = 1 - \varepsilon$, $b_1 \ge p_0 + 2$, $A(q_0) = A(q_0+1) = \varepsilon$ and A contains no 010, then Y fires at a centre $[q', b_1 - 1]$ with $q' > q_0$ exactly once, at $[y_1 - 1, b_1 - 1]$, if A has a descending kink $y \equiv b_1$, y_1 the least of them, and otherwise never.

Proof. Let Y° be the solution with the boundary values $A^\circ(q) = \varepsilon$ for all $q \ge q_0$ and B. The function $[q, p] \mapsto B(p)$ has these boundary values, and it satisfies (4), because $N(B(p-2), B(p-1), B(p)) = 0$; so $Y^\circ[q, p] = B(p)$. By (4) and induction on $q + p$, Y on $G \cap \{q \le a_1 - 1\}$ depends only on A on $[q_0, a_1 - 1]$ and on B, so $Y[q, p] = B(p)$ for $q_0 \le q \le a_1 - 1$.

Let $U = \{p \ge p_0 : p \equiv a_1\}$, with least element $p_b \in \{p_0, p_0+1\}$, and $D(p) = Y[a_1, p] \oplus B(p)$ for $p \in U$. Then $D(p_b) = A(a_1) \oplus B(p_b) = (1 - \varepsilon) \oplus \varepsilon = 1$. For $p \in U$ with $p \ge p_b + 2$, (4) at $[a_1, p]$, where $[a_1 - 2, p - 2]$, $[a_1 - 2, p]$ and $[a_1 - 1, p - 1]$ carry $B(p-2)$, $B(p)$, $B(p-1)$, gives

$$D(p) = D(p-2) \oplus n(p), \qquad n(p) = N\big(Y[a_1, p-2], B(p-1), B(p)\big),$$

and $n(p)$ is the firing at the centre $[a_1 - 1, p - 1]$; these are all the centres $[a_1 - 1, p']$ with $p' > p_0$. If $D(p-2) = 1$, then $Y[a_1, p-2] = 1 \oplus B(p-2)$, and $n(p) = 1$ if and only if $B(p-2) = B(p-1) = 1$ and $B(p) = 0$; since B contains no 010, this holds if and only if p is a descending kink of B.

Every descending kink x of B has $x \geq p_0 + 2$, as $B(p_0) = B(p_0 + 1)$. So, by induction along U, $D = 1$ and $n = 0$ on $U \cap [p_b, x_1)$, and $n(x_1) = 1$, $D(x_1) = 0$ if x_1 exists. If $D(p-2) = 0$, then $Y[a_1, p-2] = B(p-2)$ and $n(p) = N(B(p-2), B(p-1), B(p)) = 0$; so after x_1 there are no further firings on the ray $a_1 - 1$. The last statement follows by applying this to the reflection $[q, p] \mapsto [p, q]$, which maps solutions of (4) to solutions, since N is invariant under reversal of the triple, and exchanges A and B. □

For $b \in \mathbb{Z}$ and $g \geq 2$ let $T_{b,g} \in \Phi^-$ be the *train* with diagram $\theta_b(p) \oplus \theta_{b+g}(p)$: a plateau 1^g on the background 00, between the ascending kink b and the descending kink $b + g$. It lies in Φ_{94} because $g \geq 2$.

Lemma 4.8 (train)**.** *Let $f = 94$, $a, b \in \mathbb{Z}$ and $g \geq 2$ with $a \not\equiv b + g \pmod 2$. Then $(K^+_a, T_{b,g})$ is a placed pair with background 00 at every corner $[q_0, p_0]$ with $q_0 \leq a - 2$, $p_0 \leq b - 2$ and $q_0 \equiv p_0$, and*

$$\mathcal{S}(K^+_a, T_{b,g}) = (\Gamma, K^+_a),$$

where Γ has the diagram $1 \oplus \theta_b(p) \oplus \theta_{b+g}(p)$.

Proof. The first statement is clear. Choose the corner with $q_0 \equiv p_0 \equiv b$. By Lemma 4.4(i) with $m = 0$, the pair (K^+_a, K^-_b) has the diagram $Y = \theta_a(q) \oplus \theta_b(p)$ on G, and $\mathcal{S}(K^+_a, K^-_b) = (\bar{K}^-_b, \bar{K}^+_a)$ by (7). On $\{p \geq b\}$, $Y = 1 \oplus \theta_a(q)$ is the diagram of $\bar{K}^+_a$, and $\theta_b(p) = 1$ for $p \geq b$; so Lemma 4.6 applies with $X = K^-_b$, $p_2 = b$ and $X' = \bar{K}^-_{b+g}$, whose diagram is 1 for $p \leq b + 1$ since $g \geq 2$, and $X * X' = T_{b,g}$. The pair $(\bar{K}^+_a, \bar{K}^-_{b+g})$ is placed at $[q_0, b]$ with background 11, and $a \geq q_0 + 2$, $b + g \geq b + 2$, $a \not\equiv b + g$; by (7), $\mathcal{S}(\bar{K}^+_a, \bar{K}^-_{b+g}) = (K^-_{b+g}, K^+_a)$. By Lemma 4.6, $\mathcal{S}(K^+_a, T_{b,g}) = (\bar{K}^-_b * K^-_{b+g}, K^+_a)$, and $\bar{K}^-_b * K^-_{b+g}$ has the stated diagram. □

If $a \equiv b + g$, the second crossing in the proof delays the kink of the probe by one step, and the same argument gives $\mathcal{S}(K^+_a, T_{b,g}) = (\Gamma', K^+_{a+1})$ for some Γ'.

Theorem 4.9 (kink rigidity for 94)**.** *Let $h \in \mathrm{Aut}(X_{F_{94}})$, regarded as a block map commuting with F (Lemma 2.2). Then h fixes b_0 and b_1, and there are integers $\alpha^+ \equiv \alpha^- \pmod 2$ such that for all $a, b \in \mathbb{Z}$*

$$hK^+_a = K^+_{a+\alpha^+}, \qquad h\bar{K}^+_a = \bar{K}^+_{a+\alpha^+}, \qquad hK^-_b = K^-_{b+\alpha^-}, \qquad h\bar{K}^-_b = \bar{K}^-_{b+\alpha^-}.$$

Consequently, with $x = (\alpha^- - \alpha^+)/2$ and $y = (\alpha^+ + \alpha^-)/2$, the automorphism $\sigma^x F^y h$ fixes b_0, b_1 and every bare kink.

Proof. Let R be the radius of h and $r = R + 1$. By (6), h acts on the diagrams of $\Phi^\pm$ by sliding block codes of radius at most r that commute with translation.

Step 0. By the paragraph before Lemma 4.5, h maps b_0 and b_1, the backgrounds of 94, to backgrounds; since h is injective, there is $\varepsilon \in \{0, 1\}$ with $h(b_0) = b_\varepsilon$ and $h(b_1) = b_{1-\varepsilon}$.

Step 1 (a train that the probe crosses unchanged). Fix $a, b \in \mathbb{Z}$ and $g \geq 2r + 2$ with $a \not\equiv b + g \pmod 2$. By Lemma 4.8, $\mathcal{S}(K^+_a, T_{b,g}) = (\Gamma, K^+_a)$, and also $\mathcal{S}(K^+_a, T_{b+1,g-1}) = (\Gamma_1, K^+_a)$ for some Γ_1, since $(b + 1) + (g - 1) = b + g$.

Step 2 (intertwining). Let $E = hK^+_a$, $W = hT_{b,g}$ and $D = hK^-_b$, with diagrams $\pi_E(q)$, $\omega(p)$ and $\delta(p)$. By Lemma 4.5, (E, W) is a placed pair with background $\varepsilon\varepsilon$, and

$$\mathcal{S}(E, W) = (h\Gamma, E).$$

Since the diagram of hZ at p depends only on that of Z on $[p - r, p + r]$: $\delta(p) = \varepsilon$ for $p < b - r$ and $\delta(p) = 1 - \varepsilon$ for $p \geq b + r$; $\omega(p) = \delta(p)$ for $p \leq b + g - r - 1$ and $\omega(p) = \varepsilon$ for $p \geq b + g + r$; and $\pi_E(q) = \varepsilon$ for $q < a - r$, $\pi_E(q) = 1 - \varepsilon$ for $q \geq a + r$. As $b + r \leq b + g - r - 1$, every kink of δ lies in $[b - r, b + r]$ and is a kink of ω of the same kind. We may replace the corner of (E, W) by a smaller one $[q_0, p_0]$ with $q_0 \leq a - r - 2$ and $p_0 \leq b - r - 2$.

Step 3 (every ray fires an even number of times). Let $Y = Y_{E,W}$. By Lemma 4.1(i) every firing of Y is at a centre $[q', p']$ with $q' > q_0$ and $p_0 < p' < p^*$, for some p^*. Since the outgoing Φ^+ part of (E, W) is E, Lemma 4.1(ii) gives $Y[q, p] = \pi_E(q)$ for all q and all large p. Given $q \geq q_0$, take $p \equiv q$ so large that $p \geq \max(p^*, b + g + r)$ and $Y[q, p] = \pi_E(q)$. In Lemma 3.3 at $[q, p]$ we have $A(q) = \pi_E(q)$, $B(p) = \omega(p) = \varepsilon$ and $\nu = \varepsilon$, so $\pi_E(q) = \pi_E(q) \oplus \varepsilon \oplus \varepsilon \oplus \chi(q)$, where $\chi(q)$ is the parity of the number of firings at centres $[q', p']$ with $q_0 < q' < q$ and $q' \not\equiv q$. So $\chi(q) = 0$ for all $q \geq q_0$, and the number of firings on the ray $q' = q + 1$, which is $\chi(q + 2) \oplus \chi(q)$ modulo 2, is even for every $q \geq q_0$.

Step 4 (no descending kinks in D). Let a_1 be the least q with $\pi_E(q) \neq \varepsilon$; then $a - r \leq a_1 \leq a + r$ and $a_1 \geq q_0 + 2$. The pair (E, W) satisfies the hypotheses of Lemma 4.7, so the number of firings on the ray $a_1 - 1$ is 1 if ω has a descending kink $x \equiv a_1$, and 0 otherwise. By Step 3 it is 0; so ω, and hence δ, has no descending kink $x \equiv a_1$. Now repeat Steps 2–4 with $T_{b+1,g-1}$ in place of $T_{b,g}$, which is allowed by Step 1 since $g - 1 \geq 2r + 1$. The point E, and so a_1, is unchanged, and hK^-_{b+1} has the diagram $\delta(p - 1)$, because K^-_{b+1} is the translate of K^-_b by one step and h commutes with translation. So $\delta(p - 1)$ has no descending kink $\equiv a_1$, that is, δ has none $\equiv a_1 - 1$. Hence δ has no descending kink.

Step 5 (no background swap; D is bare). If $\varepsilon = 1$, then $\delta(p) = 1$ for $p < b - r$ and $\delta(p) = 0$ for $p \geq b + r$, so δ has a descending kink, contrary to Step 4. So $\varepsilon = 0$: h fixes b_0 and b_1. Then δ is 0 far to the left and 1 far to the right and has no descending kink, so it has exactly one kink, b', and $D = K^-_{b'}$. Put $\alpha^- = b' - b$. Since h commutes with translation, $hK^-_{b''} = K^-_{b''+\alpha^-}$ for every b''.

Step 6 (the mirror image). The map $h' = \operatorname{rev} h \operatorname{rev}$ is an automorphism of $X_{F_{94}}$ of radius R, by Lemma 2.3, since $f(a, b, c) = f(c, b, a)$. Since $\operatorname{rev} F \operatorname{rev} = F$, $Y_{\operatorname{rev} c}(t, i) = Y_c(t, -i)$, that is, $Y_{\operatorname{rev} c}[q, p] = Y_c[p, q]$; so rev maps the point of Φ^+ with diagram $\pi(q)$ to the point of Φ^- with diagram $\pi(p)$, and conversely, and $\operatorname{rev} K^+_a = K^-_a$. Steps 0–5 for h' give an α^+ with $h'K^-_a = K^-_{a+\alpha^+}$ for all a, so $hK^+_a = \operatorname{rev} h' K^-_a = K^+_{a+\alpha^+}$.

Step 7 (the other two kinks). Let $a, b \in \mathbb{Z}$. By (7), $\mathcal{S}(K^+_a, K^-_b) = (\bar{K}^-_b, \bar{K}^+_a)$, and by Lemma 4.5 and Steps 5 and 6,

$$(h\bar{K}^-_b, h\bar{K}^+_a) = \mathcal{S}(hK^+_a, hK^-_b) = \mathcal{S}(K^+_{a+\alpha^+}, K^-_{b+\alpha^-}) = (\bar{K}^-_{b+\alpha^-}, \bar{K}^+_{a+\alpha^+}),$$

the last by (7) again.

Step 8 (parity, from the delay on the background 11). Take $a \equiv b$. By (7), $\mathcal{S}(\bar{K}^+_a, \bar{K}^-_b) = (K^-_{b+1}, K^+_{a+1})$, so by Lemma 4.5 and Steps 5–7

$$\mathcal{S}(\bar{K}^+_{a+\alpha^+}, \bar{K}^-_{b+\alpha^-}) = (hK^-_{b+1}, hK^+_{a+1}) = (K^-_{b+1+\alpha^-}, K^+_{a+1+\alpha^+}).$$

If $\alpha^+ \not\equiv \alpha^-$, then $a + \alpha^+ \not\equiv b + \alpha^-$, and by (7) the left side is $(K^-_{b+\alpha^-}, K^+_{a+\alpha^+})$, a contradiction. So $\alpha^+ \equiv \alpha^-$, and x, y are integers. By the paragraph before Lemma 4.5, $\sigma^x F^y$ maps the diagram $\pi(q)$ of a point of Φ^+ to $\pi(q - x + y)$ and the diagram $\xi(p)$ of a point of Φ^- to $\xi(p + x + y)$; so it maps $K^+_{a+\alpha^+}$ to $K^+_{a+\alpha^+ + x - y} = K^+_a$ and $K^-_{b+\alpha^-}$ to $K^-_{b+\alpha^- - x - y} = K^-_b$, and likewise for $\bar{K}^\pm$. It fixes b_0 and b_1. □

Kink rigidity for 95. Let $f = 95$, so that $N(a, b, c) = (1 \oplus a)(1 \oplus c)$ and 11 is the only valid background. For $a, b \in \mathbb{Z}$ let $\Delta^+_a \in \Phi^+$ and $\Delta^-_b \in \Phi^-$ be the points whose diagrams are 1 except for the value 0 at $q = a$, resp. at $p = b$; we call them the *bare defects*. They lie in $\Phi^\pm$ because Φ_{95} is the set of sequences with no two zeros at distance 2.

Since N_{95} does not read its middle argument, (4) at $[q, p]$ involves only points with first coordinate $\equiv q$ and second $\equiv p \pmod 2$, and the lattice splits into the two sublattices $\{q \equiv p \equiv e\}$, $e \in \{0, 1\}$. Fix a quadrant G with corner $[q_0, p_0]$, and let Y be a solution of (4) on G with boundary values A, B and corner values 1. For $e \in \{0, 1\}$ let $q_e \in \{q_0, q_0 + 1\}$ and $p_e \in \{p_0, p_0 + 1\}$

be the elements $\equiv e$, and put, for $j,k\geq 0$,

$$u(j,k)=1\oplus Y[q_e{+}2j,p_e{+}2k],\qquad x_j=u(j,0)=1\oplus A(q_e{+}2j),\qquad y_k=u(0,k)=1\oplus B(p_e{+}2k).$$

So $x_0=y_0=0$, and $x_j=1$ (resp. $y_k=1$) exactly at the zeros of A (resp. B) in the class e. For $j,k\geq 1$, (4) at $[q_e+2j,p_e+2k]$ reads $1\oplus u_{\mathsf N}=(1\oplus u_{\mathsf W})\oplus(1\oplus u_{\mathsf S})\oplus(1\oplus u_{\mathsf E})\oplus u_{\mathsf W}u_{\mathsf E}$, with $u_{\mathsf N}=u(j,k)$, $u_{\mathsf W}=u(j,k-1)$, $u_{\mathsf E}=u(j-1,k)$, $u_{\mathsf S}=u(j-1,k-1)$; since $a\oplus b\oplus ab=a\vee b$,

$$u(j,k)=\big(u(j,k-1)\vee u(j-1,k)\big)\oplus u(j-1,k-1),\tag{8}$$

and Y fires at the centre $[q_e+2j-1,p_e+2k-1]$ if and only if $u(j,k-1)=u(j-1,k)=1$. Every centre $[q',p']$ with $q'>q_0$, $p'>p_0$ is of this form for $e\equiv q'+1$ and some $j,k\geq 1$. If A and B are diagrams of points of $\Phi^{\pm}$, then x and y contain no two consecutive 1s. Call the sequence $k\mapsto u(j,k)$ the *row* j.

Lemma 4.10 (rows for 95)**.** *Let u be the solution of (8) with boundary values x, y, where $x_0=y_0=0$, neither x nor y contains two consecutive* 1*s, and $y_k=0$ for all large k.*

(0) *If $x_j=0$ for $j\leq J$, then $u(j,k)=y_k$ for $j\leq J$ and all k, and no centre between two of these rows fires. The same holds with the roles of j and k, and of x and y, exchanged.*

Let $j_1\geq 1$ be the least j with $x_j=1$, and k_1 the least k with $y_k=1$ ($k_1=\infty$ if $y=0$).

(i) *$u(j_1,k)=1$ for $k\leq k_1$ and $u(j_1,k)=y_k$ for $k>k_1$. In particular $\lim_k u(j_1,k)=1$ if and only if $y=0$.*

(ii) *$\lim_k u(j_1+1,k)=1$ if and only if y has exactly one* 1*.*

Proof. (0) By induction on j and k: $u(j,0)=x_j=0=y_0$, and if $u(j-1,\cdot)=y$ and $u(j,k-1)=y_{k-1}$, then $u(j,k)=(y_{k-1}\vee y_k)\oplus y_{k-1}=y_k$ and the firing term $y_{k-1}y_k$ vanishes, because $y_{k-1}y_k=0$. Equation (8) is symmetric in j and k.

(i) By (0), row j_1-1 is y. Write $v(k)=u(j_1,k)$, so $v(0)=1$ and $v(k)=(v(k-1)\vee y_k)\oplus y_{k-1}$. For $1\leq k<k_1$, $y_{k-1}=y_k=0$ and $v(k)=v(k-1)=1$. If $k_1<\infty$, then $k_1\geq 1$, $y_{k_1-1}=0$ and $v(k_1)=1$; next $y_{k_1+1}=0$ and $v(k_1+1)=v(k_1)\oplus y_{k_1}=0=y_{k_1+1}$; and if $v(k-1)=y_{k-1}$ with $k\geq k_1+2$, then $v(k)=(y_{k-1}\vee y_k)\oplus y_{k-1}=y_k$. The limit follows, since y is eventually 0.

(ii) Since $x_{j_1}=1$, $x_{j_1+1}=0$. Write $w(k)=u(j_1+1,k)$, so $w(0)=0$ and $w(k)=(w(k-1)\vee v(k))\oplus v(k-1)$, with v from (i). If $y=0$, then $v\equiv 1$ and $w\equiv 0$. Let $k_1<\infty$. For $1\leq k\leq k_1$, $v(k)=v(k-1)=1$ and $w(k)=0$. Then $v(k_1+1)=0$ and $v(k_1)=1$ give $w(k_1+1)=1$. Let k_2 be the least $k>k_1$ with $y_k=1$ ($k_2=\infty$ if there is none). For $k_1+1<k<k_2$, $v(k)=y_k=0$ and $v(k-1)=y_{k-1}=0$, so $w(k)=w(k-1)=1$; if $k_2=\infty$, the limit is 1. If $k_2<\infty$, then $k_2\geq k_1+2$, $v(k_2-1)=0$ and $w(k_2)=1$; then $v(k_2+1)=y_{k_2+1}=0$ and $w(k_2+1)=1\oplus 1=0=y_{k_2+1}$, and as in (i), $w(k)=y_k$ for all $k\geq k_2+1$, so the limit is 0. □

Lemma 4.11 (crossing of two defects; cf. [4])**.** *Let $f=95$, $a,b\in\mathbb{Z}$, and let $[q_0,p_0]$ be a corner with $q_0\leq a-2$ and $p_0\leq b-2$. Then (Δ_a^+,Δ_b^-) is a placed pair with background* 11*, and*

$$\mathcal{S}(\Delta_a^+,\Delta_b^-)=\begin{cases}(\Delta_b^-,\Delta_a^+), & a\not\equiv b\pmod 2,\\ (\Delta_{b+2}^-,\Delta_{a+2}^+), & a\equiv b\pmod 2.\end{cases}$$

Proof. Let $Y=Y_{\Delta_a^+,\Delta_b^-}$, and use the notation above on the two sublattices. If $a\not\equiv b$, then on the sublattice of a we have $y=0$, and on that of b, $x=0$; by Lemma 4.10(0), $u(j,k)=x_j$ on the first and $u(j,k)=y_k$ on the second, so Y equals the diagram of Δ_a^+ on the first sublattice and that of Δ_b^- on the second, and reading Y on the half-planes of Lemma 4.1(ii) gives (Δ_b^-,Δ_a^+). Let $a\equiv b\equiv e$. On the other sublattice $x=y=0$ and $Y=1$. On the sublattice e, x and y have a single 1, at $j_0=(a-q_e)/2\geq 1$ and $k_0=(b-p_e)/2\geq 1$. By Lemma 4.10(0) and (i), the rows $j<j_0$ are y and the row j_0 is 1 on $[0,k_0]$ and 0 after it; by the proof of (ii), with $k_2=\infty$, the row j_0+1 is 0 on $[0,k_0]$ and 1 after it. For the row j_0+2, $z(k)=u(j_0+2,k)$, we

have $z(0) = x_{j_0+2} = 0$ and $z(k) = (z(k-1) \vee u(j_0+1,k)) \oplus u(j_0+1,k-1)$, so $z = 0$ on $[0,k_0]$, $z(k_0+1) = 1$, $z(k_0+2) = (1 \vee 1) \oplus 1 = 0$, and then $z(k) = (z(k-1) \vee 1) \oplus 1 = 0$ for $k > k_0+2$. So the row j_0+2 has a single 1, at k_0+1. Since $x_j = 0$ for $j \geq j_0+2$, Lemma 4.10(0), applied to the quadrant $\{j \geq j_0+2\}$ with this row as its first row, shows that every later row equals it. Hence $\lim_j u(j,k) = 1$ exactly for $k = k_0+1$, and $\lim_k u(j,k) = 1$ exactly for $j = j_0+1$. Reading Y on the half-planes of Lemma 4.1(ii) gives $\Gamma = \Delta^-_{b+2}$ and $\Omega = \Delta^+_{a+2}$. □

In the second case Y fires exactly at $[a-1,b-1]$ and $[a+3,b+3]$: both defects are delayed by two steps, the analogue of Lemma 4.4(ii).

Theorem 4.12 (kink rigidity for 95)**.** *Let $h \in \mathrm{Aut}(X_{F_{95}})$. Then h fixes b_{11}, and there are integers $\alpha^+ \equiv \alpha^-$ (mod 2) such that $h\Delta^+_a = \Delta^+_{a+\alpha^+}$ and $h\Delta^-_b = \Delta^-_{b+\alpha^-}$ for all $a,b \in \mathbb{Z}$. Consequently, with $x = (\alpha^- - \alpha^+)/2$ and $y = (\alpha^+ + \alpha^-)/2$, $\sigma^x F^y h$ fixes b_{11} and every bare defect.*

Proof. Let R be the radius of h and $r = R+1$. By the paragraph before Lemma 4.5, $h(b_{11})$ is a background, so $h(b_{11}) = b_{11}$. Fix a, let $E = h\Delta^+_a$, with diagram $\pi_E(q)$, and for $b \in \mathbb{Z}$ let $D_b = h\Delta^-_b$, with diagram $\delta_b(p)$. Since h commutes with translation, $\delta_{b+1}(p) = \delta_b(p-1)$, and $h\Delta^+_{a+2}$ has the diagram $\pi_E(q-2)$. By (6), $\pi_E(q) = 1$ for $|q-a| > r$ and $\delta_b(p) = 1$ for $|p-b| > r$. Since h is injective and $\Delta^+_a \neq b_{11}$, π_E has a zero.

Fix $b \not\equiv a$. By Lemmas 4.11 and 4.5,

$$\text{(T)}\ \ \mathcal{S}(E,D_b) = (D_b,E), \qquad \text{(}\Delta\text{)}\ \ \mathcal{S}(E,D_{b+1}) = (D_{b+3}, h\Delta^+_{a+2}),$$

and we take the corner of both pairs at some $[q_0,p_0]$ with $q_0 \leq a-r-2$ and $p_0 \leq b-r-2$. Let e be a class modulo 2 that contains a zero of π_E, let a_1 be the least such zero, and $j_1 = (a_1-q_e)/2 \geq 1$. On the sublattice e the data x of both pairs satisfy $x_j = 0$ for $j < j_1$ and $x_{j_1} = 1$, so Lemma 4.10 applies; the outgoing Φ^+ part Ω, with diagram $o(q)$, has $o(q_e+2j) = 1 \oplus \lim_k u(j,k)$ by Lemma 4.1(ii). In (T), $o = \pi_E$, so $o(a_1) = 0$, $\lim_k u(j_1,k) = 1$, and by Lemma 4.10(i) δ_b has no zero in the class e. In (Δ), $o(q) = \pi_E(q-2)$, so $o(a_1) = \pi_E(a_1-2) = 1$ and $o(a_1+2) = \pi_E(a_1) = 0$; by Lemma 4.10(i) and (ii), δ_{b+1} has exactly one zero in the class e, that is, δ_b has exactly one zero in the class $1-e$. If π_E had zeros in both classes, the first conclusion for the class $1-e$ would contradict the second for the class e. So the zeros of π_E lie in a single class e, and δ_b has no zero in the class e and exactly one in the class $1-e$: $D_b = \Delta^-_{b'}$ for some b'. With $\alpha^- = b'-b$ and translation, $h\Delta^-_{b''} = \Delta^-_{b''+\alpha^-}$ for every b''.

As in Step 6 of the proof of Theorem 4.9, $\mathrm{rev}\, h\, \mathrm{rev}$ is an automorphism of radius R and $\mathrm{rev}\,\Delta^+_a = \Delta^-_a$, so there is α^+ with $h\Delta^+_a = \Delta^+_{a+\alpha^+}$ for all a. Finally let $a \equiv b$. By Lemmas 4.11 and 4.5,

$$\mathcal{S}(\Delta^+_{a+\alpha^+}, \Delta^-_{b+\alpha^-}) = (h\Delta^-_{b+2}, h\Delta^+_{a+2}) = (\Delta^-_{b+2+\alpha^-}, \Delta^+_{a+2+\alpha^+}),$$

and if $\alpha^+ \not\equiv \alpha^-$ the left side is $(\Delta^-_{b+\alpha^-}, \Delta^+_{a+\alpha^+})$ by Lemma 4.11. So $\alpha^+ \equiv \alpha^-$, and $\sigma^x F^y$ maps $\Delta^+_{a+\alpha^+}$ to Δ^+_a and $\Delta^-_{b+\alpha^-}$ to Δ^-_b, as in Step 8 of that proof. □

5. The scattering closure and the automorphism group of 94R

The scattering closure. Let $f \in \{94,95\}$. The *bare points* are the backgrounds and the bare kinks for 94, and b_{11} and the bare defects for 95. An automorphism that fixes the bare points has two further properties: it fixes every point of $\Phi^\pm$ that agrees with points it fixes on every window of the size of its local rule, and by Lemma 4.5 and Proposition 4.2 its set of fixed points in $\Phi^+ \cup \Phi^-$ is closed under scattering, forwards and backwards. The following definition collects these properties. A *word of* a point of Φ^+ or Φ^- is a finite word that occurs in its diagram, the sequence $\pi(q)$ or $\xi(p)$.

Definition 5.1. Let $f \in \{94, 95\}$ and $L \geq 1$. The *scattering closure* C_L is the smallest subset of $\Phi^+ \cup \Phi^-$ that contains the bare points and has the following properties:

(C0) if $Z \in \Phi^+$ and every word of length L of Z is a word of some point of $C_L \cap \Phi^+$, then $Z \in C_L$; and likewise with Φ^- in place of Φ^+;
(C1) it is invariant under σ, σ^{-1}, rev and $\mu = \mathrm{rev} \circ \rho$;
(C2) if $P, X \in C_L$ and (P, X) is a placed pair, then both components of $\mathcal{S}(P, X)$ lie in C_L;
(C3) if (P, X) is a placed pair and both components of $\mathcal{S}(P, X)$ lie in C_L, then $P, X \in C_L$.

Each of these properties passes from a family of subsets of $\Phi^+ \cup \Phi^-$ to its intersection (for (C0), a point of the intersection whose words of length L are words of points of the intersection has them among the words of points of each member), so the smallest such set exists. In (C0) the words of Z may be words of different points of C_L. Since rev maps Φ^+ onto Φ^- and μ maps each of them onto itself (the paragraph before Proposition 4.2), $C_L \cap \Phi^- = \mathrm{rev}(C_L \cap \Phi^+)$. If $L \leq L'$, every word of length L of a point is part of a word of length L' of that point, so C_L has the property (C0) for L', and $C_L \supseteq C_{L'}$.

The property (C0) cannot be omitted: the set of bare points itself has the properties (C1)–(C3). Indeed, $\sigma^{\pm 1}$, rev and μ map bare points to bare points (the proof of Lemma 5.2 below lists the images). If a placed pair (P, X) of bare points contains no background, it is (K_a^+, K_b^-) or $(\bar{K}_a^+, \bar{K}_b^-)$ for 94 and (Δ_a^+, Δ_b^-) for 95, and both components of $\mathcal{S}(P, X)$ are bare points by (7), resp. Lemma 4.11. If X is a background, they are bare points by Corollary 4.3(1); and if P is a background, then, as in the proof of that corollary, $Y_{P,X} = \xi(p)$ everywhere, so $\mathcal{S}(P, X)$ consists of X and a background. Finally, if (P, X) is a placed pair and both components of $\mathcal{S}(P, X) = (\Gamma, \Omega)$ are bare points, then $(\mu\Omega, \mu\Gamma)$ is a placed pair of bare points and $\mathcal{S}(\mu\Omega, \mu\Gamma) = (\mu X, \mu P)$ by Proposition 4.2, so P and X are bare points.

Lemma 5.2. *Let $f \in \{94, 95\}$, $R \geq 0$ and $L \geq 2R + 2$, and let g be an automorphism of X_{F_f} of radius at most R that fixes every bare point. Then g fixes every point of C_L.*

Proof. Let $\mathcal{M}$ be the set of automorphisms of radius at most R that fix every bare point, and let $\mathcal{F}$ be the set of points of $\Phi^+ \cup \Phi^-$ that are fixed by every element of $\mathcal{M}$. It contains the bare points. We show that $\mathcal{F}$ has the properties (C0)–(C3), with $\mathcal{F}$ in place of C_L; then $C_L \subseteq \mathcal{F}$.

(C0) Let $Z \in \Phi^-$, with diagram ζ, be such that every word of length L of Z is a word of a point of $\mathcal{F} \cap \Phi^-$, and let $g \in \mathcal{M}$ and $p \in \mathbb{Z}$. By (6), if two points of Φ^- have the same diagram on the window $[p - R - 1, p + R]$, then their images under g have the same diagram at p. The window has length $2R + 2 \leq L$, so the word of ζ on it is part of a word of length L of Z, and hence a word of some $Z' \in \mathcal{F} \cap \Phi^-$. Since σ acts on the diagrams of points of Φ^- by $\xi(p) \mapsto \xi(p + 1)$ (the paragraph before Lemma 4.5), there is s such that $Z'' = \sigma^s Z'$ has the same diagram as Z on $[p - R - 1, p + R]$. As g commutes with σ, $gZ'' = Z''$. So the diagram of gZ at p is that of $gZ'' = Z''$ at p, which is $\zeta(p)$. Hence $gZ = Z$. For $Z \in \Phi^+$ the argument is the same, with q in place of p.

(C1) An element g of $\mathcal{M}$ commutes with $\sigma^{\pm 1}$, so it fixes $\sigma^{\pm 1} Z$ for $Z \in \mathcal{F}$. The map $\mathrm{rev}\, g\, \mathrm{rev}$ is an automorphism of radius at most R by Lemma 2.3, and rev maps bare points to bare points ($K_a^\pm \mapsto K_a^\mp$, $\bar{K}_a^\pm \mapsto \bar{K}_a^\mp$, $\Delta_a^\pm \mapsto \Delta_a^\mp$, backgrounds to themselves); so $\mathrm{rev}\, g\, \mathrm{rev} \in \mathcal{M}$, and for $Z \in \mathcal{F}$, $g(\mathrm{rev}\, Z) = \mathrm{rev}\,(\mathrm{rev}\, g\, \mathrm{rev})(Z) = \mathrm{rev}\, Z$. Likewise $\mu g \mu$ commutes with σ and, since $\mu F \mu = F^{-1}$, with F; it is invertible, it has radius at most R because ρ acts cellwise, and μ maps bare points to bare points, because it maps the diagram $\pi(q)$ to $\pi(-1 - q)$ and $\xi(p)$ to $\xi(-1 - p)$: $K_a^\pm \mapsto \bar{K}_{-a}^\pm$, $\bar{K}_a^\pm \mapsto K_{-a}^\pm$, $\Delta_a^\pm \mapsto \Delta_{-1-a}^\pm$. So $\mu g \mu \in \mathcal{M}$, and g fixes μZ for $Z \in \mathcal{F}$.

(C2), (C3) Let $g \in \mathcal{M}$, and let (P, X) be a placed pair with background β and $\mathcal{S}(P, X) = (\Gamma, \Omega)$. Since $g(b_\beta) = b_\beta$, Lemma 4.5 shows that (gP, gX) is a placed pair and $\mathcal{S}(gP, gX) = (g\Gamma, g\Omega)$. If $P, X \in \mathcal{F}$, then $(\Gamma, \Omega) = \mathcal{S}(P, X) = \mathcal{S}(gP, gX) = (g\Gamma, g\Omega)$. If $\Gamma, \Omega \in \mathcal{F}$, then $\mathcal{S}(gP, gX) =$

$(g\Gamma, g\Omega) = (\Gamma, \Omega) = \mathcal{S}(P, X)$, and since $\mathcal{S}$ is injective on placed pairs (Proposition 4.2), $gP = P$ and $gX = X$. □

For 94 the closure contains every point with finitely many kinks and hence, by (C0), is all of $\Phi^+ \cup \Phi^-$ (Theorem 5.8 below), and this determines $\mathrm{Aut}(X_{F_{94}})$ (Theorem 5.9); for 95 it is too small (Remark 5.10).

Prepending a kink. For the rest of this section, except in Remark 5.10, let $f = 94$. By an *object* we mean a Φ^- object, that is, a point of Φ^- with finitely many kinks; if it is not a background, its *first kink* is its least kink. Let U be an object that is not a background, with diagram υ and first kink u, and let c be the value of υ on $\{p < u\}$. For $b \le u - 2$ let $[b; U] \in \Phi^-$ be the point with the diagram

$$\xi(p) = 1 \oplus c \quad (p < b), \qquad \xi(p) = \upsilon(p) \quad (p \ge b).$$

It is an object whose kinks are b and the kinks of U; the two triples of ξ that are not triples of υ or constant are $(1 \oplus c, 1 \oplus c, c)$ and $(1 \oplus c, c, c)$, so $\xi \in \Phi_{94}$.

Lemma 5.3 (prepending a kink)**.** *Let $L \ge 1$, let $U \in C_L$ be an object that is not a background, with first kink u, and let $b \le u - 2$. Then $[b; U] \in C_L$.*

The proof scatters bare kinks off $[b; U]$ in such a way that the kink b is never delayed, while the first kink of the rest is delayed at every second scattering. After $2L$ scatterings the two are so far apart that (C0) applies, and (C3) leads back to $[b; U]$. What the rest becomes in the course of this is not controlled, and does not need to be: it stays in C_L by (C2).

Proof. We construct objects $U_0 = U, U_1, U_2, \dots$ in C_L, none of them a background, with first kinks $u_0 = u \le u_1 \le u_2 \le \cdots$, bare kinks $P_i \in \Phi^+$ and points $\Omega_i \in C_L$ such that, with $X_i = [b; U_i]$,

$$(P_i, X_i) \text{ is a placed pair}, \qquad \mathcal{S}(P_i, X_i) = (X_{i+1}, \Omega_i), \tag{9}$$

and $u_{i+1} \ge u_i + 1$ whenever the diagram of U_i is 1 on $\{p < u_i\}$. Let U_i be given, with diagram υ_i and $\upsilon_i(p) = c$ for $p < u_i$, and $u_i \ge b + 2$.

The case $c = 0$. Here X_i has the value 1 on $\{p < b\}$. Let $P_i = \bar{K}^+_a$ with $a \not\equiv b \pmod 2$, and choose a corner $[q_0, p_0]$ with $q_0 \le a - 2$, $p_0 \le b - 2$ and $q_0 \equiv p_0 \equiv b$. By Lemma 4.4(i) with $m = 1$ and (7), the pair $(\bar{K}^+_a, \bar{K}^-_b)$ has the diagram $Y_1 = 1 \oplus \theta_a(q) \oplus \theta_b(p)$ on G, and $\mathcal{S}(\bar{K}^+_a, \bar{K}^-_b) = (K^-_b, K^+_a)$. The diagram of $\bar{K}^-_b$ is 0 on $\{p \ge b\}$, and $Y_1[q, p] = \theta_a(q)$ for $p \ge b$, which is the diagram of K^+_a. So Lemma 4.6 applies to this pair with $p_2 = b$ and $X' = U_i$, whose diagram is 0 for $p \le b + 1$ because $u_i \ge b + 2$; and $\bar{K}^-_b * U_i = X_i$. Hence (P_i, X_i) is a placed pair, (K^+_a, U_i) is a placed pair, and

$$\mathcal{S}(P_i, X_i) = (K^-_b * \Gamma', \Omega'), \qquad (\Gamma', \Omega') = \mathcal{S}(K^+_a, U_i).$$

Since K^+_a and U_i lie in C_L, so do Γ' and Ω', by (C2). By Lemma 4.1(ii), Γ' is an object, and its diagram γ' has the value 1, the value of $\theta_a(q)$ for large q, on $\{p \le p'_0 + 1\}$ for every corner $[q'_0, p'_0]$ of the pair (K^+_a, U_i); there are corners with $p'_0 = u_i - 2$, so $\gamma'(p) = 1$ for $p < u_i$. By Corollary 4.3(2), Γ' is not a background. Put $U_{i+1} = \Gamma'$ and $\Omega_i = \Omega'$. Then $u_{i+1} \ge u_i$, and $K^-_b * \Gamma'$, whose diagram is $\theta_b(p)$ for $p \le b + 1$ and $\gamma'(p)$ for $p \ge b$, is $[b; U_{i+1}]$.

The case $c = 1$. Here X_i has the value 0 on $\{p < b\}$. Let $P_i = K^+_a$ with $a \equiv u_i \pmod 2$, and choose the corner as before. By Lemma 4.4(i) with $m = 0$ and (7), the pair (K^+_a, K^-_b) has the diagram $Y_1 = \theta_a(q) \oplus \theta_b(p)$ on G, and $\mathcal{S}(K^+_a, K^-_b) = (\bar{K}^-_b, \bar{K}^+_a)$. The diagram of K^-_b is 1 on $\{p \ge b\}$, and $Y_1[q, p] = 1 \oplus \theta_a(q)$ for $p \ge b$, which is the diagram of $\bar{K}^+_a$. So Lemma 4.6 applies with $p_2 = b$ and $X' = U_i$, whose diagram is 1 for $p \le b + 1$; and $K^-_b * U_i = X_i$. Hence (P_i, X_i) and $(\bar{K}^+_a, U_i)$ are placed pairs, and

$$\mathcal{S}(P_i, X_i) = (\bar{K}^-_b * \Gamma', \Omega'), \qquad (\Gamma', \Omega') = \mathcal{S}(\bar{K}^+_a, U_i).$$

As before, $\Gamma', \Omega' \in C_L$, and Γ' is an object that is not a background. We claim that its diagram γ' vanishes on $\{p \le u_i\}$. Let $[q_0', p_0']$ be a corner with $q_0' \le a-2$, $p_0' \le u_i - 2$ and $q_0' \equiv p_0'$, and G' its quadrant. It is a corner of the pair $(\bar{K}_a^+, U_i)$ and of the pair $(\bar{K}_a^+, \bar{K}_{u_i}^-)$, and the diagrams of U_i and of $\bar{K}_{u_i}^-$ agree on $\{p \le u_i\}$, where they are 1 for $p < u_i$ and 0 at u_i. By Goursat causality the diagrams of the two pairs agree on $G' \cap \{p \le u_i\}$. By Lemma 4.4(ii), which applies because $a \equiv u_i$, the diagram of the second pair is 0 at every point of G' with $q \ge a$ and $p \le u_i$. Reading the diagram of the first pair on a half-plane $\{q \ge q_1\}$ on which it equals $\gamma'(p)$ (Lemma 4.1(ii)) gives $\gamma'(p) = 0$ for $p_0' \le p \le u_i$, and $\gamma'(p) = 0$ for $p \le p_0' + 1$ by the same lemma, 0 being the value of $1 \oplus \theta_a(q)$ for large q. This proves the claim. Put $U_{i+1} = \Gamma'$ and $\Omega_i = \Omega'$. Then $u_{i+1} \ge u_i + 1$, and $\bar{K}_b^- * \Gamma'$, whose diagram is $1 \oplus \theta_b(p)$ for $p \le b+1$ and $\gamma'(p)$ for $p \ge b$, is $[b; U_{i+1}]$.

Conclusion. In both cases the diagram of U_{i+1} is $1 \oplus c$ on $\{p < u_{i+1}\}$, so the two cases alternate, and $u_{i+2} \ge u_i + 1$ for all i. Let $n = 2L$; then $u_n \ge u_0 + L \ge b + L + 2$. A window $[s, s+L-1]$ with $s \ge b$ lies in $\{p \ge b\}$, where the diagram of X_n is that of $U_n \in C_L$; a window with $s < b$ lies in $\{p < u_n\}$, where the diagram of X_n is that of a bare kink with its kink at b. So every word of length L of X_n is a word of a point of $C_L \cap \Phi^-$, and $X_n \in C_L$ by (C0). If $X_{i+1} \in C_L$, then by (9) both components of $\mathcal{S}(P_i, X_i)$ lie in C_L, and $X_i \in C_L$ by (C3). Hence $X_0 = [b; U]$ lies in C_L. □

Starting from a bare kink, Lemma 5.3 gives every object in which any two consecutive kinks are at distance at least 2. The remaining objects contain a 0 between two 1s. For $a, b \in \mathbb{Z}$ let $Z_a^+ \in \Phi^+$ and $Z_b^- \in \Phi^-$ be the points whose diagrams are 1 except for the value 0 at $q = a$, resp. at $p = b$; we call them the *isolated zeros*. The first of the next two lemmas produces them from objects of the first kind, and the second describes how they cross a kink and each other.

Lemma 5.4 (seed event)**.** *Let $g \ge 2$ and $a, b \in \mathbb{Z}$ with $a \equiv b + g \pmod 2$, put $r = b + g$, and let $X \in \Phi^-$ have the diagram that is 0 for $p < b$, 1 for $b \le p < r$, 0 for $p \in \{r, r+1\}$ and 1 for $p \ge r+2$; so $X = 0^\infty 1^g 00 1^\infty$. Let $[q_0, p_0]$ be a corner with $q_0 \le a-2$, $p_0 \le b-2$ and $q_0 \equiv p_0$. Then (K_a^+, X) is a placed pair with background 00, its diagram fires exactly at the centres $[a-1, r-1]$ and $[a+2, r+4]$, and $\mathcal{S}(K_a^+, X) = (\Gamma, Z_{a+1}^+)$, where the diagram of Γ is 1 for $p < b$, 0 for $b \le p \le r$, 1 for $p \in \{r+1, r+2\}$, 0 for $p = r+3$ and 1 for $p \ge r+4$. So $\Gamma = 1^\infty 0^{g+1} 11 0 1^\infty$.*

Proof. The first statement is clear. Lowering the corner, which does not change the diagram of the pair, we may assume $q_0 \equiv b$. Let $X_1 = K_b^-$, and let X' have the diagram that is 1 for $p < r$, 0 for $p \in \{r, r+1\}$ and 1 for $p \ge r+2$; since $r \ge b+2$, $X = X_1 * X'$ with the cut at $p_2 = b$. As in the second case of the proof of Lemma 5.3, Lemma 4.4(i) and Lemma 4.6 show that $Y_{K_a^+, X_1}$ does not fire, that $(\bar{K}_a^+, X')$ is a placed pair with corner $[q_0, b]$ and background 11, that $Y_{K_a^+, X}$ equals $Y_{K_a^+, X_1}$ on $\{p \le b+1\}$ and $Y' = Y_{\bar{K}_a^+, X'}$ on $\{p \ge b\}$, and that $\mathcal{S}(K_a^+, X) = (\bar{K}_b^- * \Gamma', \Omega')$ with $(\Gamma', \Omega') = \mathcal{S}(\bar{K}_a^+, X')$. A centre $[q', p']$ with $p' \le b$ has its five points in $\{p \le b+1\}$, and one with $p' \ge b+1$ has them in $\{p \ge b\}$; so the firings of $Y_{K_a^+, X}$ are those of Y', which by Lemma 4.1(i) lie at centres with $p' > b$.

We determine Y' on the quadrant $G' = \{q \ge q_0,\ p \ge b\}$ by the ansatz of the proof of Lemma 4.4. Write $x = q - a$ and $y = p - r$, so $x \equiv y \pmod 2$; the boundary values are $A(q) = 1 \oplus \alpha(x)$ and $B(p) = 1 \oplus \beta'(y)$ with $\alpha(x) = [x \ge 0]$ and $\beta'(y) = [y \in \{0, 1\}]$, and the corner value is 1. Let $\Sigma = \{(-1, -1), (2, 4)\}$ in these coordinates; both centres have $q' > q_0$ and $p' > b$. The function $Y_\Sigma = A \oplus B \oplus 1 \oplus \chi_\Sigma$ of that proof is

$$Y_\Sigma(x, y) = 1 \oplus \alpha(x) \oplus \beta'(y) \oplus [x \text{ even},\ x \ge 0,\ y \ge 0] \oplus [x \text{ odd},\ x \ge 3,\ y \ge 5],$$

that is: for $x < 0$, $Y_\Sigma = 0$ if $y \in \{0, 1\}$ and 1 otherwise; for even $x \ge 0$, $Y_\Sigma = 1$ if $y \ge 2$ and 0 if $y \le 0$; for $x = 1$, $Y_\Sigma = 1$ if $y = 1$ and 0 otherwise; for odd $x \ge 3$, $Y_\Sigma = 1$ if $y = 1$

or $y \geq 5$ and 0 if $y \leq -1$ or $y = 3$. A firing at $C = (x, y)$ needs $Y_\Sigma(C) = 1$ and $Y_\Sigma = 0$ at $\mathsf{W} = (x+1, y-1)$ and $\mathsf{E} = (x-1, y+1)$. If $x \leq -2$, W and E lie in $\{x < 0\}$ and would need $y - 1, y + 1 \in \{0, 1\}$, which is impossible. If $x = -1$, $Y_\Sigma(\mathsf{E}) = 0$ needs $y + 1 \in \{0, 1\}$, so $y = -1$, and $C = (-1, -1)$ does fire, as $Y_\Sigma(-1, -1) = 1$ and $Y_\Sigma(0, -2) = Y_\Sigma(-2, 0) = 0$. If $x = 0$, $Y_\Sigma(C) = 1$ needs $y \geq 2$, while $Y_\Sigma(\mathsf{E}) = Y_\Sigma(-1, y + 1) = 0$ needs $y \leq 0$. If $x = 1$, $Y_\Sigma(C) = 1$ needs $y = 1$, and then $Y_\Sigma(\mathsf{E}) = Y_\Sigma(0, 2) = 1$. If $x \geq 2$ is even, $Y_\Sigma(C) = 1$ needs $y \geq 2$, and $Y_\Sigma(\mathsf{W}) = 0$ then needs $y - 1 = 3$; for $x = 2$ the centre $(2, 4)$ fires, as $Y_\Sigma(3, 3) = Y_\Sigma(1, 5) = 0$, and for $x \geq 4$, $Y_\Sigma(\mathsf{E}) = Y_\Sigma(x - 1, 5) = 1$. If $x \geq 3$ is odd, $Y_\Sigma(\mathsf{W}) = 0$ needs $y \leq 1$, so $y = 1$, and $Y_\Sigma(\mathsf{E}) = Y_\Sigma(x - 1, 2) = 1$. So Y_Σ fires exactly on Σ, and $Y' = Y_\Sigma$ on G'.

For large x, Y_Σ is 0 for $y \leq 0$, 1 for $y \in \{1, 2\}$, 0 for $y = 3$ and 1 for $y \geq 4$; for large y it is 1 except at $x = 1$. Reading Y' on the half-planes of Lemma 4.1(ii), Γ' has the diagram that is 0 for $p \leq r$, 1 for $p \in \{r + 1, r + 2\}$, 0 for $p = r + 3$ and 1 for $p \geq r + 4$, and $\Omega' = Z^+_{a+1}$. With the diagram $1 \oplus \theta_b(p)$ of $\bar{K}^-_b$ for $p \leq b + 1$, this gives Γ. □

Corollary 5.5. *For every $L \geq 1$, the isolated zeros Z^+_a and Z^-_a, $a \in \mathbb{Z}$, lie in C_L.*

Proof. In Lemma 5.4 with $g = 2$, $X = [b; [r; K^-_{r+2}]]$, where $[r; K^-_{r+2}]$ has the first kink $r = b + 2$. By two applications of Lemma 5.3, starting from the bare kink K^-_{r+2}, $X \in C_L$. The probe K^+_a is a bare kink, so $Z^+_{a+1} \in C_L$ by (C2). Since σ acts on the diagrams of points of Φ^+ by $\pi(q) \mapsto \pi(q - 1)$, (C1) gives all Z^+_a, and $Z^-_a = \operatorname{rev} Z^+_a$ (Step 6 of the proof of Theorem 4.9). □

Lemma 5.6 (isolated zeros)**.** *Let $f = 94$, and let $[q_0, p_0]$ be a corner, with quadrant G.*

(i) *Let $a \equiv d \pmod 2$, $q_0 \leq a - 2$ and $p_0 \leq d - 2$. Then $(\bar{K}^+_a, Z^-_d)$ is a placed pair with background 11, its diagram Y fires exactly at the centres $[a - 1, d - 1]$ and $[a + 2, d + 2]$,*

$$\mathcal{S}(\bar{K}^+_a, Z^-_d) = (K^-_{d+2}, Z^+_{a+1}),$$

and $Y[q, p] = 1 \oplus [q = a + 1]$, the diagram of Z^+_{a+1}, at every point of G with $p \geq d + 2$.

(ii) *Let $a \equiv u \pmod 2$, $q_0 \leq a - 2$ and $p_0 \leq u - 2$. Then $(Z^+_a, \bar{K}^-_u)$ is a placed pair with background 11, and its diagram is $1 \oplus [p = u + 1]$ at every point of G with $q \geq a + 2$.*

(iii) *Let $a \not\equiv d \pmod 2$, $q_0 \leq a - 2$ and $p_0 \leq d - 2$. Then (Z^+_a, Z^-_d) is a placed pair with background 11, its diagram is $1 \oplus [q = a] \oplus [p = d]$ on G, it does not fire, and $\mathcal{S}(Z^+_a, Z^-_d) = (Z^-_d, Z^+_a)$.*

In (i) the kink of the probe and the zero of the object exchange their roles: the probe leaves as an isolated zero and the object as a bare kink. In (iii) the two zeros lie on different sublattices and do not meet.

Proof. That the pairs are placed is clear. We use the ansatz of the proof of Lemma 4.4, with the functions Y_Σ defined there.

(i) Write $x = q - a$ and $y = p - d$, so $x \equiv y \pmod 2$; the boundary values are $A(q) = 1 \oplus [x \geq 0]$ and $B(p) = 1 \oplus [y = 0]$, and the corner value is 1. Let $\Sigma = \{(-1, -1), (2, 2)\}$ in these coordinates; both centres have $q' > q_0$ and $p' > p_0$. The first contributes to χ_Σ at the points with $x \geq 0$, $y \geq 0$ and x even, the second at those with $x \geq 3$, $y \geq 3$ and x odd, so

$$Y_\Sigma(x, y) = 1 \oplus [x \geq 0] \oplus [y = 0] \oplus [x \text{ even}, x \geq 0, y \geq 0] \oplus [x \text{ odd}, x \geq 3, y \geq 3],$$

that is: for $x < 0$, $Y_\Sigma = 0$ if $y = 0$ and 1 otherwise; for even $x \geq 0$, $Y_\Sigma = 1$ if $y \geq 2$ and 0 if $y \leq 0$; for $x = 1$, $Y_\Sigma = 0$; for odd $x \geq 3$, $Y_\Sigma = 1$ if $y \geq 3$ and 0 if $y \leq 1$. A firing at $C = (x, y)$ needs $Y_\Sigma(C) = 1$ and $Y_\Sigma = 0$ at $\mathsf{W} = (x + 1, y - 1)$ and $\mathsf{E} = (x - 1, y + 1)$. If $x \leq -2$, W and E lie in $\{x < 0\}$ and would need $y - 1 = 0$ and $y + 1 = 0$. If $x = -1$, $Y_\Sigma(\mathsf{E}) = 0$ needs $y = -1$, and $C = (-1, -1)$ does fire, as $Y_\Sigma(-1, -1) = 1$ and $Y_\Sigma(0, -2) = Y_\Sigma(-2, 0) = 0$. If $x = 0$, $Y_\Sigma(C) = 1$ needs $y \geq 2$, while $Y_\Sigma(\mathsf{E}) = Y_\Sigma(-1, y + 1) = 0$ needs $y = -1$. If $x = 1$, $Y_\Sigma(C) = 0$. If $x = 2$,

$Y_\Sigma(C) = 1$ needs $y \geq 2$ and $Y_\Sigma(\mathsf{W}) = Y_\Sigma(3, y-1) = 0$ needs $y \leq 2$, and $C = (2,2)$ does fire, as $Y_\Sigma(2,2) = 1$ and $Y_\Sigma(3,1) = Y_\Sigma(1,3) = 0$. If $x \geq 3$ is odd, $Y_\Sigma(C) = 1$ needs $y \geq 3$, while $Y_\Sigma(\mathsf{W}) = 0$, at the even $x+1$, needs $y - 1 \leq 0$. If $x \geq 4$ is even, $Y_\Sigma(C) = 1$ needs $y \geq 2$, while $Y_\Sigma(\mathsf{E}) = 0$, at the odd $x - 1 \geq 3$, needs $y + 1 \leq 1$. So Y_Σ fires exactly on Σ, and $Y = Y_\Sigma$ on G. For $x \geq 3$, Y_Σ is 0 for $y \leq 1$ and 1 for $y \geq 2$; for $y \geq 2$ it is 1 except at $x = 1$, where it is 0. This is the last statement of (i), and reading Y on the half-planes of Lemma 4.1(ii) gives $\mathcal{S}(\bar{K}_a^+, Z_d^-)$.

(ii) The reflection $[q,p] \mapsto [p,q]$ maps solutions of (4) to solutions, since N is invariant under reversal of the triple, and it exchanges the two boundary rays. So the diagram of $(Z_a^+, \bar{K}_u^-)$ on G is $Y'[q,p] = Y[p,q]$, where Y is the diagram of the pair $(\bar{K}_u^+, Z_a^-)$ of (i) at the corner $[p_0, q_0]$. By the last statement of (i), $Y[p,q] = 1 \oplus [p = u+1]$ whenever $q \geq a+2$.

(iii) Take $\Sigma = \emptyset$, so $Y_\emptyset = 1 \oplus [q = a] \oplus [p = d]$. Since $q \equiv p$ at every point and $a \not\equiv d$, at most one of $q = a$, $p = d$ holds, and $Y_\emptyset[q,p] = 0$ exactly if $q = a$ or $p = d$. A firing at $C = [q,p]$ needs $Y_\emptyset = 0$ at $\mathsf{W} = [q+1, p-1]$ and at $\mathsf{E} = [q-1, p+1]$, that is, $q+1 = a$ or $p-1 = d$, and $q-1 = a$ or $p+1 = d$. Of the four combinations, two are contradictory, and the other two, $q+1 = a$ with $p+1 = d$ and $p-1 = d$ with $q-1 = a$, give $a \equiv d$. So $Y_\emptyset$ does not fire, it is the diagram on G, and reading it on the half-planes of Lemma 4.1(ii) gives $\mathcal{S}(Z_a^+, Z_d^-) = (Z_d^-, Z_a^+)$. □

Let U be an object that is not a background, with diagram υ and first kink u, such that $\upsilon = 1$ on $\{p < u\}$. For $d \leq u - 3$ let $\langle d; U\rangle \in \Phi^-$ be the point whose diagram is υ except for the value 0 at $p = d$. Its triples that contain d are 110, 101 and 011, since $\upsilon(d+1) = \upsilon(d+2) = 1$; so it is an object, with the kinks d, $d+1$ and those of U.

Lemma 5.7 (prepending an isolated zero)**.** *Let $L \geq 1$, let $U \in C_L$ be an object that is not a background, with first kink u and diagram 1 on $\{p < u\}$, and let $d \leq u - 3$. Then $\langle d; U\rangle \in C_L$.*

Proof. Let υ be the diagram of U. For every a, $Z_a^+ \in C_L$ by Corollary 5.5.

The case $u \geq d + 4$. Let $a \equiv d \pmod 2$ and $P = \bar{K}_a^+$, and choose a corner $[q_0, p_0]$ with $q_0 \leq a - 2$, $p_0 \leq d - 2$ and $q_0 \equiv p_0 \equiv d$. By Lemma 5.6(i), $\mathcal{S}(P, Z_d^-) = (K_{d+2}^-, Z_{a+1}^+)$, and the diagram of the pair (P, Z_d^-) equals that of Z_{a+1}^+ on the rays $p \in \{d+2, d+3\}$ of G; the diagram of Z_d^- is 1 on $\{p \geq d+2\}$. So Lemma 4.6 applies with $p_2 = d+2$ and $X' = U$, whose diagram is 1 for $p \leq d+3$ because $u \geq d+4$; and $Z_d^- * U = \langle d; U\rangle$. Hence $(P, \langle d; U\rangle)$ and (Z_{a+1}^+, U) are placed pairs, and

$$\mathcal{S}(P, \langle d; U\rangle) = (K_{d+2}^- * \Gamma', \Omega'), \qquad (\Gamma', \Omega') = \mathcal{S}(Z_{a+1}^+, U).$$

Since $Z_{a+1}^+, U \in C_L$, also $\Gamma', \Omega' \in C_L$ by (C2). As in the proof of Lemma 5.3, Γ' is an object that is not a background, and its diagram γ' is 1 on $\{p < u\}$, by Lemma 4.1(ii) at a corner with $p_0' = u - 2$ and Corollary 4.3(2). So the first kink of Γ' is at least $u \geq d+4$, and $K_{d+2}^- * \Gamma'$, whose diagram is $\theta_{d+2}(p)$ for $p \leq d+3$ and $\gamma'(p)$ for $p \geq d+2$, is $[d+2; \Gamma']$. By Lemma 5.3 it lies in C_L. Both components of $\mathcal{S}(P, \langle d; U\rangle)$ lie in C_L, so $\langle d; U\rangle \in C_L$ by (C3).

The case $u = d + 3$. Let $a \not\equiv d \pmod 2$, so that $a \equiv u$, and $P = Z_a^+$, and choose a corner $[q_0, p_0]$ with $q_0 \leq a - 2$, $p_0 \leq d - 2$ and $q_0 \equiv p_0 \equiv d + 1$. By Lemma 5.6(iii), $\mathcal{S}(P, Z_d^-) = (Z_d^-, Z_a^+)$, and the diagram of the pair (P, Z_d^-) is $1 \oplus [q = a]$, the diagram of Z_a^+, on $\{p \geq d+1\}$; the diagram of Z_d^- is 1 there. So Lemma 4.6 applies with $p_2 = d+1$ and $X' = U$, whose diagram is 1 for $p \leq d+2$; and $Z_d^- * U = \langle d; U\rangle$. Hence

$$\mathcal{S}(P, \langle d; U\rangle) = (Z_d^- * \Gamma', \Omega'), \qquad (\Gamma', \Omega') = \mathcal{S}(Z_a^+, U),$$

and $\Gamma', \Omega' \in C_L$ by (C2); Γ' is an object that is not a background. We claim that its diagram γ' is 1 on $\{p \leq u\}$. Let $[q_0', p_0']$ be a corner with $q_0' \leq a - 2$, $p_0' \leq u - 2$ and $q_0' \equiv p_0'$, and G' its quadrant. It is a corner of the pairs (Z_a^+, U) and $(Z_a^+, \bar{K}_u^-)$, and the diagrams of U and $\bar{K}_u^-$ agree on $\{p \leq u\}$. By Goursat causality the diagrams of the two pairs agree on $G' \cap \{p \leq u\}$,

and by Lemma 5.6(ii) the second is 1 at every point of G' with $q \geq a+2$ and $p \leq u$. Reading the first on a half-plane $\{q \geq q_1\}$ on which it equals $\gamma'(p)$ gives $\gamma'(p) = 1$ for $p'_0 \leq p \leq u$, and $\gamma'(p) = 1$ for $p \leq p'_0 + 1$ by Lemma 4.1(ii). So the first kink u' of Γ' satisfies $u' \geq u+1 = d+4$, and $Z_d^- * \Gamma'$, whose diagram is that of Z_d^- for $p \leq d+2$ and $\gamma'(p)$ for $p \geq d+1$, is $\langle d; \Gamma' \rangle$. By the first case, applied to $\Gamma' \in C_L$, it lies in C_L. Both components of $\mathcal{S}(P, \langle d; U \rangle)$ lie in C_L, so $\langle d; U \rangle \in C_L$ by (C3). □

Theorem 5.8. *Let $f = 94$ and $L \geq 1$. Every point of $\Phi^+ \cup \Phi^-$ with finitely many kinks lies in C_L. Consequently $C_L = \Phi^+ \cup \Phi^-$.*

Proof. Since rev maps the point of Φ^- with the diagram $\xi(p)$ to the point of Φ^+ with the diagram $\xi(q)$, and C_L is invariant under rev, it suffices to treat Φ^-. Let $X \in \Phi^-$ have the diagram ξ and the kinks $b_1 < \cdots < b_k$; we use induction on k. If $k \leq 1$, X is a bare point. Let $k \geq 2$.

If $b_2 - b_1 \geq 2$, let U be the point with the diagram that is $\xi(p)$ for $p \geq b_1$ and $\xi(b_1)$ for $p < b_1$. Its triples are constant, or triples of ξ, or the triple $(\xi(b_1), \xi(b_1), \xi(b_1+1))$, whose first two letters are equal; so it lies in Φ^-. It is an object with the kinks $b_2 < \cdots < b_k$, so $U \in C_L$ by the induction hypothesis, and $X = [b_1; U] \in C_L$ by Lemma 5.3.

If $b_2 - b_1 = 1$, then $\xi(b_1)$ differs from $\xi(b_1 - 1)$ and from $\xi(b_1 + 1)$, and since ξ contains no 010, $\xi(b_1) = 0$ and $\xi(b_1 \pm 1) = 1$. If $k = 2$, then $X = Z_{b_1}^-$, which lies in C_L by Corollary 5.5. If $k \geq 3$, then $b_3 - b_2 \geq 2$, since 1 0 cannot follow $\xi(b_1)\xi(b_2) = 01$. Let U be the point with the diagram that is $\xi(p)$ for $p > b_1$ and 1 for $p \leq b_1$. Its triples that are not triples of ξ begin with 11, since $\xi(b_1 + 1) = 1$, so it lies in Φ^-. It is an object with the kinks $b_3 < \cdots < b_k$ and the value 1 to the left of its first kink $b_3 \geq b_1 + 3$, so $U \in C_L$ by the induction hypothesis, and $X = \langle b_1; U \rangle \in C_L$ by Lemma 5.7.

For the last statement, let $w = w_1 \cdots w_n$ be a word of a sequence of Φ_{94}. The sequence that continues w by w_1 to the left and by w_n to the right has finitely many kinks, and its triples that are not triples of w have two equal neighbouring letters, so it lies in Φ_{94}. Hence every word of a sequence of Φ_{94} is a word of a point of $C_L \cap \Phi^-$, and of a point of $C_L \cap \Phi^+$. In particular this holds for every word of length L of an arbitrary point of Φ^- or of Φ^+, which therefore lies in C_L by (C0). □

Theorem 5.9. $\mathrm{Aut}(X_{F_{94}}) = \langle \sigma, F_{94} \rangle \cong \mathbb{Z}^2$, *and the same holds for rule* 122.

Proof. Let $h \in \mathrm{Aut}(X_{F_{94}})$. By Theorem 4.9 there are x, y such that $g = \sigma^x F^y h$ fixes every bare point. Let R be the radius of g and $L = 2R + 2$. By Lemma 5.2, g fixes every point of C_L, and $C_L = \Phi^+ \cup \Phi^-$ by Theorem 5.8; so g fixes every point of $\Phi^+ \cup \Phi^-$. By Theorem 3.1, $g = \mathrm{id}$, so $h = F^{-y}\sigma^{-x} \in \langle \sigma, F \rangle$. Finally $\langle \sigma, F \rangle \cong \mathbb{Z}^2$ by Lemma 2.4, since Φ_{94} contains sequences that are not periodic, and the statement for 122, the other rule of the class, follows by Lemma 2.3. □

The proof has two steps, as have the proofs of $\mathrm{Aut}(X_{F_f}) = \langle \sigma, F_f \rangle$ for the permutive rules and for rule 110 [12, 11]: the normalisation is Theorem 4.9, and a normalised automorphism is the identity by Lemma 5.2, Theorem 5.8 and Lemma L. Both steps rest on what happens when two kinks cross on the background 11 at equal parity (Lemma 4.4(ii)): in Theorem 4.9 the firing detects the descending kinks of the image of a bare kink (Lemma 4.7), and in Lemma 5.3 the delay moves one kink away from another.

Remark 5.10 (the closure for 95). For 95 the closure C_L is not all of $\Phi^+ \cup \Phi^-$ for $L \geq 5$: by Corollary 6.5 below, $\mathcal{S}$ translates the zeros of each class, and no point of C_L contains the word 01110 of Φ_{95} (see the paragraph after that corollary). So the argument of Theorem 5.9 does not apply to 95, and its conclusion fails: $\mathrm{Aut}(X_{F_{95}})$ is determined in Section 6 and is not $\langle \sigma, F \rangle$.

6. Rule 95R: two soliton systems, and the automorphism group

For $f = 95$ the group $\mathrm{Aut}(X_{F_{95}})$ is much larger than $\langle \sigma, F\rangle$, and we determine it. Each of the two sublattices of a diagram of 95R is a copy of the reversible automaton known as Rule 54, which is known to carry left- and right-moving particles whose only interaction is a delay by one step at each crossing [4]. We state this picture in the form that is needed below, for every point and with the fibres of the parametrisation, and prove it in that form (Theorem 6.3): the system $\mathcal{W}$ on one sublattice is the image of the product $\mathsf{G} \times \mathsf{G}$ of two golden mean shifts under a continuous map Ψ, at most two-to-one, with $\Psi(\sigma^a x, \sigma^b y) = \tau^{\theta_{x,y}(a,b)}\Psi(x,y)$ for a cocycle θ, so that the delays are removed by a change of coordinates that depends on the point (Remark 6.12(2)), and its automorphisms are the pairs of automorphisms of the golden mean shift that conserve the number of particles (Theorem 6.7). After the group is determined (Theorem 6.9), we find the smallest radius of an automorphism outside $\langle \sigma, F\rangle$ (Proposition 6.15), show that F_{95} has no blocking word and no eigenfield of any speed (Propositions 6.17 and 6.18), and identify the commuting block maps of radius at most 4 (Proposition 6.21). Throughout this section $f = 95$ and $F = F_{95}$.

Relation to the reversible automaton Rule 54. With $(t, x) = (j + k, k - j)$, equation (10) below is the update rule $s_x^{t+1} = s_x^{t-1} \oplus (s_{x-1}^t \vee s_{x+1}^t)$ on the checkerboard lattice, so the half system $\mathcal{W}$ is the spacetime subshift of Rule 54 [4, 7]. The literature on this automaton, and what is and is not contained in it, is described in Section 1.3: the particle picture of Lemma 6.1, Theorem 6.3 and Corollary 6.5 is not new, and proofs are given here because no source states or proves the parametrisation of all points of $\mathcal{W}$ by $\mathsf{G} \times \mathsf{G}$ with its fibres, which the proofs of Theorems 6.7 and 6.9 need.

The half system. Let $\mathcal{W}$ be the set of all $u\colon \mathbb{Z}^2 \to \{0,1\}$ with

$$u(j,k) \oplus u(j-1,k-1) = u(j,k-1) \vee u(j-1,k) \qquad \text{for all } (j,k) \in \mathbb{Z}^2, \tag{10}$$

which is (8) at every point of $\mathbb{Z}^2$. It is a closed subset of $\{0,1\}^{\mathbb{Z}^2}$, invariant under the translations $(\tau^n u)(P) = u(P + n)$, $n \in \mathbb{Z}^2$. Write $e_1 = (1,0)$ and $e_2 = (0,1)$. By (10), $u(j,k)$ is determined by $u(j-1,k-1)$, $u(j,k-1)$ and $u(j-1,k)$; so for every m an element of $\mathcal{W}$ is determined by its restriction to $\{j \le m\} \cup \{k \le m\}$. Let $\mathrm{Aut}(\mathcal{W})$ be the group of homeomorphisms of $\mathcal{W}$ that commute with all τ^n; by the Curtis–Hedlund–Lyndon theorem each of them is given by a local rule: there is a finite $D \subset \mathbb{Z}^2$ such that $(hu)(P)$ is a function of $(u(P + d))_{d \in D}$. For a diagram Y and $e \in \{0,1\}$ put

$$u_e(j,k) = 1 \oplus Y[2j + e, 2k + e].$$

That the two sublattices evolve independently is noted for the rules that do not read their middle argument, among them 95R, in [42, p. 386] and in [43, Section 4], and that each is a copy of RCA54 is the decomposition of 250R in [27, Section 3.1]; the next lemma states this for 95R with the formulas for σ and F that are used below, and its short proof is included.

Lemma 6.1 (splitting; cf. [42, p. 386], [43, Section 4], [27])**.** *The map $\Pi\colon c \mapsto (u_0, u_1)$, formed with $Y = Y_c$, is a homeomorphism of $\mathcal{A}^{\mathbb{Z}}$ onto $\mathcal{W} \times \mathcal{W}$, and*

$$\Pi\sigma\Pi^{-1}(u_0,u_1) = (\tau^{-e_1}u_1,\ \tau^{e_2}u_0), \qquad \Pi F\Pi^{-1}(u_0,u_1) = (u_1,\ \tau^{e_1+e_2}u_0).$$

Consequently $\Pi\,\sigma F\,\Pi^{-1} = \tau^{e_2} \times \tau^{e_2}$, $\Pi\,\sigma^{-1}F\,\Pi^{-1} = \tau^{e_1} \times \tau^{e_1}$ and $\Pi\sigma^2\Pi^{-1} = \tau^{e_2-e_1} \times \tau^{e_2-e_1}$; and $\Pi(\Phi^+) = \mathcal{W}^+ \times \mathcal{W}^+$, $\Pi(\Phi^-) = \mathcal{W}^- \times \mathcal{W}^-$, where $\mathcal{W}^+ = \{u \in \mathcal{W} : \tau^{e_2}u = u\}$ and $\mathcal{W}^- = \{u \in \mathcal{W} : \tau^{e_1}u = u\}$.

Proof. The map $c \mapsto Y_c$ is a bijection from $\mathcal{A}^{\mathbb{Z}}$ onto the set of functions on the lattice that satisfy (3) at every centre, and it is a homeomorphism for the product topologies, since $Y_c(t,i)$ depends on finitely many cells of c and $c_i = (Y_c(-1,i), Y_c(0,i))$. For $N = N_{95}$ the relation at the centre $[q,p]$ reads $Y(\mathsf{N}) \oplus Y(\mathsf{S}) \oplus Y(\mathsf{W}) \oplus Y(\mathsf{E}) = (1 \oplus Y(\mathsf{W}))(1 \oplus Y(\mathsf{E}))$, and with $u = 1 \oplus Y$ this

is $u(\mathsf{N}) \oplus u(\mathsf{S}) = u(\mathsf{W}) \vee u(\mathsf{E})$. For the centre $[2j+e-1, 2k+e-1]$ the points N, S, W, E are $[2j+e, 2k+e]$, $[2j+e-2, 2k+e-2]$, $[2j+e, 2k+e-2]$ and $[2j+e-2, 2k+e]$, so the relation is (10) for u_e at (j,k); every centre is of this form for exactly one e and (j,k). Hence Y is a diagram if and only if $u_0, u_1 \in \mathcal{W}$, and Π is a homeomorphism onto $\mathcal{W} \times \mathcal{W}$. The two formulas follow from $Y_{\sigma c}[q,p] = Y_c[q-1, p+1]$ and $Y_{F(c)}[q,p] = Y_c[q+1, p+1]$: for instance $1 \oplus Y_{\sigma c}[2j, 2k] = 1 \oplus Y_c[2(j-1)+1, 2k+1] = u_1(j-1, k)$ and $1 \oplus Y_{\sigma c}[2j+1, 2k+1] = u_0(j, k+1)$. The consequences are obtained by composing them, and the last statement holds because $\Phi^{\pm}$ is the set of fixed points of $\sigma^{\pm 1} F$. □

Let $\mathsf{G} \subset \{0,1\}^{\mathbb{Z}}$ be the golden mean shift, the set of sequences with no two consecutive 1s, with the shift $(\sigma x)_i = x_{i+1}$; a point of G is *finite* if it has finitely many 1s, and $|x|$ is their number. If $u(j,k) = x_j$ for all (j,k), then (10) says $x_j \oplus x_{j-1} = x_j \vee x_{j-1}$, that is $x_j x_{j-1} = 0$; so $\mathcal{W}^+$ is the set of these u with $x \in \mathsf{G}$, and likewise $\mathcal{W}^-$ is the set of the $u(j,k) = y_k$ with $y \in \mathsf{G}$. For $x \in \mathsf{G}$ let $A_x \colon \mathbb{Z} \to \mathbb{Z}$ be the counting function with

$$A_x(0) = 0, \qquad A_x(J+1) - A_x(J) = x_J,$$

so that $A_x(a+J) = A_x(a) + A_{\sigma^a x}(J)$. For $x, y \in \mathsf{G}$ define $\theta = \theta_{x,y} \colon \mathbb{Z}^2 \to \mathbb{Z}^2$ by

$$\theta_{x,y}(J,K) = \big(J + A_y(K),\ K + A_x(J)\big).$$

Thus $\theta(J+1, K) = \theta(J,K) + (1, x_J)$ and $\theta(J, K+1) = \theta(J,K) + (y_K, 1)$.

Lemma 6.2. *Let $x, y \in \mathsf{G}$ and $\theta = \theta_{x,y}$.*

(i) *$\theta_{x,y}(a+J, b+K) = \theta_{x,y}(a,b) + \theta_{\sigma^a x, \sigma^b y}(J,K)$ for all a, b, J, K.*
(ii) *θ is surjective. Two distinct points have the same image if and only if they are (J,K) and $(J+1, K-1)$ with $x_J = y_{K-1} = 1$; in particular every fibre has at most two points.*
(iii) *If $\theta(J,K) = (j,k)$, then $|J|, |K| \le 2(|j| + |k|) + 1$.*

Proof. (i) is the additivity of A_x and A_y. (iii) Since x has no two consecutive 1s, $|A_x(J)| \le (|J|+1)/2$, and likewise for y; so $|J| \le |j| + (|K|+1)/2$ and $|K| \le |k| + (|J|+1)/2$, which gives $|J| \le \frac{4}{3}|j| + \frac{2}{3}|k| + 1$ and the same for K with j and k exchanged. (ii) Fix (j,k). A point (J,K) has an image with second coordinate k exactly if $K = K(J) = k - A_x(J)$, and then the first coordinate is $g(J) = J + A_y(K(J))$. Since $K(J+1) = K(J) - x_J$ and $A_y(K-1) = A_y(K) - y_{K-1}$,

$$g(J+1) - g(J) = 1 - x_J\, y_{K(J)-1} \in \{0, 1\}.$$

If this difference is 0, then $x_J = 1$, so $x_{J+1} = 0$ and the next difference is 1. Hence g is nondecreasing, unbounded in both directions and takes every integer value, at one J or at two consecutive ones; the latter happens at J, $J+1$ exactly if $x_J = y_{K(J)-1} = 1$, and then $K(J+1) = K(J) - 1$. □

By (ii), if a point P has the two preimages (J,K) and $(J+1, K-1)$, then $x_J = 1$ and $y_{K-1} = 1$. So there is a unique function $u_{x,y} \colon \mathbb{Z}^2 \to \{0,1\}$ with

$$u_{x,y}\big(\theta_{x,y}(J,K)\big) = x_J \vee y_K \qquad \text{for all } (J,K) \in \mathbb{Z}^2. \tag{11}$$

For $y = 0$ we have $\theta(J,K) = (J, K + A_x(J))$ and $u_{x,0}(j,k) = x_j$; likewise $u_{0,y}(j,k) = y_k$. In general $u_{x,y}$ is the superposition $x_J \vee y_K$ of a family of right-moving particles, the lines $J = \text{const}$ with $x_J = 1$, and a family of left-moving ones, the lines $K = \text{const}$ with $y_K = 1$, drawn in coordinates in which every particle is delayed by one step each time it crosses a particle of the other family.

Theorem 6.3 (structure of the half system; cf. [4, 19])**.**

(a) *$u_{x,y} \in \mathcal{W}$ for all $x, y \in \mathsf{G}$.*
(b) *$\tau^n u_{x,y} = u_{\sigma^a x, \sigma^b y}$ whenever $n = \theta_{x,y}(a,b)$.*

(c) *The map* $\Psi\colon \mathsf{G}\times\mathsf{G}\to\mathcal{W}$, $(x,y)\mapsto u_{x,y}$, *is continuous and surjective.*
(d) $u_{x,y}=u_{x',y'}$ *if and only if* $(x',y')=(\sigma^a x,\sigma^b y)$ *for some* (a,b) *with* $\theta_{x,y}(a,b)=(0,0)$*; that is, if and only if* $(x',y')=(x,y)$*, or* $x_0=y_{-1}=1$ *and* $(x',y')=(\sigma x,\sigma^{-1}y)$*, or* $x_{-1}=y_0=1$ *and* $(x',y')=(\sigma^{-1}x,\sigma y)$.

The proof of (c) uses the following consequence of Proposition 3.7.

Lemma 6.4 (incoming pairs)**.** *Let* c *be a finite perturbation of* b_{11}*. Then there are* $t\ge 0$ *and a placed pair* (P,X) *with corner* $[0,0]$ *and background* 11*, such that the diagrams of* P *and* X *have finitely many zeros and* $Y_{F^{-t}(c)}=Y_{P,X}$.

Proof. The configuration μc is a finite perturbation of $\mu b_{11}=b_{11}$. By Proposition 3.7, applied to μc with $\psi=\varphi=b_{11}$, there are C and T such that for $t\ge T$ the configuration $F^t(\mu c)$ agrees with a point $d^-\in\Phi^-$ on $(-\infty,t/2-C]$ and with a point $d^+\in\Phi^+$ on $[-t/2+C,\infty)$. Fix such a t with $m=\lfloor t/2-C\rfloor\ge 4$, and let $\xi'(p)$ and $\pi'(q)$ be the diagrams of d^- and d^+, so that $d^-_i=(\xi'(i-1),\xi'(i))$ and $d^+_i=(\pi'(-1-i),\pi'(-i))$. For $|i|\le m$ both descriptions apply, so $\xi'(i)=\pi'(-i)$ and $\xi'(i-1)=\pi'(-1-i)$; hence $\xi'(i-1)=\xi'(i+1)$ for $-m\le i\le m-1$, and ξ' is 2-periodic on $[-m-1,m]$. A triple (a,b,a) of a sequence of Φ_{95} has $a=1$, so $\xi'=1$ on $[-m-1,m]$, and then $\pi'=1$ on $[-m-1,m]$. Replacing ξ' by 1 on $[0,\infty)$ and π' by 1 on $[0,\infty)$ gives sequences of Φ_{95}, because every triple of the new sequences is a triple of the old one or consists of 1s; let $D^-\in\Phi^-$ and $D^+\in\Phi^+$ be the corresponding points. Then $F^t(\mu c)$ equals D^- on the cells $i\le 0$ and D^+ on the cells $i\ge 0$, and $D^-_i=D^+_i=(1,1)$ for $|i|\le 2$. Since $\mu F^t\mu=F^{-t}$, the configuration $F^{-t}(c)=\mu F^t(\mu c)$ equals $P=\mu D^+\in\Phi^+$ on the cells $i\le 0$ and $X=\mu D^-\in\Phi^-$ on the cells $i\ge 0$, and $P_i=X_i=(1,1)$ for $|i|\le 2$. The diagrams of P and X are $\pi(q)=\pi''(-1-q)$ and $\xi(p)=\xi''(-1-p)$, where π'' and ξ'' are the diagrams of D^+ and D^-, which are 1 on $[-m-1,\infty)$; so $\pi(q)=1$ for $q\le m$ and $\xi(p)=1$ for $p\le m$. Since $F^{-t}(c)$ is a finite perturbation of b_{11}, they are also 1 for all large q, resp. p. So (P,X) is a placed pair with corner $[0,0]$ and background 11, and $F^{-t}(c)$ is the configuration that is P on the left and X on the right; by the paragraph on placed pairs, $Y_{F^{-t}(c)}=Y_{P,X}$. □

Proof of Theorem 6.3. Let $\theta=\theta_{x,y}$ and $u=u_{x,y}$.

(a) Fix (j,k), write $\mathsf{S}=(j-1,k-1)$, $\mathsf{W}=(j,k-1)=\mathsf{S}+e_1$, $\mathsf{E}=(j-1,k)=\mathsf{S}+e_2$ and $\mathsf{N}=(j,k)$, and choose a preimage (J,K) of S, by Lemma 6.2(ii). Then $\theta(J+1,K)=\mathsf{S}+(1,x_J)$ and $\theta(J,K+1)=\mathsf{S}+(y_K,1)$. If $x_J=y_K=0$, then $\mathsf{W}=\theta(J+1,K)$, $\mathsf{E}=\theta(J,K+1)$ and $\mathsf{N}=\theta(J+1,K+1)$, so by (11) the four values at $\mathsf{S},\mathsf{W},\mathsf{E},\mathsf{N}$ are 0, x_{J+1}, y_{K+1}, $x_{J+1}\vee y_{K+1}$, and (10) holds. If $x_J=1$ and $y_K=0$, then $x_{J+1}=0$, $\mathsf{N}=\theta(J+1,K)$ and $\mathsf{E}=\theta(J,K+1)$, so $u(\mathsf{S})=1$, $u(\mathsf{N})=x_{J+1}\vee y_K=0$ and $u(\mathsf{E})=1$, and both sides of (10) are 1. The case $x_J=0$, $y_K=1$ is symmetric: $\mathsf{N}=\theta(J,K+1)$, $\mathsf{W}=\theta(J+1,K)$, $u(\mathsf{S})=u(\mathsf{W})=1$ and $u(\mathsf{N})=0$. Finally let $x_J=y_K=1$, so $x_{J\pm1}=y_{K\pm1}=0$. Then $\mathsf{N}=\theta(J+1,K)$ and $u(\mathsf{N})=x_{J+1}\vee y_K=1=u(\mathsf{S})$; moreover $\theta(J+1,K-1)=\theta(J+1,K)-(y_{K-1},1)=\mathsf{S}+e_1=\mathsf{W}$ and $\theta(J-1,K+1)=\theta(J,K+1)-(1,x_{J-1})=\mathsf{S}+e_2=\mathsf{E}$, so $u(\mathsf{W})=x_{J+1}\vee y_{K-1}=0$ and $u(\mathsf{E})=x_{J-1}\vee y_{K+1}=0$, and both sides of (10) are 0.

(b) By Lemma 6.2(i) and (11), for all (J,K),

$$u_{\sigma^a x,\sigma^b y}\big(\theta_{\sigma^a x,\sigma^b y}(J,K)\big)=x_{a+J}\vee y_{b+K}=u\big(\theta(a+J,b+K)\big)=(\tau^n u)\big(\theta_{\sigma^a x,\sigma^b y}(J,K)\big),$$

and $\theta_{\sigma^a x,\sigma^b y}$ is surjective.

(c) By Lemma 6.2(iii), the restriction of $u_{x,y}$ to a finite set depends on finitely many coordinates of x and y; so Ψ is continuous, and its image $\mathcal{U}$ is compact. By (b) and the surjectivity of $\theta_{x,y}$, $\mathcal{U}$ is invariant under all τ^n. Let c be a finite perturbation of b_{11}, and take t and (P,X) from Lemma 6.4, with diagrams $\pi(q)$ and $\xi(p)$. Let $(u_0,u_1)=\Pi(F^{-t}(c))$, and for $e\in\{0,1\}$ put $x^e_j=1\oplus\pi(2j+e)$ and $y^e_k=1\oplus\xi(2k+e)$. These are finite points of G, because Φ_{95} is the set of

sequences with no two zeros at distance 2, and they vanish on $(-\infty,0]$. Since $Y_{P,X}=\pi(q)$ on $\{p\le 1\}$ and $Y_{P,X}=\xi(p)$ on $\{q\le 1\}$, we have $u_e(j,k)=x_j^e$ for $k\le 0$ and $u_e(j,k)=y_k^e$ for $j\le 0$. On the other hand, if $x,y\in\mathsf{G}$ vanish on $(-\infty,0]$, then $A_x(J)\ge 0$ for all J and $A_y(K)=0$ for $K\le 1$, so for $K\le 0$ we get $\theta_{x,y}(J,K)=(J,K+A_x(J))$ and $u_{x,y}(J,K+A_x(J))=x_J$; thus $u_{x,y}(j,k)=x_j$ whenever $k\le 0$, and symmetrically $u_{x,y}(j,k)=y_k$ whenever $j\le 0$. An element of $\mathcal{W}$ is determined by its restriction to $\{j\le 0\}\cup\{k\le 0\}$, so $u_e=u_{x^e,y^e}\in\mathcal{U}$. By Lemma 6.1, the two components of $\Pi(c)=\Pi(F^t(F^{-t}(c)))$ are translates of u_0 and u_1, so they lie in $\mathcal{U}$. The finite perturbations of b_{11} are dense in $\mathcal{A}^{\mathbb{Z}}$ and Π is a homeomorphism onto $\mathcal{W}\times\mathcal{W}$, so $\mathcal{U}$ is dense in $\mathcal{W}$; being compact, it is $\mathcal{W}$.

(d) We first describe u near a point $P=\theta(J,K)$ in terms of the *label* (x_J,y_K) of the preimage. If the label is $(0,0)$, then $u(P)=0$. If it is $(1,1)$, then $u(P)=1$ and $u(P+e_1)=u(P+e_2)=0$, by the last case of (a) applied to $\mathsf{S}=P$. Let the label be $(1,0)$. Then $P+e_2=\theta(J,K+1)$ and $u(P+e_2)=1$. If $y_{K-1}=0$, then P has no second preimage, and $P+e_1=\theta(J+1,K)-(y_{K-1},1)=\theta(J+1,K-1)$ carries $x_{J+1}\vee y_{K-1}=0$. If $y_{K-1}=1$, then $P=\theta(J+1,K-1)$ as well, with the label $(0,1)$, and $P+e_1=\theta(J+2,K-1)$ carries 1. The label $(0,1)$ is symmetric. Hence the triple $(u(P),u(P+e_1),u(P+e_2))$ determines the labels of the preimages of P:

(12)

$(u(P),u(P+e_1),u(P+e_2))$	$(0,\cdot,\cdot)$	$(1,0,0)$	$(1,0,1)$	$(1,1,0)$	$(1,1,1)$
labels	$(0,0)$	$(1,1)$	$(1,0)$	$(0,1)$	$(1,0)$ and $(0,1)$

where in the last column P has two preimages, (J,K) with the label $(1,0)$ and $(J+1,K-1)$ with the label $(0,1)$, and in the other columns it has one.

Next we show that for each of the four neighbours (J',K') of (J,K) in $\mathbb{Z}^2$, the point $\theta(J',K')$ and the label $(x_{J'},y_{K'})$ are determined by u, the point $P=\theta(J,K)$ and the label (x_J,y_K), by a rule that does not involve x and y otherwise. For $(J+1,K)$: the point is $Q=P+(1,x_J)$, and its label is (x_{J+1},y_K), where x_{J+1} is read off from (12) at Q, in the last column as the first entry of the label whose second entry is y_K. For $(J-1,K)$ we need x_{J-1}; then the point is $P-(1,x_{J-1})$ and the label is (x_{J-1},y_K). If $x_J=1$, then $x_{J-1}=0$. If $x_J=0$ and $y_K=0$, then $x_{J-1}=u(P-e_1)$: if $x_{J-1}=0$, then $P-e_1=\theta(J-1,K)$ carries 0, and if $x_{J-1}=1$, then $P-e_1=\theta(J-1,K)+(y_K,1)=\theta(J-1,K+1)$ carries 1. If $x_J=0$ and $y_K=1$, then by Lemma 6.2(ii) $x_{J-1}=1$ if and only if P has the second preimage $(J-1,K+1)$, that is, if and only if the triple at P is $(1,1,1)$. The neighbours $(J,K\pm 1)$ are symmetric.

Now let $u_{x,y}=u_{x',y'}=u$ and $\theta'=\theta_{x',y'}$. The point $(0,0)$ is a preimage of $(0,0)$ under θ'; by (12) there is a preimage (a,b) of $(0,0)$ under θ with the same label. By the previous paragraph and induction on $|J|+|K|$, $\theta'(J,K)=\theta(a+J,b+K)$ and $(x'_J,y'_K)=(x_{a+J},y_{b+K})$ for all (J,K). So $(x',y')=(\sigma^a x,\sigma^b y)$ with $\theta(a,b)=(0,0)$. The converse is (b) with $n=(0,0)$, and the list of cases is the description of the fibre of θ over $(0,0)=\theta(0,0)$ in Lemma 6.2(ii). □

Corollary 6.5 (translation property; cf. [4])**.** *Let (P,X) be a placed pair for $f=95$ whose diagrams $\pi(q)$ and $\xi(p)$ have finitely many zeros, and for $e\in\{0,1\}$ let $n_e(P)$ and $n_e(X)$ be the numbers of zeros of π, resp. of ξ, in the class e modulo 2. Then $\mathcal{S}(P,X)=(\Gamma,\Omega)$, where the diagrams γ of Γ and o of Ω are*

$$\gamma(p)=\xi\big(p-2n_e(P)\big)\quad(p\equiv e),\qquad o(q)=\pi\big(q-2n_e(X)\big)\quad(q\equiv e).$$

So $\mathcal{S}$ translates the zeros of X in each class by twice the number of zeros of P in that class, and conversely.

Proof. The translation $[q,p]\mapsto[q+2a,p+2b]$ of the lattice maps diagrams to diagrams and placed pairs to placed pairs, and the corner may be replaced by a smaller one; so we may assume that the corner is $[0,0]$. Define u_e, x^e and y^e from $Y_{P,X}$, π and ξ as in the proof of Theorem 6.3(c); then $u_e=u_{x,y}$ with $x=x^e$, $y=y^e$, which vanish on $(-\infty,0]$, and $|x|=n_e(P)$,

$|y| = n_e(X)$. For K larger than every position of a 1 of y we have $A_y(K) = |y|$ and $y_K = 0$, so $u_{x,y}(J + |y|, K + A_x(J)) = x_J$; hence $u_e(j,k) = x_{j-|y|}$ for all large k, and symmetrically $u_e(j,k) = y_{k-|x|}$ for all large j. Reading $Y_{P,X}$ on the half-planes of Lemma 4.1(ii) gives $1 \oplus o(2j+e) = x^e_{j-|y|} = 1 \oplus \pi(2j + e - 2n_e(X))$ and $1 \oplus \gamma(2k+e) = 1 \oplus \xi(2k + e - 2n_e(P))$. □

Lemma 4.11 is the case of one zero on each side. The corollary applies to every placed pair for 95, because 11 is the only valid background and so the diagrams of a placed pair have finitely many zeros. It shows that $\mathcal{S}$ and $\mathcal{S}^{-1}$ do not change the distances between the zeros of a point in one class; neither do μ, rev and $\sigma^{\pm 1}$. Two zeros at distance 4 lie in the same class and in a word of length 5. Hence the set of points of $\Phi^+ \cup \Phi^-$ that have no two zeros at distance 4 contains the bare points and has the properties (C1)–(C3) of Definition 5.1, and the property (C0) for every $L \geq 5$. So for $L \geq 5$ no point of the scattering closure C_L contains the word 01110 of Φ_{95}, and $C_L \neq \Phi^+ \cup \Phi^-$: for 95 the argument of Theorem 5.9 does not apply, and in fact its conclusion is false (Corollary 6.10).

Particle-conserving automorphisms of the golden mean shift. Let $\mathcal{N}$ be the set of automorphisms φ of (G, σ) with $|\varphi x| = |x|$ for every finite $x \in \mathsf{G}$. An automorphism of G fixes 0^∞, the only fixed point of σ, and maps finite points to finite points; so $\mathcal{N}$ is a subgroup of $\mathrm{Aut}(\mathsf{G}, \sigma)$, and $\sigma \in \mathcal{N}$.

That a block map conserves an additive quantity if and only if the change of the density is the difference of a flux and its translate is known: for cellular automata on full shifts it is in [24, Theorems 2.1, 2.2, 2.6 and 2.7] and in [5, 38, 36]; the framework on shifts of finite type is in [29, Section 4.1], where the implication from conserved to locally conserved is stated for full shifts. The next lemma is this criterion for automorphisms of the golden mean shift, together with the cocycle identities for the flux that are used below; it is not claimed as new, and its proof is included because the statement is needed in this form on the subshift G.

Lemma 6.6 (flux)**.** *For every $\varphi \in \mathcal{N}$ there is a unique continuous $\lambda_\varphi \colon \mathsf{G} \to \mathbb{Z}$ with $\lambda_\varphi(0^\infty) = 0$ and*

$$(\varphi x)_0 - x_0 = \lambda_\varphi(x) - \lambda_\varphi(\sigma x) \qquad (x \in \mathsf{G}).$$

It satisfies $A_{\varphi x}(J) - A_x(J) = \lambda_\varphi(x) - \lambda_\varphi(\sigma^J x)$ for all $J \in \mathbb{Z}$, $\lambda_{\varphi\varphi'} = \lambda_\varphi \circ \varphi' + \lambda_{\varphi'}$ for $\varphi, \varphi' \in \mathcal{N}$, and $\lambda_\sigma(x) = -x_0$. Conversely, if for an automorphism φ of (G, σ) there is a continuous $\lambda \colon \mathsf{G} \to \mathbb{Z}$ with $(\varphi x)_0 - x_0 = \lambda(x) - \lambda(\sigma x)$ for all x, then $\varphi \in \mathcal{N}$.

Proof. Let r be a radius of φ. For $x \in \mathsf{G}$ and $a < b$ in $\mathbb{Z} \cup \{\infty\}$ let $x^{[a,b)}$ be the sequence that equals x on $[a,b)$ and 0 elsewhere; it lies in G. Put $x^+ = x^{[0,\infty)}$ and

$$\varrho(x) = \sum_{-r \leq i < r} (\varphi x^+)_i - \sum_{0 \leq i < r} (\varphi x)_i.$$

This is a continuous function of x, and $\varrho(0^\infty) = 0$. Note that $(\varphi x^+)_i = 0$ for $i < -r$ and $(\varphi x^+)_i = (\varphi x)_i$ for $i \geq r$. Let $M > 2r + 2$. The points $x^{[0,M)}$ and $x^{[1,M)}$ are finite, and φ preserves the number of 1s, so

$$x_0 = \sum_i \Big((\varphi x^{[0,M)})_i - (\varphi x^{[1,M)})_i \Big).$$

The terms with $i \geq r + 1$ vanish, since the two points agree on $[1, \infty)$. For $i \leq r$ we have $(\varphi x^{[0,M)})_i = (\varphi x^+)_i$ and $(\varphi x^{[1,M)})_i = (\varphi(\sigma x)^+)_{i-1}$, because $x^{[1,\infty)} = \sigma^{-1}((\sigma x)^+)$. Hence

$$x_0 = \sum_{i<r} (\varphi x^+)_i + (\varphi x)_r - \sum_{i<r} (\varphi(\sigma x)^+)_i,$$

and therefore $\varrho(x) - \varrho(\sigma x) = x_0 - (\varphi x)_r - (\varphi x)_0 + (\varphi \sigma x)_{r-1} = x_0 - (\varphi x)_0$. So $\lambda_\varphi = -\varrho$ has the two properties. If λ' is a second such function, then $\lambda_\varphi - \lambda'$ is continuous and σ-invariant, hence

constant on the closure of every orbit; G has a dense orbit, so it is constant, and it vanishes at 0^∞. The formula for $A_{\varphi x} - A_x$ follows by summing the defining relation along the orbit of x. The function $\lambda_\varphi \circ \varphi' + \lambda_{\varphi'}$ is continuous, vanishes at 0^∞ and satisfies the defining relation for $\varphi\varphi'$, because φ' commutes with σ; and $(\sigma x)_0 - x_0 = x_1 - x_0$. For the converse, let x be finite; summing the relation along the orbit of x gives $|\varphi x| - |x| = \lim_{n\to\infty}\big(\lambda(\sigma^{-n}x) - \lambda(\sigma^n x)\big) = 0$, because $\sigma^{\pm n}x \to 0^\infty$ and λ is continuous. □

Theorem 6.7 (automorphisms of the half system)**.**

(a) *For $\varphi, \psi \in \mathcal{N}$ there is a unique map $H_{\varphi,\psi}\colon \mathcal{W} \to \mathcal{W}$ with*

$$H_{\varphi,\psi}(u_{x,y}) = \tau^{(\lambda_\psi(y),\, \lambda_\varphi(x))}\, u_{\varphi x, \psi y} \qquad (x, y \in \mathsf{G}),$$

and $H_{\varphi,\psi} \in \mathrm{Aut}(\mathcal{W})$. It maps $u_{x,0}$ to $u_{\varphi x,0}$ and $u_{0,y}$ to $u_{0,\psi y}$.

(b) *The map $(\varphi, \psi) \mapsto H_{\varphi,\psi}$ is an isomorphism of groups $\mathcal{N} \times \mathcal{N} \to \mathrm{Aut}(\mathcal{W})$, and $H_{\sigma,\mathrm{id}} = \tau^{e_1}$, $H_{\mathrm{id},\sigma} = \tau^{e_2}$.*

(c) *A continuous map $\mathcal{W} \to \mathcal{W}$ that commutes with all τ^n and fixes every point of $\mathcal{W}^+ \cup \mathcal{W}^-$ is the identity.*

Proof. (c) Let g be such a map. Then $\Pi^{-1}(g \times g)\Pi$ is a continuous map of $\mathcal{A}^{\mathbb{Z}}$ that commutes with σ and F, by the formulas of Lemma 6.1, so it is a block map commuting with F; and it fixes every point of $\Phi^+ \cup \Phi^-$. By Corollary 3.9 it is the identity, so g is the identity.

(a) Write $\ell(x, y) = (\lambda_\psi(y), \lambda_\varphi(x))$. Let $x, y \in \mathsf{G}$, $(a, b) \in \mathbb{Z}^2$ and $n = \theta_{x,y}(a, b)$. By Lemma 6.6,

$$\theta_{\varphi x,\psi y}(a, b) = \big(a + A_{\psi y}(b),\, b + A_{\varphi x}(a)\big) = n + \ell(x, y) - \ell(\sigma^a x, \sigma^b y),$$

so by Theorem 6.3(b), applied to $(\varphi x, \psi y)$,

$$\tau^{\ell(\sigma^a x,\sigma^b y)} u_{\varphi\sigma^a x,\psi\sigma^b y} = \tau^{\ell(\sigma^a x,\sigma^b y)} \tau^{\theta_{\varphi x,\psi y}(a,b)} u_{\varphi x,\psi y} = \tau^n\, \tau^{\ell(x,y)} u_{\varphi x,\psi y}. \tag{13}$$

For $n = (0, 0)$ this shows, with Theorem 6.3(d), that $\tau^{\ell(x,y)} u_{\varphi x,\psi y}$ depends only on $u_{x,y}$; with Theorem 6.3(c), $H = H_{\varphi,\psi}$ is well defined on $\mathcal{W}$, and unique. For general (a, b), (13) and Theorem 6.3(b) say $H(\tau^n u_{x,y}) = \tau^n H(u_{x,y})$, and every n occurs, since $\theta_{x,y}$ is surjective. The map $H \circ \Psi$ is continuous, and Ψ is a continuous surjection of compact metric spaces, hence a quotient map; so H is continuous. By Lemma 6.6, ℓ is additive under composition, and since the maps H commute with translations, $H_{\varphi,\psi} H_{\varphi',\psi'} = H_{\varphi\varphi',\psi\psi'}$; as $H_{\mathrm{id},\mathrm{id}} = \mathrm{id}$, every $H_{\varphi,\psi}$ is a bijection, hence a homeomorphism. Since $u_{\varphi x,0}(j, k) = (\varphi x)_j$ does not depend on k, $H(u_{x,0}) = u_{\varphi x,0}$, and likewise $H(u_{0,y}) = u_{0,\psi y}$.

(b) The map is a homomorphism by the proof of (a), and it is injective by the last statement of (a). By Theorem 6.3(b), $u_{\sigma x,y} = \tau^{\theta_{x,y}(1,0)} u_{x,y} = \tau^{(1,x_0)} u_{x,y}$, so $H_{\sigma,\mathrm{id}}(u_{x,y}) = \tau^{(0,-x_0)} u_{\sigma x,y} = \tau^{e_1} u_{x,y}$, and likewise for $H_{\mathrm{id},\sigma}$.

It remains to show surjectivity. Let $h \in \mathrm{Aut}(\mathcal{W})$, with a local rule on $D \subseteq [-r, r]^2$. The zero function is the only element of $\mathcal{W}$ fixed by all translations, so h fixes it. Since h commutes with τ^{e_2} and τ^{e_1}, it maps $\mathcal{W}^+$ and $\mathcal{W}^-$ onto themselves, and there are bijections φ, ψ of G with $h(u_{x,0}) = u_{\varphi x,0}$ and $h(u_{0,y}) = u_{0,\psi y}$; they are continuous, and they commute with σ because $\tau^{e_1} u_{x,0} = u_{\sigma x,0}$ and $\tau^{e_2} u_{0,y} = u_{0,\sigma y}$. So $\varphi, \psi \in \mathrm{Aut}(\mathsf{G}, \sigma)$. We show that $\psi \in \mathcal{N}$. Let $y \in \mathsf{G}$ be finite, and let $x \in \mathsf{G}$ be finite and not 0^∞; since ψ commutes with σ, we may assume that x and y vanish on $(-\infty, 0]$. Let $u = u_{x,y}$ and $v = hu$. As in the proofs of Theorem 6.3(c) and Corollary 6.5, $u(j, k) = x_j$ for $k \le 0$, $u(j, k) = y_k$ for $j \le 0$, and $u(j, k) = x_{j-|y|}$ for $k \ge k_1$, for some k_1. On these three sets u agrees with $u_{x,0}$, with $u_{0,y}$ and with $\tau^{-|y|e_1} u_{x,0}$, so with $x' = \varphi x$ and $y' = \psi y$,

$$v(j, k) = x'_j \ \ (k \le -r), \qquad v(j, k) = y'_k \ \ (j \le -r), \qquad v(j, k) = x'_{j-|y|} \ \ (k \ge k_1 + r).$$

The points x' and y' are finite and vanish on $(-\infty,-r]$, and $x'\neq 0^\infty$. Let $x''=\sigma^{-r-1}x'$ and $y''=\sigma^{-r-1}y'$; they vanish on $(-\infty,0]$, and $v''=\tau^{-(r+1)(e_1+e_2)}v$ satisfies $v''(j,k)=x''_j$ for $k\le 0$ and $v''(j,k)=y''_k$ for $j\le 0$. Hence $v''=u_{x'',y''}$, and so $v''(j,k)=x''_{j-|y''|}$ for all large k; that is, $v(j,k)=x'_{j-|y'|}$ for all large k. Comparing with the third formula, $x'_{j-|y'|}=x'_{j-|y|}$ for all j; since x' is finite and not zero, $|y'|=|y|$. So $\psi\in\mathcal{N}$, and by the symmetry of (10) in j and k, $\varphi\in\mathcal{N}$. The map $H^{-1}_{\varphi,\psi}h$ fixes every point of $\mathcal{W}^+\cup\mathcal{W}^-$, by (a), so it is the identity, by (c). □

Many particle-conserving automorphisms. Let $x\in\mathsf{G}$. A 1 of x at z is *isolated* if $x_{z\pm1}=x_{z\pm2}=0$. A *box* of x of size $m\ge1$ is an interval $I=[a+1,a+m+4]$ such that x has the word 10101 at $[a-4,a]$ and at $[a+m+5,a+m+9]$, and I contains exactly one 1 of x, at a position $a+2+i$ with $1\le i\le m$. For a finitely supported family $\varpi=(\varpi_m)_{m\ge1}\in\bigoplus_m S_m$ let $\varphi_\varpi(x)$ be obtained from x by moving, in every box, the 1 at $a+2+i$ to $a+2+\varpi_m(i)$, where m is the size of the box.

Lemma 6.8. *The map $\varpi\mapsto\varphi_\varpi$ is an injective homomorphism $\bigoplus_{m\ge1}S_m\to\mathcal{N}$.*

Proof. The 1 in a box is isolated, and the 1s of the two words 10101 next to the box are not. Two boxes $[a+1,a+m+4]$ and $[a+1,a+m'+4]$ with $m<m'$ cannot both occur, since the second would contain the non-isolated 1 at $a+m+5$. Let I and I' be boxes of x, with $a<a'$. The three 1s at $a'-4,a'-2,a'$ are not isolated, so they do not lie in I; since $a'>a$ and $a'\notin I$, $a'\ge a+m+5$; then $a'-4>a$ and $a'-4\notin I$ give $a'-4\ge a+m+5$; so I' lies to the right of I, and distinct boxes are disjoint. Let $x'=\varphi_\varpi(x)$. A 1 that is moved lies, before and after the move, in its box at distance at least 2 from both ends of the box, so it is isolated in x' and at distance at least 3 from every 1 of x outside the box; since boxes are disjoint, $x'\in\mathsf{G}$, the 1s of x outside the boxes are 1s of x', and each of them is isolated in x' if and only if it is isolated in x. A word 10101 consists of non-isolated 1s and the zeros between them, so x and x' have the word 10101 at the same places. Every box of x is therefore a box of x'. Conversely, let $I'=[a'+1,a'+m'+4]$ be a box of x'. Its 1 is isolated in x', and the 1s at a' and $a'+m'+5$ are not. If I' meets a box I of x, then $I'=I$, since I is a box of x' and distinct boxes of a point are disjoint. If I' meets no box of x, then $x=x'$ on I', and I' is a box of x as well. So x and x' have the same boxes, of the same sizes, and $\varphi_{\varpi'}(\varphi_\varpi(x))=\varphi_{\varpi'\varpi}(x)$. Each φ_ϖ commutes with σ, is given by a local rule (only the boxes of the finitely many sizes m with $\varpi_m\neq\mathrm{id}$ matter), and is a bijection with inverse $\varphi_{\varpi^{-1}}$; it preserves the number of 1s. If $\varpi_m(i)\neq i$, then φ_ϖ moves the finite point with the word $10101\,00\,0^{i-1}10^{m-i}\,00\,10101$. □

The automorphism group of $95R$. For $g\in\mathrm{Aut}(\mathcal{W})$ put $\hat g=\Pi^{-1}(g\times g)\Pi$.

Theorem 6.9. *The map $g\mapsto\hat g$ is an injective homomorphism $\mathrm{Aut}(\mathcal{W})\to\mathrm{Aut}(X_{F_{95}})$ whose image $\hat{\mathcal{H}}$ has index 2, and*

$$\mathrm{Aut}(X_{F_{95}})=\hat{\mathcal{H}}\cup\hat{\mathcal{H}}\,\sigma.$$

The shift σ commutes with every element of $\hat{\mathcal{H}}$, and $\sigma^2=\widehat{\tau^{e_2-e_1}}$, $\sigma F=\widehat{\tau^{e_2}}$, $\sigma^{-1}F=\widehat{\tau^{e_1}}$. Consequently, with $\mathcal{N}$ as above,

$$\mathrm{Aut}(X_{F_{95}})\;\cong\;\big(\mathcal{N}\times\mathcal{N}\times\mathbb{Z}\big)\big/\big\langle(\sigma^{-1},\sigma,-2)\big\rangle,$$

where (φ,ψ,n) corresponds to $\widehat{H_{\varphi,\psi}}\,\sigma^n$; under this isomorphism σ is the class of $(\mathrm{id},\mathrm{id},1)$ and F the class of $(\mathrm{id},\sigma,-1)$.

Proof. For $g\in\mathrm{Aut}(\mathcal{W})$ the map $g\times g$ commutes with $\Pi\sigma\Pi^{-1}$ and $\Pi F\Pi^{-1}$, by the formulas of Lemma 6.1; so $\hat g$ is a homeomorphism of $\mathcal{A}^{\mathbb{Z}}$ commuting with σ and F, that is, an automorphism of $X_{F_{95}}$ (Lemma 2.2), and it commutes with σ. The map $g\mapsto\hat g$ is an injective homomorphism, and the formulas for σ^2 and $\sigma^{\pm1}F$ are those of Lemma 6.1. The shift is not in $\hat{\mathcal{H}}$: if c is the

bare defect Δ_0^-, then $\Pi(c) = (u, 0)$ with $u \neq 0$, while $\Pi(\sigma c) = (0, \tau^{e_2} u)$, and no map $g \times g$ with $g(0) = 0$ maps the former to the latter.

Let $h \in \mathrm{Aut}(X_{F_{95}})$. By Theorem 4.12 there are $\alpha^+ \equiv \alpha^- \pmod 2$ with $h\Delta_a^\pm = \Delta_{a+\alpha^\pm}^\pm$. Since $\sigma\Delta_a^+ = \Delta_{a+1}^+$ and $\sigma\Delta_b^- = \Delta_{b-1}^-$, replacing h by $h\sigma$ changes the parity of both; so we may assume that α^+ and α^- are even, and we show that then $h \in \hat{\mathcal{H}}$. Let $\tilde{h} = \Pi h \Pi^{-1}$; it commutes with $\tau^n \times \tau^n$ for all n, it has a local rule on some $D \subseteq [-r, r]^2$ in both components, and it fixes $(0, 0) = \Pi(b_{11})$. It maps $\mathcal{W}^+ \times \mathcal{W}^+$ and $\mathcal{W}^- \times \mathcal{W}^-$ onto themselves, so there are homeomorphisms Ξ and Υ of $\mathsf{G} \times \mathsf{G}$, commuting with $\sigma \times \sigma$, with

$$\tilde{h}(u_{x^0,0}, u_{x^1,0}) = (u_{\Xi(x)^0,0}, u_{\Xi(x)^1,0}), \qquad \tilde{h}(u_{0,y^0}, u_{0,y^1}) = (u_{0,\Upsilon(y)^0}, u_{0,\Upsilon(y)^1})$$

for $x = (x^0, x^1)$ and $y = (y^0, y^1)$ in $\mathsf{G} \times \mathsf{G}$. Let x^0, x^1, y^0, y^1 be finite and vanish on $(-\infty, 0]$. Exactly as in the proof of Theorem 6.7(b), applied to both components of $\tilde{h}(u_{x^0,y^0}, u_{x^1,y^1})$, comparing the two descriptions of this pair on $\{k \geq k_1\}$ for large k_1 gives

$$\Xi\big(\sigma^{-|y^0|}x^0,\ \sigma^{-|y^1|}x^1\big) = \big(\sigma^{-|\Upsilon(y)^0|}\Xi(x)^0,\ \sigma^{-|\Upsilon(y)^1|}\Xi(x)^1\big), \tag{14}$$

and this extends to all finite x^e, y^e, because Ξ and Υ commute with $\sigma \times \sigma$ and the numbers $|y^e|$ and $|\Upsilon(y)^e|$ do not change when y is replaced by $(\sigma \times \sigma)^s y$. Let $\mathbf{1}_b \in \mathsf{G}$ be the point with a single 1, at b. The bare defect Δ_{2b}^- has $\Pi(\Delta_{2b}^-) = (u_{0,\mathbf{1}_b}, 0)$, and $h\Delta_{2b}^- = \Delta_{2b+\alpha^-}^-$ with α^- even; so $\Upsilon(\mathbf{1}_b, 0^\infty) = (\mathbf{1}_{b+\alpha^-/2}, 0^\infty)$. With $y = (\mathbf{1}_b, 0^\infty)$, (14) says that Ξ commutes with $\sigma \times \mathrm{id}$ on the finite points, hence everywhere, since the finite points are dense and Ξ is continuous. So Ξ commutes with $\sigma \times \mathrm{id}$ and with $\mathrm{id} \times \sigma$. The coordinate $\Xi(x)_0^0$ depends only on x^0 and x^1 on some window $[-r', r']$, and it does not change if x^1 is replaced by $\sigma^s x^1$; any two words of G of length $2r' + 1$ occur in one point of G, so $\Xi(x)_0^0$ does not depend on x^1. Hence $\Xi(x)^0 = \varphi_0(x^0)$ and likewise $\Xi(x)^1 = \varphi_1(x^1)$, with φ_0, φ_1 continuous and commuting with σ; applying the same to Ξ^{-1} shows that $\varphi_0, \varphi_1 \in \mathrm{Aut}(\mathsf{G}, \sigma)$. The same argument with the roles of j and k exchanged, using Δ^+ and α^+, gives $\Upsilon = \psi_0 \times \psi_1$. Now (14) with a finite $x^e \neq 0^\infty$ gives $|\psi_e(y^e)| = |y^e|$ for all finite y^e, and symmetrically for φ_e; so $\varphi_e, \psi_e \in \mathcal{N}$. Finally, $\Pi\sigma\Pi^{-1}$ acts on $\mathcal{W}^+ \times \mathcal{W}^+$ by $(x^0, x^1) \mapsto (\sigma^{-1}x^1, x^0)$ and on $\mathcal{W}^- \times \mathcal{W}^-$ by $(y^0, y^1) \mapsto (y^1, \sigma y^0)$, and $\tilde{h}$ commutes with it; this gives $\varphi_0 = \varphi_1$ and $\psi_0 = \psi_1$. Let $g = H_{\varphi_0,\psi_0}$. Then $\hat{g}^{-1}h$ is an automorphism of $X_{F_{95}}$ that fixes every point of $\Phi^+ \cup \Phi^-$, so it is the identity, by Corollary 3.9; thus $h = \hat{g}$.

So $\mathrm{Aut}(X_{F_{95}}) = \hat{\mathcal{H}} \cup \hat{\mathcal{H}}\sigma$, the map $(g, n) \mapsto \hat{g}\sigma^n$ is a surjective homomorphism $\mathrm{Aut}(\mathcal{W}) \times \mathbb{Z} \to \mathrm{Aut}(X_{F_{95}})$, and $\hat{g}\sigma^n = \mathrm{id}$ if and only if $n = 2s$ is even and $g = \tau^{s(e_1 - e_2)}$. By Theorem 6.7(b), $\tau^{e_2 - e_1} = H_{\sigma^{-1},\sigma}$, which gives the last statement; $F = \sigma^{-1} \cdot \sigma F = \sigma^{-1}\widehat{H_{\mathrm{id},\sigma}}$. □

Corollary 6.10. *There is an injective homomorphism $\bigoplus_{m \geq 1} S_m \to \mathrm{Aut}(X_{F_{95}})$, namely $\varpi \mapsto \widehat{H_{\varphi_\varpi,\mathrm{id}}}$. So $\mathrm{Aut}(X_{F_{95}})$ contains every finite group and is not virtually $\mathbb{Z}^n$ for any n, and $\mathrm{Aut}(X_{F_{95}}) \neq \langle \sigma, F \rangle$.*

Proof. Lemma 6.8, Theorem 6.7(b), Theorem 6.9, and Lemma 2.4, by which $\langle \sigma, F \rangle \cong \mathbb{Z}^2$ is torsion-free. □

Corollary 6.11. *The subgroup $\langle \sigma, F \rangle$ is central in $\mathrm{Aut}(X_{F_{95}})$, and*

$$\mathrm{Aut}(X_{F_{95}})/\langle \sigma, F \rangle \ \cong\ \big(\mathcal{N}/\langle\sigma\rangle\big) \times \big(\mathcal{N}/\langle\sigma\rangle\big).$$

In particular $\langle \sigma, F \rangle$ has infinite index in $\mathrm{Aut}(X_{F_{95}})$.

Proof. In the presentation of Theorem 6.9, $\langle \sigma, F \rangle$ is the image of the subgroup of $\mathcal{N} \times \mathcal{N} \times \mathbb{Z}$ generated by $(\mathrm{id}, \mathrm{id}, 1)$, $(\mathrm{id}, \sigma, -1)$ and $(\sigma^{-1}, \sigma, -2)$. This subgroup contains $(\mathrm{id}, \sigma, 0)$ and $(\sigma^{-1}, \mathrm{id}, 0)$, so it is $\langle\sigma\rangle \times \langle\sigma\rangle \times \mathbb{Z}$; it is central, because σ commutes with every element of $\mathcal{N}$, and it contains the kernel $\langle(\sigma^{-1}, \sigma, -2)\rangle$. This gives the first two statements. The group $\bigoplus_m S_m$ is an

infinite torsion group, and $\langle\sigma\rangle \subset \mathcal{N}$ is torsion-free, since σ has infinite order on G; so the image of $\bigoplus_m S_m$ under the embedding of Lemma 6.8 meets $\langle\sigma\rangle$ trivially, and $\mathcal{N}/\langle\sigma\rangle$ is infinite. □

Remark 6.12. (1) The construction of Theorem 6.7(a) does not use injectivity: if $\varphi, \psi\colon \mathsf{G} \to \mathsf{G}$ are block maps that preserve the number of 1s of finite points, the same formula defines a continuous map $H_{\varphi,\psi}\colon \mathcal{W} \to \mathcal{W}$ commuting with translations, and $\Pi^{-1}(H_{\varphi,\psi} \times H_{\varphi,\psi})\Pi$, which we again denote by $\widehat{H_{\varphi,\psi}}$, is a block map commuting with F_{95} and with σ, which is not injective if φ or ψ is not. An example is the map φ that moves every 1 one step to the right unless there is a 1 two steps to its right. By Proposition 6.21 these maps, composed with powers of σ, are all the non-constant block maps of radius at most 4 that commute with F_{95}; in particular they account for the 20 and the 220 further maps of the census of Lemma 6.20.

(2) By Theorem 6.3, $\mathcal{W}$ is the quotient of $\mathsf{G} \times \mathsf{G}$ by the identification of (x, y) with $(\sigma x, \sigma^{-1} y)$ when $x_0 = y_{-1} = 1$, and $\sigma^a \times \sigma^b$ acts on the class of (x, y) as the translation by $\theta_{x,y}(a, b)$. So the delays are removed by a change of coordinates that depends on the point, and in the new coordinates the two families of particles move freely.

(3) Theorem 4.12 is consistent with Theorem 6.9: an automorphism $\widehat{H_{\varphi,\psi}}\sigma^n$ maps bare defects to bare defects, because φ and ψ map points with one 1 to points with one 1. The automorphisms that fix every bare defect are the $\widehat{H_{\varphi,\psi}}$ with $\varphi(\mathbf{1}_0) = \psi(\mathbf{1}_0) = \mathbf{1}_0$; they form a subgroup that contains the image of $\bigoplus_m S_m$, and for them the conclusion of Lemma 5.2 holds while they are not the identity.

The smallest radius. A block map $\varphi\colon \mathsf{G} \to \mathsf{G}$ *depends on* w *consecutive cells* if there are $a \in \mathbb{Z}$ and a function $\hat{\varphi}$ on the words of G of length w with $(\varphi x)_i = \hat{\varphi}(x_{i+a}, \dots, x_{i+a+w-1})$ for all x and i; it *conserves particles* if $|\varphi x| = |x|$ for every finite x. By Remark 6.12(1) the maps $\widehat{H_{\varphi,\psi}} = \Pi^{-1}(H_{\varphi,\psi} \times H_{\varphi,\psi})\Pi$ are defined for all block maps $\varphi, \psi\colon \mathsf{G} \to \mathsf{G}$ that conserve particles, and they commute with F and with σ. By Lemma 6.1 and Theorem 6.7(a), in terms of the points $x^e_j = 1 \oplus \pi(2j + e)$ of a point of Φ^+ with diagram $\pi(q)$ and $y^e_k = 1 \oplus \xi(2k + e)$ of a point of Φ^- with diagram $\xi(p)$,

$$\begin{aligned} \widehat{H_{\varphi,\psi}}\colon\ & (x^0, x^1) \mapsto (\varphi x^0, \varphi x^1), && (y^0, y^1) \mapsto (\psi y^0, \psi y^1), \\ \sigma\colon\ & (x^0, x^1) \mapsto (\sigma^{-1} x^1, x^0), && (y^0, y^1) \mapsto (y^1, \sigma y^0). \end{aligned} \tag{15}$$

Lemma 6.13 (windows)**.** *Let h be a block map of radius $R \geq 1$ that commutes with F_{95}.*

(i) *Let $Z \in \Phi^+$ have the diagram $\pi(q)$, and let $\pi'(q)$ be the diagram of hZ. Then $\pi'(q)$ depends only on $\pi(q - R), \dots, \pi(q + R)$. The same holds for Φ^- and the diagrams $\xi(p)$.*

(ii) *Let $h = \widehat{H_{\varphi,\psi}}\,\sigma^n$ with block maps $\varphi, \psi\colon \mathsf{G} \to \mathsf{G}$ that conserve particles, and let $s = \lfloor n/2 \rfloor$. Then $\varphi\sigma^{-s}$ and $\psi\sigma^s$ depend on $2\lfloor R/2 \rfloor + 1$ consecutive cells if n is even, and on $2\lceil R/2 \rceil$ consecutive cells if n is odd. In both cases they depend on $R + 1$ consecutive cells.*

Proof. (i) Let $[q, p] = (t, i)$. The value $Y_{h(c)}(t, i)$ is the second component of $F^t(h(c))_i = h(F^t(c))_i$, so it depends only on the values $Y_c(t', j)$ with $t' \in \{t - 1, t\}$ and $|j - i| \leq R$; it is also the first component of $h(F^{t+1}(c))_i$, so it depends only on the values $Y_c(t', j)$ with $t' \in \{t, t + 1\}$ and $|j - i| \leq R$. For $c = Z$ these values are $\pi(t' - j)$, and $t' - j$ ranges over $[q - R - 1, q + R]$ in the first case and over $[q - R, q + R + 1]$ in the second. Let $\pi, \tilde{\pi} \in \Phi_{95}$ agree on $[q - R, q + R]$, and let π'' be equal to π on $(-\infty, q + R]$ and to $\tilde{\pi}$ on $[q + R + 1, \infty)$. Two positions at distance 2 of which one is at least $q + R + 1$ are both at least $q + R - 1 \geq q - R$, where $\pi'' = \tilde{\pi}$; so π'' has no two zeros at distance 2, that is, $\pi'' \in \Phi_{95}$. It agrees with π on $[q - R - 1, q + R]$ and with $\tilde{\pi}$ on $[q - R, q + R + 1]$, so the three sequences give the same value $\pi'(q)$. For Φ^- exchange the roles of q and p.

(ii) By (15), h acts on Φ^+ by $(x^0,x^1)\mapsto(\varphi' x^0,\varphi' x^1)$ if $n=2s$ and by $(x^0,x^1)\mapsto(\varphi'\sigma^{-1}x^1,\varphi' x^0)$ if $n=2s+1$, where $\varphi'=\varphi\sigma^{-s}$. In the first case $(\varphi' x^e)_j = 1\oplus\pi'(2j+e)$ is a function of x^e alone, and by (i), applied with $x^{1-e}=0^\infty$, it depends only on the $x^e_{j'}$ with $|2j'-2j|\le R$; these are $2\lfloor R/2\rfloor+1$ consecutive cells. In the second case $(\varphi' x^0)_j = 1\oplus\pi'(2j+1)$ depends only on the $x^0_{j'}$ with $|2j'-2j-1|\le R$, which are $2\lceil R/2\rceil$ consecutive cells. On Φ^-, h acts by $(y^0,y^1)\mapsto(\psi' y^0,\psi' y^1)$, resp. by $(y^0,y^1)\mapsto(\psi' y^1,\psi'\sigma y^0)$, with $\psi'=\psi\sigma^s$, and the same count applies to $(\psi' y^e)_k = 1\oplus\xi'(2k+e)$, resp. to $(\psi' y^1)_k = 1\oplus\xi'(2k)$. □

Lemma 6.14 (computer-assisted)**.** *For $w\le 8$ the numbers of block maps $\mathsf{G}\to\mathsf{G}$ of the form $(\varphi x)_i=\hat\varphi(x_i,\dots,x_{i+w-1})$ that conserve particles, and of the injective ones among them, are those of Table 1. For $w\le 7$ the injective ones are the w shifts $\sigma^0,\dots,\sigma^{w-1}$. For $w=8$ they are the eight shifts $\sigma^0,\dots,\sigma^7$ and the two maps $\sigma^4\varphi_*$ and $\sigma^3\bar\varphi_*$, where $\varphi_*\in\mathcal{N}$ is the involution that exchanges the words* 0001001 *and* 0000101 *wherever they occur, and $\bar\varphi_*$ the one that exchanges* 1001000 *and* 1010000*.*

w	1	2	3	4	5	6	7	8
words of G of length w	2	3	5	8	13	21	34	55
particle-conserving block maps	1	2	3	6	15	54	476	13516
injective	1	2	3	4	5	6	7	10

TABLE 1. Block maps of the golden mean shift that depend on w consecutive cells and conserve the number of 1s of finite points (Lemma 6.14).

Two distinct occurrences of the words 0001001 and 0000101 in a point of G are disjoint, because no proper suffix of one of them is a prefix of one of them; so φ_* is well defined, and $(\varphi_* x)_i$ depends on $x_{i-4},\dots,x_{i+3}$. It is an involution: $(\varphi_*\varphi_* x)_i$ depends on $x_{i-8},\dots,x_{i+6}$, and it equals x_i for each of the 1597 words of G of length 15. The same holds for $\bar\varphi_*$, the conjugate of φ_* by the reflection. The conservation of particles is decided exactly: for a block map φ as in the lemma, with $w\ge 2$, let B_w be the graph whose vertices are the words of G of length $w-1$ and whose edges are the words $\epsilon=\epsilon_0\cdots\epsilon_{w-1}$ of G of length w, from $\epsilon_0\cdots\epsilon_{w-2}$ to $\epsilon_1\cdots\epsilon_{w-1}$. A finite point is a closed walk through the vertex 0^{w-1}, and $|\varphi x|-|x|$ is the sum of the weights $\hat\varphi(\epsilon)-\epsilon_0$ along it. Since B_w is strongly connected, the weight of every closed walk through 0^{w-1} vanishes if and only if the weight of every cycle vanishes, that is, if and only if there is a function λ on the vertices with

$$\hat\varphi(\epsilon)-\epsilon_0=\lambda(\epsilon_0\cdots\epsilon_{w-2})-\lambda(\epsilon_1\cdots\epsilon_{w-1})\qquad\text{for every edge }\epsilon;$$

this is the local form of Lemma 6.6. So the particle-conserving block maps correspond to the integer-valued functions λ with $\lambda(0^{w-1})=0$ for which $\epsilon_0+\lambda(\epsilon_0\cdots\epsilon_{w-2})-\lambda(\epsilon_1\cdots\epsilon_{w-1})\in\{0,1\}$ on every edge and the resulting $\hat\varphi$ does not take the value 1 on two consecutive edges. An injective block map of G into itself is an automorphism: G is an irreducible shift of finite type, the image of an injective block map has the entropy of G [35, Theorem 8.1.16], and a proper subshift of G has smaller entropy [35, Corollary 4.4.9], so the map is surjective; so the injective maps of the lemma are the elements of $\mathcal{N}$ that depend on w consecutive cells.

Proposition 6.15 (the smallest radius; computer-assisted)**.**

(a) *Every automorphism of $X_{F_{95}}$ of radius at most* 6 *lies in $\langle\sigma,F\rangle$.*

(b) *The automorphism $h_*=\sigma^{-1}\widehat{H_{\varphi_*,\mathrm{id}}}$ is a block map of radius* 7*. It is not in $\langle\sigma,F\rangle$, and $(\sigma h_*)^2=\mathrm{id}$. The bit y'_i of $h_*(c)_i=(x'_i,y'_i)$ depends exactly on the y_{i+m} with m odd,*

$-7 \le m \le 5$, *and the* x_{i+m} *with* m *even,* $|m| \le 6$*; the bit* x'_i *depends exactly on the* x_{i+m} *with* m *odd,* $-5 \le m \le 7$*, and the* y_{i+m} *with* m *even,* $|m| \le 6$.

(c) *The automorphisms of* $X_{F_{95}}$ *of radius at most* 7 *that are not in* $\langle \sigma, F\rangle$ *are the* 36 *maps* $\widehat{H_{\varphi,\psi}}\,\sigma$ *with* $\varphi \in \{\sigma^{-3}, \dots, \sigma^4,\ \varphi_*\sigma,\ \bar\varphi_*\}$ *and* $\psi \in \{\sigma^{-4}, \dots, \sigma^3,\ \varphi_*,\ \bar\varphi_*\sigma^{-1}\}$*, not both powers of* σ*. They are pairwise distinct, each has the neighbourhood* $[-7, 7]$*, and* h_* *is the one with* $(\varphi, \psi) = (\varphi_*\sigma, \sigma^{-1})$.

So the smallest radius of an automorphism of $X_{F_{95}}$ *outside* $\langle \sigma, F\rangle$ *is* 7.

Proof. (a) Let h be an automorphism of radius $R \le 6$. By Theorem 6.9, $h = \widehat{H_{\varphi,\psi}}\sigma^n$ with $\varphi, \psi \in \mathcal{N}$ and $n \in \mathbb{Z}$. By Lemma 6.13(ii), $\varphi\sigma^{-s}$ and $\psi\sigma^s$, with $s = \lfloor n/2 \rfloor$, are elements of $\mathcal{N}$ that depend on 7 consecutive cells. By Lemma 6.14 they are powers of σ; so φ and ψ are powers of σ, and $h \in \langle \sigma, F\rangle$, because $\widehat{H_{\sigma,\mathrm{id}}} = \sigma^{-1}F$ and $\widehat{H_{\mathrm{id},\sigma}} = \sigma F$ by Theorems 6.7(b) and 6.9.

(b) Let h be the block map of radius 7 defined by two tables T_x, T_y of 2^{15} entries: $y'_i = T_y(b)$ with $b_m = y_{i+m}$ for odd m and $b_m = x_{i+m}$ for even m, $|m| \le 7$, and $x'_i = T_x(b)$ with $b_m = x_{i+m}$ for odd m and $b_m = y_{i+m}$ for even m. The following are verified exhaustively (Appendix A.2.4). First, $hF = Fh$: each bit of $hF(c)_0$ and of $Fh(c)_0$ is a function of the 17 bits of one of the two classes $\{y_k : k \text{ even}\} \cup \{x_k : k \text{ odd}\}$ and $\{x_k : k \text{ even}\} \cup \{y_k : k \text{ odd}\}$ in the cells $|k| \le 8$, and the two sides agree for all $2 \cdot 2^{17}$ values. Second, $(\sigma h)^2 = \mathrm{id}$: each bit of $(\sigma h)^2(c)_0$ is a function of the 29 bits of one class in the cells $-12 \le k \le 16$, and it equals the corresponding bit of c_0 for all $2 \cdot 2^{29}$ values. Third, on the points of Φ^-, h is σ^{-1}, and on the points of Φ^+ it is σ^{-1} composed with $(x^0, x^1) \mapsto (\varphi_* x^0, \varphi_* x^1)$; a window of 15 cells of a point of $\Phi^\pm$ is given by a word of length 16 of Φ_{95}, and all $55^2 = 3025$ of them are checked. Fourth, the tables depend on the bits listed in the statement and on no others. By the first and third facts and (15), h and h_* are block maps that commute with F and agree on $\Phi^+ \cup \Phi^-$, so $h = h_*$ by Corollary 3.9. The first two facts show, without Theorems 6.3–6.9, that h is an automorphism of $X_{F_{95}}$ and that σh is an involution; it is not the identity, by the fourth, and $\langle \sigma, F\rangle \cong \mathbb{Z}^2$ (Lemma 2.4) has no element of order 2. So $h_* \notin \langle \sigma, F\rangle$. Explicitly, for the configuration c of spatial period 5 with the cells $(x_i, y_i) = 00, 00, 00, 01, 10$, whose F-orbit has 15 points, $h_*(c)$ has the cells $10, 00, 01, 00, 01$ and is not of the form $\sigma^a F^b(c)$. By the fourth fact the neighbourhood of h_* is $[-7, 7]$ and no smaller interval.

(c) Let $h = \widehat{H_{\varphi,\psi}}\sigma^n$, with $\varphi, \psi \in \mathcal{N}$ (Theorem 6.9), be an automorphism of radius at most 7. If $n = 2s$, then $\varphi\sigma^{-s}$ and $\psi\sigma^s$ depend on 7 consecutive cells by Lemma 6.13(ii), and $h \in \langle \sigma, F\rangle$ as in (a). Let $n = 2s+1$. Since $\sigma^2 = \widehat{H_{\sigma^{-1},\sigma}}$, we have $h = \widehat{H_{\varphi',\psi'}}\sigma$ with $\varphi' = \varphi\sigma^{-s}$ and $\psi' = \psi\sigma^s$, and by the proof of Lemma 6.13(ii), $(\varphi' x)_j$ depends only on $x_{j-3}, \dots, x_{j+4}$ and $(\psi' y)_k$ only on $y_{k-4}, \dots, y_{k+3}$. By Lemma 6.14, $\varphi'\sigma^3$ and $\psi'\sigma^4$ belong to $\{\sigma^0, \dots, \sigma^7, \sigma^4\varphi_*, \sigma^3\bar\varphi_*\}$, which gives the two lists; and $h \in \langle \sigma, F\rangle$ if both are powers of σ. Conversely, for each of the 36 pairs a block map of radius 7 given by two tables as in (b) is verified, exhaustively, to commute with F and to have the restrictions (15) of $\widehat{H_{\varphi,\psi}}\sigma$ to Φ^+ and Φ^-; by Corollary 3.9 it is $\widehat{H_{\varphi,\psi}}\sigma$. The 36 pairs of tables are distinct, each depends on a bit of the cell $i-7$ and on a bit of the cell $i+7$, and by Corollary 6.11 none of the 36 maps is in $\langle \sigma, F\rangle$, because φ_* and $\bar\varphi_*$ are not powers of σ. Finally $h_* = \sigma^{-1}\widehat{H_{\varphi_*,\mathrm{id}}} = \widehat{H_{\varphi_*\sigma,\sigma^{-1}}}\sigma$. □

Blocking words and eigenfields. Blocking words, the graphs $W_M^{(k,p)}$ and eigenfields are defined in Section 2.4. In [13] the automorphisms outside $\langle \sigma, F\rangle$ of rules that are not affine are built from blocking words. We show that F_{95} has none, and that it has no eigenfield of any speed. For speed 0 and words of at most 6 cells the second statement is known: 95R is among the rules without a local conservation law in [42, Section 4, Table IV]. We use the following consequences of Theorem 6.3.

Lemma 6.16.

(i) *If c is a finite perturbation of b_{11}, then there are finite points x^0, y^0, x^1, y^1 of G with $\Pi(c) = (u_{x^0,y^0}, u_{x^1,y^1})$.*

(ii) *Let $x, y \in \mathsf{G}$ be finite, with all 1s of x at positions $\le M_x$ and all 1s of y at positions $\le M_y$. Then $u_{x,y}(j,k) = 0$ whenever $j > M_x + |y|$ and $k > M_y + |x|$.*

(iii) *Let $x, y \in \mathsf{G}$, let y be finite, and let n be the number of 1s of y at positions ≥ 0. Then $\tau^{te_2} u_{x,y} \to u_{\sigma^{-n}x,0}$ as $t \to \infty$.*

Proof. (i) The proof of Theorem 6.3(c) gives $t \ge 0$ and finite points x^e, y^e with $\Pi(F^{-t}(c)) = (u_{x^0,y^0}, u_{x^1,y^1})$. By Lemma 6.1 the components of $\Pi(c)$ are translates of these two functions, and by Theorem 6.3(b) and the surjectivity of $\theta_{x,y}$ every translate of $u_{x,y}$ is $u_{\sigma^a x, \sigma^b y}$ for some (a,b).

(ii) Let $(j,k) = \theta_{x,y}(J,K) = (J + A_y(K), K + A_x(J))$. Since $A_y(K) \le |y|$ and $A_x(J) \le |x|$, we get $J > M_x$ and $K > M_y$, so $u_{x,y}(j,k) = x_J \vee y_K = 0$.

(iii) Let $K > 0$ be larger than every position of a 1 of y. Then $A_y(K) = n$ and $y_K = 0$, so $u_{x,y}(J + n, K + A_x(J)) = x_J$ for every J. Hence for every j we have $u_{x,y}(j,k) = x_{j-n} = u_{\sigma^{-n}x,0}(j,k)$ for all large k, which is the convergence of $\tau^{te_2} u_{x,y}$ in the product topology. □

Proposition 6.17. *F_{95} has no blocking word.*

Proof. Let w be a word on an interval I of cells, let $i_0 \in I$, and let c be the configuration that carries w on I and $(1,1)$ elsewhere. By Lemma 6.16(i), $\Pi(c) = (u_{x,y}, u_{x^1,y^1})$ with finite points. The point (j,k) of the sublattice e is the point $(t,i) = (j + k + e, k - j)$ of the diagram, so the points (t, i_0) and $(t-1, i_0)$ have $j, k \ge (t - |i_0| - 2)/2$; by Lemma 6.16(ii) there is t_1 with $Y_c(t, i_0) = Y_c(t-1, i_0) = 1$, that is $F^t(c)_{i_0} = (1,1)$, for all $t \ge t_1$.

Let $J_0 \ge 2$ exceed every position of a 1 of x by at least 2, and let $\tilde{x} = x + \mathbf{1}_{J_0} \in \mathsf{G}$ and $c' = \Pi^{-1}(u_{\tilde{x},y}, u_{x^1,y^1})$. As $J_0 \to \infty$ we have $\tilde{x} \to x$ and therefore $c' \to c$, by Theorem 6.3(c) and Lemma 6.1; so c' carries w on I if J_0 is large. The number $m = A_{\tilde{x}}(J_0) = A_x(J_0)$ does not depend on J_0, and the points

$$\theta_{\tilde{x},y}(J_0, K) = \big(J_0 + A_y(K),\, K + m\big), \qquad K \in \mathbb{Z},$$

carry the value $\tilde{x}_{J_0} \vee y_K = 1$ of $u_{\tilde{x},y}$. Their columns $i = K - A_y(K) + m - J_0$ are nondecreasing in K with steps $1 - y_K \in \{0,1\}$ and unbounded in both directions, so one of these points lies in the column i_0; its time is $t = 2j + i_0$ with $j = J_0 + A_y(K) \ge J_0 - |y|$. There $Y_{c'}(t, i_0) = 0$, so $F^t(c')_{i_0} \ne (1,1)$, and $t \ge \max(t_1, 0)$ if J_0 is large. Hence $F^t(c)_{i_0} \ne F^t(c')_{i_0}$ for two configurations that carry w on I, and w is not blocking with the column i_0. □

Equivalently, F_{95} is sensitive to initial conditions and has no equicontinuity point [34, Theorem 3 and Corollary 2], [3, Proposition 2.1]. In words: the column i_0 of c is eventually $(1,1)$, and a single right-moving particle placed far to the left of w crosses everything in its way, with finitely many delays, and passes through the column i_0 at a late time.

Proposition 6.18. *Let $(k,p) \in \mathbb{Z}^2$ with $p \ge 1$. Every continuous function $v\colon \mathcal{A}^{\mathbb{Z}} \to \{0,1\}$ with $v \circ \sigma^k F_{95}^p = v$ is constant. Consequently the graphs $W_M^{(k,p)}$ of $95R$ are strongly connected for all (k,p) and all M; in particular there is no nonconstant block map $\vartheta\colon \mathcal{A}^{\mathbb{Z}} \to \{0,1\}^{\mathbb{Z}}$ with $\vartheta \circ F_{95} = \sigma^{\mp 1} \circ \vartheta$ or with $\vartheta \circ F_{95} = \vartheta$.*

Proof. The consequences follow from Lemma 2.5. Let v be continuous with $v \circ \sigma^k F^p = v$, and let c be a finite perturbation of b_{11}, equal to $(1,1)$ outside $[-L, L]$. The cell i of $(\sigma^k F^p)^t(c)$ is $F^{pt}(c)_{i+kt} = (Y_c(pt-1, i+kt), Y_c(pt, i+kt))$.

If $|k| > p$, these values are 1 for large t, because $Y_c(s,j)$ depends only on $c_{[j-s, j+s]}$ for $s \ge 0$. If $|k| < p$, the points (s,j) with $s \in \{pt-1, pt\}$ and $j = i + kt$ are the points (j', k') of their sublattices with $j' + k' + e = s$ and $k' - j' = j$, so j' and k' tend to infinity with t, and the values

are 1 for large t by Lemma 6.16(i) and (ii). In both cases $(\sigma^k F^p)^t(c) \to b_{11}$, so $v(c) = v(b_{11})$; the finite perturbations of b_{11} are dense, and v is constant.

Let $k = p$; the case $k = -p$ follows by conjugation with the reflection rev, which commutes with F and conjugates σ to σ^{-1}. For $x^0, x^1 \in \mathsf{G}$ put

$$V(x^0, x^1) = v\big(\Pi^{-1}(u_{x^0,0}, u_{x^1,0})\big);$$

this is a continuous function on $\mathsf{G} \times \mathsf{G}$, by Theorem 6.3(c) and Lemma 6.1. Let $x^0, x^1, y^0, y^1 \in \mathsf{G}$ be finite, let $c = \Pi^{-1}(u_{x^0,y^0}, u_{x^1,y^1})$, and let n_e be the number of 1s of y^e at positions ≥ 0. Since $\Pi(\sigma F)^t\Pi^{-1} = \tau^{te_2} \times \tau^{te_2}$ by Lemma 6.1, Lemma 6.16(iii) gives

$$(\sigma F)^{pt}(c) \;\longrightarrow\; \Pi^{-1}\big(u_{\sigma^{-n_0}x^0,0},\, u_{\sigma^{-n_1}x^1,0}\big) \qquad (t \to \infty).$$

Since v is continuous and invariant under $(\sigma F)^p$,

$$v(c) = V\big(\sigma^{-n_0}x^0, \sigma^{-n_1}x^1\big). \tag{16}$$

Now let $y^1 = 0^\infty$, and compare $y^0 = 0^\infty$ with $y^0 = \mathbf{1}_K$, $K > 0$. The configurations $c_K = \Pi^{-1}(u_{x^0,\mathbf{1}_K}, u_{x^1,0})$ converge to $c_\infty = \Pi^{-1}(u_{x^0,0}, u_{x^1,0})$ as $K \to \infty$, so $v(c_K) = v(c_\infty)$ for large K; by (16), $v(c_K) = V(\sigma^{-1}x^0, x^1)$ and $v(c_\infty) = V(x^0, x^1)$. So $V \circ (\sigma^{-1} \times \mathrm{id}) = V$ on the pairs of finite points, hence everywhere, because these are dense and V is continuous; in the same way $V \circ (\mathrm{id} \times \sigma^{-1}) = V$. The shift G has a point with a dense orbit, so the action of $\mathbb{Z}^2$ on $\mathsf{G} \times \mathsf{G}$ by $\sigma^a \times \sigma^b$ has a dense orbit, and V is constant, equal to $V(0^\infty, 0^\infty) = v(b_{11})$. By Lemma 6.16(i) and (16), $v(c) = v(b_{11})$ for every finite perturbation c of b_{11}; these are dense, so v is constant. □

For $k = p$ the mechanism is that of Corollary 6.5: the right-moving part into which a finite perturbation decays is translated, class by class, by every left-moving particle that it has crossed, and a left-moving particle can be added far away without changing a given window.

Corollary 6.19. *The rule* 95 *is not affine, F_{95} has no blocking word, all the graphs $W_M^{(k,p)}$ of F_{95} are strongly connected, and* $\mathrm{Aut}(X_{F_{95}})$ *is not virtually $\mathbb{Z}^2$. So for an elementary rule f that is not affine, the absence of blocking words of F_f together with the strong connectivity of all the graphs $W_M^{(k,p)}$ does not imply that* $\mathrm{Aut}(X_{F_f})$ *is virtually $\mathbb{Z}^2$.*

Proof. $f_{95}(a, b, c) = 1 \oplus ac$ is not affine. The other statements are Propositions 6.17 and 6.18 and Corollary 6.10: a group that is virtually $\mathbb{Z}^2$ has a torsion-free subgroup of finite index, so the orders of its finite subgroups are bounded. □

The non-injective commuting maps. We call a block map h of radius R of *sublattice type* if each of the two bits of $h(c)_i$ depends only on the bits of one of the two classes $\{y_k : k \equiv i\} \cup \{x_k : k \not\equiv i\}$ and $\{x_k : k \equiv i\} \cup \{y_k : k \not\equiv i\}$ (congruences modulo 2) in the cells $|k - i| \leq R$. The maps $\widehat{H_{\varphi,\psi}}\sigma^n$ are of sublattice type, because $\Pi\widehat{H_{\varphi,\psi}}\Pi^{-1} = H_{\varphi,\psi} \times H_{\varphi,\psi}$ and $\Pi\sigma\Pi^{-1}$ exchanges the two components.

The commuting block maps of small radius are known from an exhaustive enumeration.

Lemma 6.20 (census for 95; computer-assisted)**.** *For $R \leq 4$ the block maps of radius at most R on $\mathcal{A}^\mathbb{Z}$ that commute with F_{95} are the $2R^2 + 2R + 1$ maps $\sigma^a F^b$ with $|a| + |b| \leq R$, which are pairwise distinct; the constant map with value b_{11}; and* 0, 0, 20 *and* 220 *further maps for $R = 1, 2, 3, 4$. In all there are* 6, 14, 46 *and* 262 *of them.*

The lemma is the result of an enumeration, with a SAT solver, of all tables $\mathcal{A}^{2R+1} \to \mathcal{A}$ whose block map commutes with F_{95}, with no restriction on their form; every table found is checked again by direct evaluation, and the tables are compared with those of the maps $\sigma^a F^b$ (Appendix A.1). The same programs are applied to all 88 classes in [14]. The next proposition identifies the further maps.

Proposition 6.21 (computer-assisted)**.**

(a) *Every block map of radius at most* 4 *that commutes with* F_{95} *is the constant map with value* b_{11} *or one of the maps* $\widehat{H_{\varphi,\psi}}\sigma^n$, $n \in \{0,1\}$, *where* $\varphi, \psi \colon \mathsf{G} \to \mathsf{G}$ *are block maps that conserve particles. For radius* 3 *these are* 9 *maps with* $n = 0$ *and* 36 *with* $n = 1$*; for radius* 4, 225 *maps with* $n = 0$ *and* 36 *with* $n = 1$*. Such a map is injective if and only if* φ *and* ψ *are, and then it lies in* $\langle \sigma, F \rangle$*; so the* 20 *further maps of radius* 3 *and the* 220 *of radius* 4 *of Lemma 6.20 are not injective, and they are exactly the maps* $\widehat{H_{\varphi,\psi}}\sigma^n$ *of these radii in which* φ *or* ψ *is not injective.*

(b) *The block maps of sublattice type and radius at most* 5 *that commute with* F_{95} *are the constant map and* 3141 *maps* $\widehat{H_{\varphi,\psi}}\sigma^n$ *with* φ, ψ *as in (a),* 225 *with* $n = 0$ *and* 2916 *with* $n = 1$*: the* 61 *maps* $\sigma^a F^b$ *with* $|a| + |b| \leq 5$ *and* 3080 *maps that are not injective.*

(c) *For* $R \leq 6$*, every block map of sublattice type and radius at most* R *that commutes with* F_{95} *and is injective on* Φ^+ *and on* Φ^- *is one of the maps* $\sigma^a F^b$*. For* $R = 7$ *there are, besides the* 113 *maps* $\sigma^a F^b$ *with* $|a| + |b| \leq 7$*, exactly* 36 *such maps, and they are the* 36 *automorphisms of Proposition 6.15(c).*

In (a) and (b) the number of maps with $n = 0$ is m_w^2 with $w = 2\lfloor R/2 \rfloor + 1$, and the number with $n = 1$ is m_w^2 with $w = 2\lceil R/2 \rceil$, where m_w is the number of particle-conserving block maps of Table 1: every pair (φ, ψ) allowed by Lemma 6.13(ii) occurs. Statement (c) does not use Lemma 6.14; with Theorem 6.9, by which every automorphism is of sublattice type, it gives Proposition 6.15(a) and the list in Proposition 6.15(c) a second time. We do not know whether every block map that commutes with F_{95} is constant or of the form $\widehat{H_{\varphi,\psi}}\sigma^n$.

Proof. The computations are described in Appendix A.2.4. For (a), each table of the census of Lemma 6.20 of radius 3 and 4 is of sublattice type, and its restrictions to Φ^+ and Φ^-, read off from the table on all words of Φ_{95}, are of the form (15) for $\widehat{H_{\varphi,\psi}}\sigma^n$ with block maps φ, ψ that satisfy the criterion stated after Lemma 6.14; by Corollary 3.9, two block maps that commute with F and agree on $\Phi^+ \cup \Phi^-$ are equal, so the table is that of $\widehat{H_{\varphi,\psi}}\sigma^n$. If φ or ψ is not injective, neither is $\widehat{H_{\varphi,\psi}}\sigma^n$, by (15); if both are injective they are powers of σ by Lemma 6.14. For (b) and (c) the commutation condition is solved for the tables of sublattice type, and in (b) each solution is identified in the same way. □

Remark 6.22 (the class of 175; computer-assisted). The rules that do not read their middle argument are the 16 rules of the classes of 0, 5, 10, 15, 90, 95, 165, 175 and 255 [42, p. 386]. Of these classes, those of 0, 5, 10, 15, 90, 165 and 255 are treated in [13], and for the class of 175 the automorphism group is not known [14]. For 175 the two sublattices of a diagram evolve independently, as for 95, with $Y(\mathsf{N}) \oplus Y(\mathsf{S}) = \neg Y(\mathsf{W}) \vee Y(\mathsf{E})$ in place of (10). The search of Appendix A.2.4 for block maps of sublattice type that commute with F_{175} finds, for every $R \leq 7$, only the $2R^2 + 2R + 1$ maps $\sigma^a F^b$ (computer-assisted, by two programs, each with two solvers; Appendix A.2.5); in particular it finds no non-injective maps of the kind that the class of 95 has from radius 3 on. We do not know whether every automorphism of $X_{F_{175}}$ is of sublattice type.

7. Rule 91R

For $f = 91$ Lemma L holds (Theorem 3.1), so that $\mathrm{Aut}(X_{F_{91}})$ embeds in $\mathrm{Aut}(\Phi^+, \sigma) \times \mathrm{Aut}(\Phi^-, \sigma)$ (Corollary 3.2), but the group is not determined. This section records how far the arguments of Sections 4 and 5 go for this rule.

What the bare defects give. Let $f = 91$, so that N is the indicator of 000, the valid backgrounds are 11, 01 and 10, and Φ_{91} is the set of sequences without 000. As for 95, let $\Delta_a^+ \in \Phi^+$ and $\Delta_b^- \in \Phi^-$ be the *bare defects*, the points whose diagrams are 1 except for the value 0 at $q = a$,

resp. at $p = b$. A sequence is 3-*sparse* if any two of its zeros are at distance at least 3, and a point of $\Phi^{\pm}$ is 3-sparse if its diagram is.

Lemma 7.1 (free superposition)**.** *Let $f = 91$ and let (P, X) be a placed pair with background 11 and corner $[q_0, p_0]$, with diagrams $\pi(q)$ and $\xi(p)$. Suppose that there is no centre $[q, p]$ with $q > q_0$, $p > p_0$ and*

$$\pi(q+1) \neq \xi(p-1), \qquad \pi(q) \neq \xi(p), \qquad \pi(q-1) \neq \xi(p+1). \tag{17}$$

Then $Y_{P,X}[q, p] = \pi(q) \oplus \xi(p) \oplus 1$ on the quadrant G, $Y_{P,X}$ does not fire, and if the left tail of P and the right tail of X are 11, then $\mathcal{S}(P, X) = (X, P)$. This applies whenever π and ξ are 3-sparse; in particular $\mathcal{S}(\Delta_a^+, \Delta_b^-) = (\Delta_b^-, \Delta_a^+)$ for all $a, b \in \mathbb{Z}$.

Proof. Let $Y_\emptyset = \pi(q) \oplus \xi(p) \oplus 1$ on G. As in the proof of Lemma 4.4, $Y_\emptyset$ has the boundary values π and ξ, and it is the solution of (4) if it fires at no centre $[q, p]$ with $q > q_0$, $p > p_0$. It fires at $[q, p]$ exactly if its values at $\mathsf{W} = [q+1, p-1]$, $[q, p]$ and $\mathsf{E} = [q-1, p+1]$ are 0, which is (17). If the tails are 11, then $Y_\emptyset = \xi(p)$ for all large q and $Y_\emptyset = \pi(q)$ for all large p, so $\mathcal{S}(P, X) = (X, P)$. If π and ξ are 3-sparse, each of the triples $(\pi(q+1), \pi(q), \pi(q-1))$ and $(\xi(p-1), \xi(p), \xi(p+1))$ contains at most one 0, so they cannot differ in all three places. □

So, in contrast to 94 and 95 (Lemmas 4.4 and 4.11), two bare defects of 91 cross without a delay in both parities. The next two lemmas describe the first two rays of a probe whose first zero is isolated.

Lemma 7.2 (one exact ray for 91)**.** *Let $f = 91$, let G be the quadrant with corner $[q_0, p_0]$, and let Y be the solution of (4) on G with boundary values A, B such that*

$$A(q) = 1 \quad (q_0 \leq q < a_1), \qquad A(a_1) = 0, \qquad a_1 \geq q_0 + 2, \qquad B(p_0) = B(p_0 + 1) = 1,$$

and B contains no 000. Then $Y[q, p] = B(p)$ for $q_0 \leq q \leq a_1 - 1$. If there is no $x \equiv a_1 \pmod 2$ with $B(x-1) = B(x) = 0$, then Y fires at no centre $[a_1 - 1, p']$ with $p' > p_0$, and $Y[a_1, p] = 1 \oplus B(p)$ for all $p \equiv a_1$. Otherwise it fires at exactly one such centre, $[a_1 - 1, x_1 - 1]$, where x_1 is the least such x.

Proof. The function $[q, p] \mapsto B(p)$ satisfies (4), because B contains no 000, and it has the boundary values 1 and B; by Goursat causality, $Y[q, p] = B(p)$ for $q_0 \leq q \leq a_1 - 1$. Let $U = \{p \geq p_0 : p \equiv a_1\}$, with least element p_b, and $D(p) = Y[a_1, p] \oplus B(p)$ for $p \in U$. Then $D(p_b) = A(a_1) \oplus B(p_b) = 1$, and for $p \in U$, $p \geq p_b + 2$, (4) at $[a_1, p]$ gives $D(p) = D(p-2) \oplus n(p)$ with $n(p) = N(Y[a_1, p-2], B(p-1), B(p))$, the firing at the centre $[a_1 - 1, p-1]$. If $D(p-2) = 1$, then $Y[a_1, p-2] = 1 \oplus B(p-2)$, and $n(p) = 1$ if and only if $B(p-2) = 1$ and $B(p-1) = B(p) = 0$; since B contains no 000 and $B(p_0) = B(p_0 + 1) = 1$, this holds if and only if $B(p-1) = B(p) = 0$. If $D(p-2) = 0$, then $n(p) = N(B(p-2), B(p-1), B(p)) = 0$. So $D = 1$ and $n = 0$ on U below x_1, $n(x_1) = 1$ and $D(x_1) = 0$ if x_1 exists, and after x_1 there are no further firings. □

Lemma 7.3 (the second ray)**.** *In the setting of Lemma 7.2 suppose in addition that $A(a_1 + 1) = 1$, that B has no two adjacent zeros, and that $B(p) = 1$ for all large p. Then Y fires at finitely many centres $[a_1, p']$ with $p' > p_0$, and their number is odd if and only if B has a zero $x \equiv a_1 + 1 \pmod 2$ and the last such zero x satisfies $B(x-2) = 0$. More precisely, with $D'(p) = Y[a_1 + 1, p] \oplus B(p)$ for $p \equiv a_1 + 1$: D' vanishes below the first zero of B in this class, $D'(p) = 1 \oplus B(p-2)$ at every zero p of B in this class, and $D'(p) = D'(p-2)$ at all other p.*

Proof. Since B has no two adjacent zeros, Lemma 7.2 gives $Y[a_1 - 1, p] = B(p)$ and $Y[a_1, p] = 1 \oplus B(p)$. Let $U' = \{p \geq p_0 : p \equiv a_1 + 1\}$, with least element p_b'. Then $D'(p_b') = A(a_1 + 1) \oplus A(a_1 - 1) = 0$, and (4) at $[a_1 + 1, p]$ gives, for $p \in U'$, $p \geq p_b' + 2$,

$$D'(p) = D'(p-2) \oplus n'(p), \qquad n'(p) = N\big(B(p-2) \oplus D'(p-2),\, 1 \oplus B(p-1),\, B(p)\big),$$

where $n'(p)$ is the firing at the centre $[a_1, p-1]$. If $B(p) = 1$, then $n'(p) = 0$. If $B(p) = 0$, then $B(p-1) = 1$, and $n'(p) = 1$ if and only if $B(p-2) = D'(p-2)$; in both cases $D'(p-2) \in \{0, 1\}$ this gives $D'(p) = 1 \oplus B(p-2)$. The number of firings on the ray has the parity of $D'(p) \oplus D'(p'_b) = D'(p)$ for large p, which is $1 \oplus B(x-2)$ for the last zero $x \in U'$, and 0 if there is none. □

Proposition 7.4. *Let $h \in \mathrm{Aut}(X_{F_{91}})$, of radius R, and $r = R + 1$. Then $h(b_{11}) = b_{11}$, and for all $a, b \in \mathbb{Z}$ the points $E = h\Delta_a^+$ and $D = h\Delta_b^-$ are different from b_{11}, their diagrams π_E and δ are 1 outside $[a - r, a + r]$, resp. $[b - r, b + r]$,*

$$\mathcal{S}(h\Delta_a^+, h\Delta_b^-) = (h\Delta_b^-, h\Delta_a^+),$$

and each of the sequences $\omega \in \{\pi_E, \delta\}$ has the following properties:

(i) *ω has no two adjacent zeros;*
(ii) *if x is the last zero of ω in its class modulo 2, then $\omega(x-2) = 1$, and if x is the first zero of ω in its class, then $\omega(x+2) = 1$.*

Proof. By the paragraph before Lemma 4.5, $h(b_{11}) = b_{\beta''}$ for a valid background β''; since $\sigma b_{11} = b_{11}$ and h commutes with σ, $\beta''_0 = \beta''_1$, so $\beta'' = 11$. By (6), $\pi_E(q) = 1$ for $|q - a| > r$ and $\delta(p) = 1$ for $|p - b| > r$, and $E, D \neq b_{11}$ because h is injective. Let $D_b = h\Delta_b^-$, with diagram δ_b; then $\delta_{b+1}(p) = \delta_b(p-1)$. By Lemmas 7.1 and 4.5, $\mathcal{S}(E, D_b) = (D_b, E)$ for every b, and we take the corner at $[q_0, p_0]$ with $q_0 \le a - r - 2$, $p_0 \le b - r - 2$. Let $Y = Y_{E,D_b}$. As in Step 3 of the proof of Theorem 4.9, Lemma 4.1(ii) and Lemma 3.3, with $A(q) = \pi_E(q)$, $B(p) = \delta_b(p) = 1$ for large p and $\nu = 1$, show that for every $q \ge q_0$ the number of firings of Y on the ray $q' = q + 1$ is even.

Let a_1 be the least zero of π_E. By Lemma 7.2, the number of firings on the ray $a_1 - 1$ is 1 if δ_b has two adjacent zeros $x - 1, x$ with $x \equiv a_1$, and 0 otherwise; so there are no such zeros. Applying this to b and $b + 1$ shows that δ has no two adjacent zeros. The map $\mathrm{rev}\, h\, \mathrm{rev}$ is an automorphism of radius R (Step 6 of that proof), and $(\mathrm{rev}\, h\, \mathrm{rev})\Delta_a^- = \mathrm{rev}\, E$ has the diagram $\pi_E(p)$; so π_E has no two adjacent zeros either. This is (i). Now $A = \pi_E$ and $B = \delta_b$ satisfy the hypotheses of Lemma 7.3, so the last zero $x \equiv a_1 + 1$ of δ_b, if there is one, has $\delta_b(x-2) = 1$; with b and $b+1$ this gives the first half of (ii) for δ, and with $\mathrm{rev}\, h\, \mathrm{rev}$ for π_E. Finally $\mu h \mu$ is an automorphism of radius R (proof of Lemma 5.2), μ maps $\Delta_a^\pm$ to $\Delta_{-1-a}^\pm$, and $(\mu h \mu)\Delta_b^- = \mu D_{-1-b}$ has the diagram $p \mapsto \delta_{-1-b}(-1-p)$, a reflection of δ; the first half of (ii) for $\mu h \mu$ is the second half for h. □

Remark 7.5 (kink rigidity for 91 is open). The analogue of Theorems 4.9 and 4.12 for 91 would say that $h\Delta_a^\pm = \Delta_{a+\alpha^\pm}^\pm$. Proposition 7.4 is as far as the argument of those theorems goes, for the following reason. The exact-ray lemma has an analogue (Lemma 7.2), and no train is needed, since a bare defect already crosses a bare defect unchanged; but by Lemma 7.1 the relation $\mathcal{S}(E, D_b) = (D_b, E)$, which is all that the crossing of two bare defects gives, holds for every pair of 3-sparse points with tails 11, so it cannot show that E and D have a single zero, and it gives no relation between α^+ and α^-. A computation (computer-assisted: exact, two independent programs; Appendix A.2) shows that among the 778 objects of Φ^- on the background 11 with core of length at most 12, those that the bare defect crosses unchanged in both parities are exactly the 41 that are 3-sparse; so one expects that (i) and (ii) can be sharpened to: π_E and δ are 3-sparse. For the same reason the analogue for 91 of the scattering closure of Definition 5.1, the smallest set that contains b_{11} and the bare defects and has the properties (C0)–(C3), is not all of $\Phi^+ \cup \Phi^-$: for $L \ge 3$ it is the set $\mathcal{D}_L$ of the points of $\Phi^+ \cup \Phi^-$ whose zeros are pairwise at distance at least L. Indeed, every word of length L of a point of $\mathcal{D}_L$ is a word of b_{11} or of a bare defect, so $\mathcal{D}_L$ is contained in the closure by (C0); and $\mathcal{D}_L$ has the four properties: (C0) and (C1) are clear; a point of $\mathcal{D}_L$ whose diagram is eventually 2-periodic has both tails 11 and is 3-sparse, so (C2) holds by Lemma 7.1; and if (P, X) is a placed pair with $\mathcal{S}(P, X) = (\Gamma, \Omega)$ and

$\Gamma, \Omega \in \mathcal{D}_L$, then $(\mu\Omega, \mu\Gamma)$ is a placed pair of 3-sparse points, so $(\mu X, \mu P) = \mathcal{S}(\mu\Omega, \mu\Gamma) = (\mu\Gamma, \mu\Omega)$ by Proposition 4.2 and Lemma 7.1, and $X = \Gamma$, $P = \Omega$ lie in $\mathcal{D}_L$. A proof of $\mathrm{Aut}(X_{F_{91}}) = \langle \sigma, F \rangle$ along the lines of Sections 4 and 5 therefore has to start from points that are not 3-sparse: the object $1^\infty 001^\infty$ and the kinks between the three backgrounds. Their crossings with a bare defect are not free (computed): for instance the bare defect Δ_a^+ leaves the object with zeros at $b, b+1$, $b \equiv a$, unchanged and emerges with two zeros at distance 2, and for $b \not\equiv a$ the crossing creates a region of the background $(01)^\infty$ that grows at light speed in both directions.

Unlike 95R (Section 6), 91R is not a system of freely moving particles with delays: the crossings just described change the shapes and the tails of the two points; see also [44, pp. 113–116], where the collisions of the quasi-particles of 91R are analysed, with the conclusion that 91R is not a soliton automaton. Two of the additive conserved quantities of F_{91}, with densities on two adjacent cells, are $Q_2(c) = \#\{i : y_i \neq x_{i+1}\} - \#\{i : x_i \neq y_{i+1}\}$, which is conserved for every rule with $f(a, b, c) = 1$ whenever $a \neq c$, in particular for 91, 94 and 95, because at every firing the values at W and E agree, and $Q_1(c) = \#\{i : c_i \neq (1,1)\} - \#\{i : c_i \neq c_{i+1},\ c_i \neq (1,1) \neq c_{i+1}\}$ (Lemma 7.6 below), which on a point of $\Phi^\pm$ with tails 11 counts the maximal blocks of its diagram that contain a 0 and no 11. Both are in the literature. Here $-Q_2$ is the energy (d) of [42, Table IV], in the second of the two forms given there, and the quantity (4.12) of [43]; and Q_1 is, up to an affine change, the energy (b) of 91R and 123R in [42, Table IV], whose sum over a period N is $2N - 2Q_1$, and the quantity [44, (40)]. The additive conserved quantities of the second-order elementary automata are determined for densities on two adjacent cells in [42, Table II] and on at most three adjacent cells in [24, Tables 3–5], and counted for densities on at most four adjacent cells in [44, Table 1]; 91R has a third quantity, with density on three cells, found in [24, Table 4 and p. 321]; see also [43].

The conservation of Q_1 is a finite local identity. In [42, 24, 44] this quantity results from the solution of the linear conservation conditions, and no verification is written out; since [44] is in Japanese and uses a different normalisation, we include one.

Lemma 7.6 ([42, Table IV(b)]; [24, Table 4]; [44, (40)])**.** *Let $f = 91$, and let c be a spatially periodic configuration, the two counts in Q_1 being taken over one period, or a finite perturbation of b_{11}. Then $Q_1(F(c)) = Q_1(c)$.*

Proof. Write $c_i = (u_i, m_i)$, so that $F(c)_i = (m_i, v_i)$ with $v_i = u_i \oplus s_i$ and $s_i = f(m_{i-1}, m_i, m_{i+1})$. The two conditions that define Q_1 are invariant under the exchange of the two bits in every cell, so $Q_1(F(c))$ is the value of Q_1 on the configuration with the cells (v_i, m_i). Since $f(a, b, c) = a \oplus c \oplus N(a, b, c)$ with N the indicator of 000,

(α) if $m_i = 1$, then $s_i = 1$ if and only if exactly one of m_{i-1}, m_{i+1} is 0;
(β) if $m_i = 0$, then $s_i = 1$ if and only if at least one of m_{i-1}, m_{i+1} is 0.

We express $Q_1(c)$ through the rows u and m. A cell with $m_i = 0$ is different from $(1,1)$, and a cell with $m_i = 1$ is different from $(1,1)$ if and only if $u_i = 0$. Let c_i and c_{i+1} both be different from $(1,1)$. If $m_i = m_{i+1} = 1$, both are $(0,1)$, and they are equal. If exactly one of m_i, m_{i+1} is 1, they are different. If $m_i = m_{i+1} = 0$, they are different if and only if $u_i \neq u_{i+1}$. So a pair of adjacent cells of which exactly one, the cell j, has $m_j = 1$ contributes $-(1 - u_j)$ to Q_1, and collecting, for each j with $m_j = 1$, the terms that contain u_j gives

$$Q_1(c) = \#\{i : m_i = 0\} + \sum_{j:\, m_j = 1} (1 - e_j)(1 - u_j) - \#\{i :\ m_i = m_{i+1} = 0,\ u_i \neq u_{i+1}\},$$

where $e_j \in \{0, 1, 2\}$ is the number of zeros among m_{j-1} and m_{j+1}; in the periodic case the indices are taken over one period, and for a finite perturbation of b_{11} all three terms are finite. The same formula with v in place of u gives $Q_1(F(c))$. The first term does not contain u. In the

second, $v_j \neq u_j$ only if $e_j = 1$, by (α), and then the coefficient $1 - e_j$ vanishes. In the third, if $m_i = m_{i+1} = 0$ then $s_i = s_{i+1} = 1$ by (β), so $v_i \neq v_{i+1}$ if and only if $u_i \neq u_{i+1}$. Hence $Q_1(F(c)) = Q_1(c)$. □

For 94 and 95 the quantity Q_1 is not conserved (Appendix A.2).

7.1. **Responses to probes.** Let $f \in \{91, 94\}$. An automorphism h of X_{F_f} is determined by its restrictions to Φ^+ and Φ^- (Corollary 3.2), and the following remark relates the restriction to Φ^- of an automorphism that is the identity on Φ^+ to the scattering of probes. For 94 the group is determined by Theorem 5.9, and nothing in the other sections depends on this subsection; we state the results for both rules because the proofs are the same.

Remark 7.7 (responses). Let $h \in \mathrm{Aut}(X_{F_f})$ fix every point of Φ^+, let X be a Φ^- object and P a probe for X. By Lemma 4.5, $\mathcal{S}(P, hX) = \mathcal{S}(hP, hX) = (h\Gamma_P(X), \Omega_X(P))$. Here hX is again a Φ^- object with the tails of X, by (6) and because h fixes the backgrounds, which lie in Φ^+; so P is a probe for hX. So hX has the same responses as X, and h commutes with the maps Γ_P. The automorphism σF is of this kind, and it acts on Φ^- as σ^2. A description of the objects that have the same responses as a given one therefore bears on $\mathrm{Aut}(X_{F_{91}})$. Two statements in this direction are proved below. Proposition 7.8 shows that no finite family of probes distinguishes all the objects that differ in the length of a stretch of the background inside their cores. Corollary 7.9 concerns the following statement.

(Sh) Every Φ^- object X with $\operatorname{supp} X \geq 1$ has a probe P with $\operatorname{supp} \Gamma_P X < \operatorname{supp} X$.

If (Sh) holds and h in addition fixes the bare kinks of Φ^-, then $h = \mathrm{id}$: by Corollary 7.9 every object X with $\operatorname{supp} X \geq 1$ has a probe P with $\Gamma_P(X)$ a bare kink, so $\mathcal{S}(P, hX) = (\Gamma_P(X), \Omega_X(P)) = \mathcal{S}(P, X)$ and $hX = X$, since $\mathcal{S}$ is injective (Proposition 4.2); every word of Φ_f is a word of an object, so h fixes every point of Φ^-, as in the proof of (C0) in Lemma 5.2, and $h = \mathrm{id}$ by Theorem 3.1. In a search over probes (computer-assisted; Appendix A.2), (Sh) holds for 94 for every object with core of length at most 23, with probes with core of length at most 24, and for 91 for every object with core of length at most 16, with probes with core of length at most 20. We do not know whether (Sh) holds for 91 or for 94.

Proposition 7.8 (gap pumping). *Let $f \in \{91, 94, 95\}$ and let β be a valid background. Let $s \in \mathbb{Z}$, let u be a sequence on $(-\infty, s)$ and v a sequence on $[s, \infty)$, both eventually 2-periodic, and for $g \geq 0$ let*

$$\xi_g(p) = u(p) \quad (p < s), \qquad \xi_g(p) = \beta_{p \bmod 2} \quad (s \leq p < s + g), \qquad \xi_g(p) = v(p - g) \quad (p \geq s + g).$$

Let $\mathcal{I}$ be the set of $g \geq 2$ with $\xi_g \in \Phi_f$, and for $g \in \mathcal{I}$ let $X_g \in \Phi^-$ be the point with diagram $\xi_g(p)$; so $X_g = \cdots U\, w_g\, V \cdots$, where w_g is a word of length g of the background β between fixed words U and V. Let $\mathcal{P} \subset \Phi^+$ be finite, and suppose that for every $P \in \mathcal{P}$ there is a corner $[q_0, p_0]$ with $p_0 + 1 < s$ at which (P, X_g) is a placed pair; this condition does not depend on g. Then there is g_0 such that for all $g \geq g_0$ with $g, g + 2 \in \mathcal{I}$ and all $P \in \mathcal{P}$, writing $\mathcal{S}(P, X_g) = (\Gamma_g, \Omega_g)$ and γ_g for the diagram of Γ_g,

$$\Omega_{g+2} = \Omega_g, \qquad \gamma_{g+2}(p) = \gamma_g(p) \quad (p < s + g), \qquad \gamma_{g+2}(p) = \gamma_g(p - 2) \quad (p \geq p_2),$$

for some $p_2 < s + g$ that depends only on P. In particular, for probes, X_g and X_{g+2} have the same responses to every probe in $\mathcal{P}$.

Proof. It suffices to treat one $P \in \mathcal{P}$, with diagram $\pi(q)$ and corner $[q_0, p_0]$, and to take the largest of the resulting g_0. We may assume $\mathcal{I} \neq \emptyset$. Let $\xi_\infty(p) = u(p)$ for $p < s$ and $\xi_\infty(p) = \beta_{p \bmod 2}$ for $p \geq s$. Every triple of ξ_∞ is a triple of ξ_g for any $g \in \mathcal{I}$ or a triple of the background β, so $\xi_\infty \in \Phi_f$; let $X_\infty \in \Phi^-$ be the point with diagram $\xi_\infty(p)$. Then (P, X_∞) is a placed pair with

corner $[q_0, p_0]$, whose right tail is β. Let Y_∞ and Y_g be the diagrams of (P, X_∞) and (P, X_g). By Lemma 4.1(i) there is b^* such that every firing of Y_∞ is at a centre $[q', p']$ with $p' \le b^*$.

Causality. Since $\xi_g = \xi_\infty$ on $(-\infty, s+g)$, Goursat causality gives $Y_g = Y_\infty$ on $G \cap \{p < s+g\}$, and outside G the two diagrams are given by the same values of π, ξ_g and ξ_∞ there. So $Y_g = Y_\infty$ on $\{p < s+g\}$.

A strip without firings. Let $p_1 = \max(s, b^*+1)$ and $[q,p] \in G$ with $p \ge p_1$. By Lemma 3.3,

$$Y_\infty[q,p] = \pi(q) \oplus \beta_{p \bmod 2} \oplus \nu \oplus n(q),$$

where ν depends only on q mod 2 and $n(q)$ is the parity of the number of firings at centres $[q', p']$ with $q' < q$ and $q' \not\equiv q$; it does not depend on p, because every firing has $p' \le b^* < p$. Points with $q < q_0$ and $p \ge s$ carry $\beta_{p \bmod 2}$. Hence $Y_\infty[q, p+2] = Y_\infty[q,p]$ whenever $p \ge p_1$, and by causality $Y_g[q, p+2] = Y_g[q,p]$ whenever $p \ge p_1$ and $p+2 < s+g$. In this strip Y_g agrees with Y_∞, which does not fire at centres with $p' > b^*$, and it is 2-periodic along the rays $q = \text{const}$.

Translation by 2. Choose $p_2 \ge p_1 + 2$ with $p_2 \equiv q_0 \pmod 2$, let $g_0 = p_2 + 2 - s$, and let $g \ge g_0$ with $g, g+2 \in \mathcal{I}$; so $p_2 + 1 < s + g$. The function $Z[q,p] = Y_g[q, p-2]$ is a diagram, since $[q,p] \mapsto [q, p+2]$ is the translation $(t,i) \mapsto (t+1, i+1)$. On the quadrant $G_1 = \{q \ge q_0,\ p \ge p_2\}$ both Z and Y_{g+2} satisfy (4). They have the same values on its two boundary rays. On the ray $p \in \{p_2, p_2+1\}$, $q \ge q_0$: here $p < s+g$ and $p - 2 \ge p_1$, so $Y_{g+2}[q,p] = Y_\infty[q,p] = Y_\infty[q, p-2] = Y_g[q, p-2]$. On the ray $q \in \{q_0, q_0+1\}$, $p \ge p_2$: here $Y_{g+2}[q,p] = \xi_{g+2}(p)$ and $Z[q,p] = \xi_g(p-2)$, and $\xi_{g+2}(p) = \xi_g(p-2)$ for $p \ge p_2$, since both are $v(p-g-2)$ when $p \ge s+g+2$ and both are $\beta_{p \bmod 2}$ when $s+2 \le p < s+g+2$. Hence $Y_{g+2} = Z$ on G_1, and also on $\{q < q_0,\ p \ge p_2\}$, where both are $\xi_{g+2}(p) = \xi_g(p-2)$. So

$$Y_{g+2}[q,p] = Y_g[q, p-2] \qquad (p \ge p_2).$$

Reading o on a half-plane $\{p \ge p_1'\}$ with p_1' large (Lemma 4.1(ii)) gives $\Omega_{g+2} = \Omega_g$. Reading γ on a half-plane $\{q \ge q_1\}$ gives $\gamma_{g+2}(p) = \gamma_g(p-2)$ for $p \ge p_2$, and $\gamma_{g+2}(p) = \gamma_g(p)$ for $p < s+g$ by causality. □

The proposition concerns stretches of the background inside the core, between fixed words U and V; then $\operatorname{supp} X_g$ grows with g, and the X_g are pairwise distinct modulo σ^2. (If the stretch is part of a tail, X_{g+2} is X_g or $\sigma^{-2}X_g$, and the statement is immediate.) It shows that no finite family of probes separates all pairs (X_g, X_{g+2}), but it gives no bound on g_0 in terms of the probes, and so no bound on the length of a separating probe in terms of g. Since moving a probe by two steps along its rays changes its responses only by a translation (Lemma 4.1(iii)), the probes on a given background of support at most ℓ give finitely many responses up to translation. Hence, for every ℓ and every family (X_g) as in the proposition with the stretch inside the core, the pairs (X_g, X_{g+2}) with g large have the same responses to all probes of support at most ℓ: an argument that separates objects by their responses must use probes of unbounded support. Proposition 7.8 holds for 91, 94 and 95 alike.

The statement (Sh) can be restated with a single probe.

Corollary 7.9. *Let $f \in \{91, 94\}$. The statement (Sh) holds if and only if every Φ^- object X with $\operatorname{supp} X \ge 1$ has a probe P such that $\Gamma_P(X)$ is a bare kink.*

Proof. If $\Gamma_P(X)$ is a bare kink, then $\operatorname{supp} \Gamma_P(X) = 0 < \operatorname{supp} X$. Conversely, assume (Sh), and let $X_0 = X$. As long as $\operatorname{supp} X_k \ge 1$, choose a probe P_{k+1} for X_k with $\operatorname{supp} X_{k+1} < \operatorname{supp} X_k$, where $X_{k+1} = \Gamma_{P_{k+1}}(X_k)$. By Lemma 4.1(ii), Γ_P preserves the left tail, so X_{k+1} has the left tail λ of X, and the probes for X_{k+1} are again the links of Φ^+ with both tails λ. The process stops at some X_n with $\operatorname{supp} X_n = 0$. Since $\operatorname{supp} X \ge 1$, X is not a background, and by Corollary 4.3(2) and induction on k no X_k is a background; so X_n is a bare kink. By Lemma 4.1(iii) we may move each P_{k+1} to the left by an even number of steps without changing X_{k+1}, and then Lemma 4.1(iv),

applied $n-1$ times, shows that the single probe $P = P_n * \cdots * P_1$, the composite of these probes, satisfies $\Gamma_P(X) = X_n$. □

8. Rule 155R: where the method stops

The method of Section 3 fails for the class of 155, for which Φ_f also has positive entropy; the automorphism group of this class is not known [14]. For $f = 155$ we have $f(a,b,c) = a \oplus c \oplus N(a,b,c)$ with N the indicator of $\{000, 110, 111\}$; so Φ_{155} is the set of sequences with none of these triples, and, since 11 can be followed by neither letter, no point of Φ_{155} contains 11. With the definitions above, the valid backgrounds are 01 and 10.

Proposition 8.1 (the edge bit of 155R)**.** *Let $f = 155$, let β be a valid background, $s \in \mathbb{Z}$, and let c agree with b_β on $(-\infty, s)$. Then $Y_c(t, s-t) = Y_c(0,s)$ for all $t \geq 0$. If moreover $\beta_{(s-1) \bmod 2} = 1$ and $Y_c(0,s) = 1$, then Y_c fires at the centre $(t, s-t)$ for every $t \geq 0$, and $F^t(c)_{s-t} = (1,1)$ for every $t \geq 1$, while no point of $\Phi^+ \cup \Phi^-$ has a cell $(1,1)$. In particular the conclusion of Proposition 3.6 fails for $f = 155$ on a dense set of finite perturbations of the valid backgrounds: for each of them and for every $R \geq 0$ and every T there are $t \geq T$ and i such that $F^t(c)_{[i-R,i+R]}$ is not the window at the same position of any point of $\Phi^+ \cup \Phi^-$.*

Proof. Write $Y = Y_c$. For $t \geq 0$ and $i < s - t$, $Y(t,i)$ depends only on $c_{[i-t,i+t]}$, so $Y(t,i) = \beta_{(t+i) \bmod 2}$. The relation (3) at the centre $(t, s-t-1)$ reads

$$Y(t+1, s-t-1) = Y(t-1, s-t-1) \oplus Y(t, s-t-2) \oplus Y(t, s-t) \\ \oplus N\big(Y(t, s-t-2), Y(t, s-t-1), Y(t, s-t)\big).$$

The first two terms on the right are background values at points with $t + i \equiv s$, so they cancel, and the first two arguments of N are $(\beta_{s \bmod 2}, \beta_{(s+1) \bmod 2})$, which is 01 or 10; $N(0,1,\cdot) = N(1,0,\cdot) = 0$. So $Y(t+1, s-(t+1)) = Y(t, s-t)$, and induction gives the first statement. Under the further hypotheses the triple at the centre $(t, s-t)$ is $(Y(t, s-t-1), Y(t, s-t), \cdot) = (\beta_{(s-1) \bmod 2}, 1, \cdot) = (1,1,\cdot)$, which fires, and for $t \geq 1$ the cell $F^t(c)_{s-t} = (Y(t-1, s-t), Y(t, s-t))$ is $(\beta_{(s-1) \bmod 2}, 1) = (1,1)$. The points of $\Phi^\pm$ have cells $(\varphi(i \pm 1), \varphi(i))$ with $\varphi \in \Phi_{155}$ (Lemma 2.4), and φ contains no 11. For every $t \geq 1$, every window of radius $R \geq 0$ of $F^t(c)$ at $s - t$ contains the cell $(1,1)$; so for every T, with $t = \max(T, 1)$ and $i = s - t$, the window $F^t(c)_{[i-R,i+R]}$ is not a window of $\Phi^+ \cup \Phi^-$. Given a finite perturbation of b_β and a finite set of cells, choosing s to the left of these cells and of every cell where it differs from b_β, with $\beta_{(s-1) \bmod 2} = 1$ and changing only c_s gives such a configuration that agrees with the given one on the set; so these configurations are dense. □

Remark 8.2 (155R). So Lemma 3.5 and Proposition 3.6 are false for 155: a front carrying a fixed pair 11 fires at every step, and the windows it leaves behind are not windows of $\Phi^+ \cup \Phi^-$. The method of Section 3 cannot give Lemma L for 155. Lemma L itself, the statement of Theorem 3.1 with $f = 155$, is not refuted, since a configuration that fails to separate may still be fixed by every h that fixes $\Phi^+ \cup \Phi^-$. For $R \leq 6$ it holds (computer-assisted): the window closure of Appendix A.2.1, with the windows of $\Phi^+ \cup \Phi^-$ as base, forces every window of radius $R \leq 6$. In the base, words of Φ_{155} must be pruned to those that extend to bi-infinite points, because 011 is an allowed triple that cannot be continued. The firing of such a front is confined to the left of the perturbation: on the right the rule is transparent (Proposition 8.3 below).

Proposition 8.3 (right transparency of 155R)**.** *Let $f = 155$, $w \in \mathbb{Z}$, and let c agree on $[w, \infty)$ with a point of Φ^+. Put $Y = Y_c$ and $E(t) = Y(t, w)$ for $t \geq -1$. Then for every $t \geq 0$:*

(a) *$Y(t,i) = Y(t-1, i-1)$ for every $i \geq w + 1$;*
(b) *Y does not fire at any centre (t,i) with $i \geq w + 1$;*

(c) $E(t+1) = 0$ *if* $E(t) = 1$; $E(t+1) = 1$ *if* $E(t-1) = E(t) = 0$; *and* $E(t+1) = Y(t, w-1)$ *if* $E(t-1) = 1$ *and* $E(t) = 0$;
(d) $F^t(c)$ *agrees on* $[w, \infty)$ *with a point of* Φ^+; *in particular, for every* $R \geq 0$ *and every* $i \geq w + R$, *the window* $F^t(c)_{[i-R,i+R]}$ *is the window at the same position of a point of* Φ^+.

Proof. A sequence on $\mathbb{Z}$ lies in Φ_{155} if and only if it contains neither 11 nor 000: the triples 110 and 111 contain 11, and a bi-infinite sequence that contains 11 contains 110 or 111. Let $\varphi \in \Phi_{155}$ be the sequence of the given point of Φ^+, so that $c_i = (\varphi(i+1), \varphi(i))$ for $i \geq w$ (Lemma 2.4). Define u on $\mathbb{Z}$ by $u(k) = \varphi(k)$ for $k \geq w+1$ and $u(w-t) = E(t)$ for $t \geq -1$; the two definitions agree at $k = w+1$, since $E(-1) = Y(-1, w) = \varphi(w+1)$, and $u(w) = E(0) = \varphi(w)$. We show by induction on $t \geq 0$ that

(P_t) $Y(\tau, i) = u(i - \tau)$ for $-1 \leq \tau \leq t$ and $i \geq w$, and
(Q_t) the word $u|_{[w-t,\infty)}$ contains neither 11 nor 000.

(P_0) is the hypothesis on c, and (Q_0) holds because $\varphi \in \Phi_{155}$. Assume (P_t) and (Q_t). For $i \geq w+1$ the relation (3) at the centre (t, i) and (P_t) give

$$Y(t+1, i) = u(i-t+1) \oplus u(i-t-1) \oplus u(i-t+1) \oplus N\big(u(i-t-1), u(i-t), u(i-t+1)\big).$$

The triple starts at $i-t-1 \geq w-t$, so by (Q_t) it is not in the support of N; hence Y does not fire at (t, i), which is (b), and $Y(t+1, i) = u(i-t-1)$. Together with $Y(t+1, w) = E(t+1) = u(w-t-1)$ this is (P_{t+1}). At the centre (t, w) put $a = Y(t, w-1)$; by (P_t), $Y(t-1, w) = E(t-1)$ and $Y(t, w+1) = u(w-t+1) = E(t-1)$, so (3) gives

$$E(t+1) = a \oplus N\big(a, E(t), E(t-1)\big).$$

If $E(t) = 1$, then $N(a, 1, \cdot) = a$ and $E(t+1) = 0$. If $E(t) = 0$, then $N(a, 0, x) = 1$ exactly when $a = x = 0$, so $E(t+1) = 1$ if $E(t-1) = 0$ and $E(t+1) = a$ if $E(t-1) = 1$. This is (c). The only triple of $u|_{[w-t-1,\infty)}$ that is not a triple of $u|_{[w-t,\infty)}$ is $E(t+1)E(t)E(t-1)$, and by (c) neither $E(t+1)E(t) = 11$ nor $E(t+1)E(t)E(t-1) = 000$; this is (Q_{t+1}). Statement (a) is $u(i-t) = u((i-1)-(t-1))$ for $i - 1 \geq w$.

For (d), a word on $[k, \infty)$ with neither 11 nor 000 extends to a sequence on $\mathbb{Z}$ with neither, by putting $\cdots 0101$ in front of it if it begins with 0 and $\cdots 1010$ if it begins with 1. Let $\varphi' \in \Phi_{155}$ be such an extension of $u|_{[w-t,\infty)}$, and let $d \in \Phi^+$ be the point with diagram $\varphi'(i-\tau)$. By (P_t), $F^t(d)_i = (\varphi'(i-t+1), \varphi'(i-t)) = (Y(t-1, i), Y(t, i)) = F^t(c)_i$ for $i \geq w$, and $F^t(d) \in \Phi^+$. A window centred at $i \geq w + R$ lies in $[w, \infty)$. □

Corollary 8.4. *Let* $f = 155$.

(1) *If* c *agrees on* $[w, \infty)$ *with a point of* Φ^-, *then for every* $t \leq 0$ *the configuration* $F^t(c)$ *agrees on* $[w, \infty)$ *with a point of* Φ^-, *and* Y_c *does not fire at any centre* (t, i) *with* $t \leq -1$ *and* $i \geq w+1$.
(2) *Let* c *be a finite perturbation of a valid background* β *that differs from* b_β *only in cells of* $[0, w)$. *Then* Y_c *fires only at centres* (t, i) *with* $i \leq w$, *for all* $t \in \mathbb{Z}$. *If* h *is a block map of radius* R *that commutes with* F *and fixes every point of* $\Phi^+ \cup \Phi^-$, *then* $Y_{h(c)}(t, i) = Y_c(t, i)$ *for all* $t \in \mathbb{Z}$ *and* $i \geq w + R$.

Proof. (1) The time reversal $\rho(x, y) = (y, x)$ satisfies $\rho F \rho = F^{-1}$, and $Y_{\rho(c)}(t, i) = Y_c(-1-t, i)$. By Lemma 2.4, ρ maps Φ^- onto Φ^+. So $\rho(c)$ agrees on $[w, \infty)$ with a point of Φ^+, and Proposition 8.3(b),(d) for $\rho(c)$ give the claim, since $F^{-t}(c) = \rho F^t(\rho(c))$ and Y_c fires at $(-1-t, i)$ exactly when $Y_{\rho(c)}$ fires at (t, i), the relation (3) being symmetric in $t \pm 1$.

(2) The diagram $\beta_{(t+i) \bmod 2} = \beta_{(i-t) \bmod 2}$ of b_β is of both forms $\varphi(i-t)$ and $\varphi(i+t)$, so $b_\beta \in \Phi^+ \cap \Phi^-$, and c agrees with it on $[w, \infty)$. Proposition 8.3(b) excludes firings at $i \geq w+1$

for $t \geq 0$, and (1) for $t \leq -1$. For $t \geq 0$, by Proposition 8.3(d), $F^t(c)$ agrees on $[w, \infty)$ with some $e \in \Phi^+$, so for $i \geq w + R$, $F^t(h(c))_i = h(F^t(c))_i = h(e)_i = e_i = F^t(c)_i$. For $t \leq 0$ the same argument applies with (1) and Φ^-, since h also commutes with F^{-1}. The y-bits of these cells are the values of the two diagrams at (t, i). □

Proposition 8.3 and Proposition 8.1 show an asymmetry between the two sides of a perturbation. To the left, a front carrying a fixed pair 11 fires at every step; to the right, the medium Φ^+ is transparent: nothing fires, the profile is transported at light speed, and every block map that fixes $\Phi^\pm$ acts as the identity there (Corollary 8.4(2)). For 91 and 94 the reflection $i \mapsto -i$ commutes with F, because N is invariant under reversal of the triple, and Proposition 3.7 uses it to pass from one side to the other. For 155 the reversal of $\operatorname{supp} N = \{000, 110, 111\}$ is $\{000, 011, 111\}$, so there is no such symmetry. Lemma L for 155 is therefore a question about the left fronts only. Appendix B describes the eventually periodic left fronts (Lemma B.1) and formulates a hypothesis on them that Lemma L implies (Remark B.2).

9. Questions

(1) Is every block map commuting with F_{95} a constant or of the form $\widehat{H_{\varphi,\psi}}\sigma^n$ with block maps φ, ψ of the golden mean shift that conserve the number of particles (Remark 6.12(1))? This holds for the block maps of radius at most 4, and for those of sublattice type of radius 5 (Proposition 6.21).

(2) Describe the group $\mathcal{N}$ of Section 6: generators, abelianisation, centre; is it finitely generated, and does it contain a free group? By Lemma 6.6 it is the stabiliser in $\mathrm{Aut}(\mathsf{G}, \sigma)$ of the class of the function $x \mapsto x_0$ modulo continuous coboundaries. (The boxes of Lemma 6.8 carry one particle each, which suggests that the marker constructions of [6] for free products of finite groups can be adapted to $\mathcal{N}$; this is a suggestion and not a claim.)

(3) Determine $\mathrm{Aut}(X_{F_{91}})$. Lemma L holds at every radius; what is missing is a normalisation of the automorphisms as in Theorem 4.9, which has to use points that are not 3-sparse (Proposition 7.4 and Remark 7.5), and a proof that a normalised automorphism fixes every point of $\Phi^+ \cup \Phi^-$.

(4) Does Lemma L hold for 155 at every radius? It would follow from the front hypothesis (Fr) and the closure (Cl) of Remark B.2.

(5) Is the class of 95 the only class of elementary rules that are not affine for which F_f has no blocking word and $\mathrm{Aut}(X_{F_f})$ is not virtually $\mathbb{Z}^2$? This is Question 1.6 of [13]; by the classification of [14] it concerns the classes whose automorphism group is not known. For the class of 175, whose rules also do not read their middle argument, see Remark 6.22.

Appendix A. Computations

This appendix describes the finite computations. Every result marked as computer-assisted in the text is the output of a terminating exact search over a finite set; where random sampling is used, this is said, and such runs serve only as checks. The directory `code/` contains the programs, with instructions for running them and their expected outputs, and the logs of the runs; the programs, or their directories, are named below next to each computation. Solution tables and other data files of more than 5 MB are not included; the programs that produce them are. Section A.1 describes the census of Lemma 6.20, Section A.2 the other computations whose results are stated in the text without a proof by hand, and Section A.3 computational confirmations of results that are proved by hand.

A.1. **The census for rule 95.** Let g be a block map of radius R with local rule γ, and $F = F_{95}$. Both $g \circ F$ and $F \circ g$ are block maps of radius $R+1$, so they are equal if and only if their local rules agree on every window $w \in \mathcal{A}^{2R+3}$. The unknowns are the $2 \cdot 4^{2R+1}$ bits of the table of γ. For a window w let e_1, e_2, e_3 be its three subwords of length $2R+1$, and e_L the word of length $2R+1$ obtained by applying the local rule of F to w; writing X_e, Y_e for the two bits of the table entry e, the window contributes the two constraints

$$X_{e_L} = Y_{e_2}, \qquad Y_{e_L} \oplus X_{e_2} = f(Y_{e_1}, Y_{e_2}, Y_{e_3}),$$

which are encoded as clauses by their truth tables. All solutions are enumerated by repeated solving with blocking clauses, until the instance is unsatisfiable, using PySAT [28] (`code/census/centralizer.py` for $R \le 3$, `enum_r4.py` for $R = 4$). Each enumeration is carried out twice, with the solvers CaDiCaL 1.5.3 [2] and MiniSat 2.2 [16], and for every $R \le 4$ the two solvers return the same set of tables. Every table returned by the solver is checked again, independently of the clauses, by evaluating $g \circ F$ and $F \circ g$ on all 4^{2R+3} windows. It is then compared with the tables of the maps $\sigma^a F^b$ with $|a| + |b| \le R$, which are built by composing local rules, and with the constant tables; the program checks that every $\sigma^a F^b$ with $|a| + |b| \le R$ occurs among the solutions. The instances have 128, 2048, 32768 and 524288 variables and 17542, 292998, 4714182 and 75486342 clauses for $R = 1, 2, 3, 4$, and 6, 14, 46 and 262 solutions: 5, 13, 25 and 41 maps $\sigma^a F^b$, one constant table, and 0, 0, 20 and 220 others (the four logs `census_95_R123_*.txt` and `census_95_R4_*.txt` in `code/census/logs`, one for each solver). That an enumeration is complete rests on the final answer of the solver that the instance with all blocking clauses is unsatisfiable. These answers are not certified by proof logs; the evidence for them is the agreement of the two solvers, and, for $R \le 4$, the agreement with the search of Section A.2.4 over tables of sublattice type, which has a different encoding. The same holds for the other searches with a SAT solver in this appendix. The solutions themselves do not depend on the solver, since each is verified directly. For $R \le 3$ the program also decides the injectivity of each solution on the graph of pairs of words: the injective ones are the maps $\sigma^a F^b$. This test is not used in a proof here, since the further maps are identified in Proposition 6.21(a). The same programs are applied to all 88 classes in [14].

A.2. **Computer-assisted statements.** The following results are stated in the text as computations. For each we give the method in brief.

A.2.1. *Rule* 155*: window closures and front strips.* (Remarks 8.2 and B.2, Lemma B.1.) The *window closure* of a rule is a monotone closure on the set of windows $w \in \mathcal{A}^{2R+1}$ at which a block map h commuting with F is known to act as the identity. It starts from the windows of configurations that h fixes by hypothesis, and it uses the locally permutive readings of the spacetime relation for $\Delta = Y_{hc} \oplus Y_c$: a window is added if, for some extension of it by two cells on each side and one of the five instances of the relation through the centre, the relation has no solution with $\Delta = 1$ at the centre and $\Delta = 0$ at those of its other four points whose windows are already in the set, the remaining values being arbitrary; this includes the instances in which the rule does not depend on an argument in the given context. Words of Φ_f are admitted to the base if they extend by six letters on each side. For 155 the base consists of the windows of $\Phi^+ \cup \Phi^-$, and the admitted words are those that extend to bi-infinite points; the closure is complete for $R \le 6$ (`code/closure/dc.c` for $R \le 5$; `code/rule155/mkbase.py` and `lc.c` for $R = 6$, with the log `lc155_R6.log`). The weaker closure of Remark B.2 uses only the reading that determines the value at the point S (`sonly.py` for $R = 2, 3$; `mkseed.py` and `sc.c` for $R = 4$). With the base $\Phi^+ \cup \Phi^-$ alone it leaves 968, 16140 and 261168 windows for $R = 2, 3, 4$. It is complete for $2 \le R \le 4$ when the base also contains the windows of the periodic front strips of both kinds of front, centred at the offsets $-R \le s \le R$ from the edge ($-4 \le s \le 5$ for $R = 4$), and of their time

reversals; for $R = 2$ and offsets $-2 \le s \le 1$ it leaves 875 windows. The periodic continuations of Lemma B.1(i) are enumerated line by line from (18), by classifying each line as slaved, free or doubling (`strip.py`; `s4.py` for the 00-front at width 18).

A.2.2. *Rule* 91. (Remark 7.5 and Lemma 7.6.) The objects are the points of Φ^- on the background 11 whose core, a word of Φ_{91} that begins and ends with 0, has length at most 12; there are 778. For each of them the scattering of a bare defect Δ_a^+ off it is computed at both relative parities, and the object counts as crossed unchanged if $\mathcal{S}(\Delta_a^+, X) = (X, \Delta_a^+)$ at both. Two programs do this (`code/rule91/t91b.py` and `t91c.py`), one with the recurrence (4) and one by direct simulation of the cells in (t, i), in which the configuration after the crossing is compared with the free superposition $\pi(q) \oplus \xi(p) \oplus 1$; both find that the objects crossed unchanged are exactly the 41 whose core is 3-sparse. The two example crossings with the object $1^\infty 001^\infty$ at the end of the first paragraph of the remark are computed with the recurrence (4) (`s91.py`). For the conserved quantities, the additive conserved quantities of F_f with a density h on r adjacent cells are the solutions of the linear system $h(F(w)) - h(w_1 \cdots w_r) = g(w_1 \cdots w_{r+1}) - g(w_0 \cdots w_r)$, $w \in \mathcal{A}^{r+2}$, in the unknowns h and g, where $F(w)$ is the image of the r inner cells; the system is solved exactly over $\mathbb{Q}$ (`cq.py`, `cq_check.py`). That the densities of Q_1 and Q_2 solve it for $r = 2$ is an exact rank computation, for Q_2 with $f \in \{91, 94, 95\}$ and for Q_1 with $f = 91$; for 94 and 95 the density of Q_1 does not solve it. Both quantities are also evaluated on c and $F(c)$ for all configurations c of spatial period at most 7 (`q91.py`). These quantities are known [43, 44]; the computation is a check of the formulas in the present notation. The identity in the proof of Lemma 7.6 is checked on all cases as well: the regrouping of the density, the cases (α) and (β), and the invariance of each term (`q1_identity.py`).

A.2.3. *The statement* (Sh). (Remark 7.7; program in `code/sh`.) All scatterings are computed by direct simulation in (t, i). For each object with a core of the stated length and each pair of tails, the probes with cores up to the stated length are searched for one that shortens the object (`sh.c`; 452060 and 793310 probes with cores of length at most 24 on the two backgrounds of 94, and 167914 and 203804 with cores of length at most 20 on the backgrounds 01 and 11 of 91, each core being counted at both of its placements modulo 2; the background 10 of 91 is the image of 01 under the shift and has the same number of probes; log `logs/sh_all.log`). For an object with two different tails only those probes are used that send the two bare kinks with these tails to bare kinks, so the search establishes more than (Sh) in the stated range.

A.2.4. *Rule* 95*: the smallest radius and the commuting maps of small radius.* (Lemma 6.14, Propositions 6.15 and 6.21; the programs, the tables and the logs are in `code/rule95`.) *Particle-conserving block maps.* For $w \le 8$ the functions λ of the criterion after Lemma 6.14 are enumerated by a depth-first search over the vertices of B_w in breadth-first order from 0^{w-1}; a value $\lambda(v)$ has at most two candidates once a neighbour of v is assigned, and every edge and every pair of consecutive edges is checked as soon as its vertices are assigned. The program `enum_golden.py` gives the numbers of Table 1 (log `logs/enum_golden_8.log`). For $w \le 7$ a second computation (`enum_golden_linear.py`) imposes conservation as the linear equations $\sum_\epsilon m_\epsilon(z)\hat{\varphi}(\epsilon) = |z|$ over all cyclic words z of G of length at most the number of vertices, where the sum is over the edges ϵ of B_w and $m_\epsilon(z)$ counts the occurrences of ϵ in z (these include all simple cycles of B_w), reduces them over $\mathbb{Q}$, and tries all 0/1 values of the free coordinates (2^{20} for $w = 7$); it gives the same sets, and the numbers for $w = 7, 8$ are obtained again by the SAT search below. Injectivity is decided on the graph whose vertices are the pairs of vertices of B_w and whose edges are the pairs of edges with equal values of $\hat{\varphi}$: the map is injective if and only if no vertex off the diagonal lies on a bi-infinite path, that is, survives the repeated removal of vertices without a successor or without a predecessor. For each of the 469 maps with $w = 7$ and the 13506 maps with $w = 8$

that are not injective, two distinct points of G of the same period, at most 10, with the same image are found as well (`inj_witness.py`).

The automorphism of radius 7. The map $\widehat{H_{\varphi,\psi}}\sigma^n$ is evaluated on a finite perturbation c of b_{11} from the properties that characterise it: F^{-t} is applied until the configuration consists of a point of Φ^+ on the left and a point of Φ^- on the right, separated by at least 64 cells $(1,1)$; there φ is applied to the two classes of zeros of the left part and ψ to those of the right part, as in (15); then F^t and σ^n are applied. With $(\varphi,\psi) = (\sigma,\mathrm{id})$, (id,σ) and (σ^{-1},σ) this reproduces $\sigma^{-1}F$, σF and σ^2 on 300 pseudo-random finite perturbations each (the self-test of `core.py`). The tables T_x, T_y of Proposition 6.15(b) are filled by evaluating h_* on configurations in which the two classes of bits run through a binary de Bruijn sequence of order 15, so that every entry is hit, and every sample is required to be consistent with the entries already filled (`gen7.py`). The table of h_* found by the SAT search of the paragraph on tables of sublattice type below is identical. The four facts of the proof are then checked on the tables alone: the commutation on $2 \cdot 2^{17}$ windows and the involution on $2 \cdot 2^{29}$ windows by a program that evaluates F and the table map cell by cell, with the bits of the other class drawn at random, and in addition on 2^{26} windows with all bits random (`verify7.c`; logs `verify7.log` and `verify7_negative_control.log` in `code/rule95/logs`); the restrictions on the 3025 words of length 16 of Φ_{95}; and the dependence by comparing every entry with the 15 entries at Hamming distance 1 (`props7.py`). With one table entry complemented, the commutation check reports 12 mismatches and the involution check 49152. The commutation and the involution are also confirmed on all configurations of spatial period at most 9; σh_* moves 310 of those of period 5 and 7168 of those of period 7, and for each of these $h_*(c)$ is not in the orbit of c under $\langle \sigma, F\rangle$ (`props7.py`; `standalone7.py` for the configuration of period 5 in the proof).

Tables of sublattice type. For a radius R and each of the two ways in which the two bits of the image can be assigned to the two classes, the unknowns are the $2 \cdot 2^{2R+1}$ entries of two tables, and $hF = Fh$ at the cell 0 consists of 2^{2R+3} equalities between two entries and 2^{2R+3} relations $a = b \oplus 1 \oplus cd$ between four entries. For $R \le 5$ all solutions are enumerated with a SAT solver and blocking clauses (`sat_sublattice.py`; CaDiCaL 1.5.3, and a second time with MiniSat 2.2, with the same sets of solutions); the numbers of solutions are $2+5$, $10+5$, $10+37$, $226+37$ and $226+2917$ for $R = 1,\dots,5$, the constant map being counted in both types, which for $R \le 4$ are the totals 6, 14, 46, 262 of the census of Lemma 6.20, made without this restriction. For $R \le 7$ the restrictions to Φ^+ that occur among the solutions are enumerated by blocking on the entries at windows of Φ^+ (at most 110 variables), likewise for Φ^-, the injective ones are selected by the test above, and for every pair of an injective restriction to Φ^+ and one to Φ^- all solutions with these restrictions are enumerated. An automorphism of sublattice type is among them, since its restrictions to $\Phi^\pm$ are injective. The numbers of restrictions to Φ^+ (and to Φ^-) that occur are $\mathsf{m}_w + 1$, with m_w from Table 1 and w as in Lemma 6.13(ii), the additional one belonging to the constant map: 477 for the type without exchange of the classes at $R = 6$ and $R = 7$, and 55 and 13517 for the type with exchange; the injective ones number 7, 7, 6 and 10, and the class maps of all of them satisfy the criterion after Lemma 6.14. This reproduces the last two columns of Table 1 by a different method. For $R \le 6$ the solutions with injective restrictions are the $2R^2 + 2R + 1$ maps $\sigma^a F^b$; for $R = 7$ they are the 113 maps $\sigma^a F^b$ and 36 others, all of the type with exchange. The tables of the 36 maps of Proposition 6.15(c) are also generated from the definition, as for h_* (`gen36.py`), and verified as in the proof of that proposition (`verify36.py`: commutation on all $2 \cdot 2^{17}$ windows, restrictions on all words, dependence on the cells $i \pm 7$; `sat_post.py` writes the tables found by the search); the two sets of 36 pairs of tables are identical. The instance for $R = 7$ has 65536 variables and about $1.2 \cdot 10^6$ clauses. The searches for $R \le 7$ are carried out with both

solvers, with the same sets of solutions; the logs are in `code/rule95/logs`, those of the search for $R = 7$ being `sat_sublattice_aut_R7.log` and `sat_sublattice_aut_R7_minisat22.log`.

Census tables. For each of the 46 tables of the census of radius 3 and the 262 of radius 4 (Lemma 6.20), the program `census_cmp.py` rechecks the commutation on all 4^{2R+3} windows, verifies on all 4^{2R+1} entries that the table is of sublattice type, extracts the four class maps of its restrictions to Φ^+ and Φ^- from the entries at the windows of $\Phi^{\pm}$, checks that they have the form (15) of $\widehat{H_{\varphi,\psi}\sigma^n}$ with $n \in \{0, 1\}$ and that φ and ψ satisfy the criterion after Lemma 6.14, and decides their injectivity. As a second check, the table of $\widehat{H_{\varphi,\psi}\sigma^n}$ is generated by the evaluation described above and compared with the census table in all entries. All 45, resp. 261, non-constant tables pass both checks; 25, resp. 41, have φ and ψ injective, and these are the maps $\sigma^a F^b$. The same two checks are applied to the 3141 non-constant solutions of sublattice type of radius 5 (`verify_sub.py`), and all pass. The census tables of radius 3 are in `code/rule95`; the 262 tables of radius 4, a file of 69 MB, are not included and are produced by `code/census/enum_r4.py`.

A.2.5. *Rule* 175. (Remark 6.22.) The program `sat_sublattice.py` takes the rule as a parameter: for an elementary rule f that does not read its middle argument, the relations between four entries are $a = b \oplus f(c, \cdot, d)$. For $f = 175$ and each $R \leq 7$ all solutions of both types are enumerated: they are the $2R^2 + 2R + 1$ maps $\sigma^a F^b$, with no constant map (no constant configuration is fixed by F_{175}) and no other solution, and every solution is checked again to commute with F_{175} on random periodic configurations. A second program with its own encoder (`sublattice_any.py`) gives the same numbers of clauses and of solutions for every $R \leq 7$. Both programs are run with CaDiCaL 1.5.3 and with MiniSat 2.2, with the same results. As a control, for $f = 250$ and $R = 3$ the first program finds the 25 maps $\sigma^a F^b$, the constant map, counted in both types, and 20 others, as for $f = 95$ (log `logs/sat_sublattice_rule250_R3.log`). The instances for $R = 7$ have 65536 variables and about $1.2 \cdot 10^6$ clauses for each type. The logs of both programs are in `code/rule95/logs`.

A.3. **Confirmations of results proved by hand.** The following computations check results that are proved by hand. None of them is used in a proof.

Lemma L (Theorem 3.1; programs in `code/lemmaL` and `code/rule155`). Lemma 3.4(i)–(iii) is checked over all words of length at most 22 for 91 and 94 (`climb.c`); the formula of Lemma 3.3 at the 72 lattice points of a block of 12×12 in each of 200 pseudo-random diagrams per rule, 14,400 points (`dalembert.py`); and Proposition 3.6 for 91 and 94, for all 4^7 fillings of 7 adjacent cells in every valid background with $R = 4$, all of which separate by $t = 18$ (`separ.py`). Working with (4) directly, the two-step gains of the climbing lemmas are 4, 3, 4 for 91, 94, 95 (`climb_gains.c`). The window closure of Section A.2 with base $\Phi^+ \cup \Phi^-$ is complete for $R \leq 4$ for 91 and 94 (`code/closure/dc.c`). For 155, Proposition 8.1 is checked on 3000 pseudo-random configurations (`edge155.py`); Proposition 8.3 on all fillings of at most 4 cells next to the medium with three pseudo-random media each, 1023 runs of 40 rows (`checkW.py`), and on the 99,059 rows of 3000 pseudo-random runs of at most 60 steps (`thmW.py`); and (18) and the eventual periodicity of the strips by simulation (`stripsim.py`).

Kink rigidity (Theorems 4.9 and 4.12; programs in `code/kink`). The following checks use a direct simulation in (t, i) of the configuration that is P on the left and X on the right, independent of (4). Lemma 4.4 is checked in all eight cases (`t_pair.py`). Lemma 4.7 is checked exhaustively for both front values and for the version with q and p exchanged, over all continuations of length at most 10 of A and B, at two values of a_1 and two lengths of the front of B: 1,465,860 cases, with the number and the position of the firings on the ray $a_1 - 1$ as stated (`t_R1.py`; log `code/kink/logs/t_R1_10.log`). Lemma 3.3, in the form used in Step 3, is checked on 800 random pairs (`t_dal.py`), and Lemma 4.8 for $2 \leq g \leq 8$ and $g \in \{15, 22, 31\}$ at the four

placements $a, b \in \{8, 9\}$ (`t_train.py`). As an end-to-end check of Steps 2–5 (`t_step4.py`), among the 351 words w of length at most 12 that are empty or begin with 1 and end with 0, and for which $00\,w\,11$ contains no 010, only the empty word gives a point W (the word w after a front 0^b, followed by a plateau of 1s and a bare descending kink at a fixed position) such that the outgoing Φ^+ part of (K_a^+, W) is K_a^+ for both b and $b+1$. For 95, Lemma 4.11 is checked at both parities (`t95a.py`), the two parts of Lemma 4.10 are checked on all pairs of cores of length at most 9 at two placements and both classes, 6000 cases each, and among all pairs of cores of length at most 9 only the pair of bare defects satisfies both relations (T) and (Δ) of the proof of Theorem 4.12 (`t95b.py`).

The translation property for 95 (Corollary 6.5). The property is checked exhaustively for cores of length at most 9 with (4), and for cores of length at most 6 also by direct simulation in (t, i) (`code/translation95/t95ex.py`: 3200 placed pairs, resp. 200).

Rule 95*: blocking words and window graphs* (Propositions 6.17 and 6.18; the programs are in `code/rule95`). For every word w of length at most 6 and every cell i_0 of it, the configurations c and c' of the proof of Proposition 6.17, with the additional particle realised as a bare defect of Φ^+ at distance $4|w| + 12$ or $4|w| + 13$ to the left, are evolved, and a time with $F^t(c)_{i_0} \neq F^t(c')_{i_0}$ is found in every case, the first such time being at most 47 (`block_test.py`). The graphs $W_M^{(1,1)}$, $W_M^{(-1,1)}$ and $W_M^{(0,1)}$ of 95R are strongly connected for $M \leq 7$ (`window_graph.py`), in accordance with Proposition 6.18; as a control, $W_M^{(\pm 1,1)}$ of 90R is not strongly connected for $1 \leq M \leq 4$, while $W_M^{(0,1)}$ is.

The scattering closure for 94 (Lemmas 4.4, 5.3, 5.4, 5.6 and 5.7, Corollary 5.5, Theorem 5.8). Two sets of programs (`code/rule94`), which share no code, check these results. Sequences are counted up to translation, and the core is that of Section 4, so that a bare kink has a core of length 0: there are 153, 467, 818, 1434 and 2515 sequences of Φ_{94} with constant tails and cores of length at most 8, 10, 11, 12 and 13, the two backgrounds included.

The first set (`lib.py`, `cc.py`, `verify1.py`, `verify2.py`, `verify3.py`, `closure_bf.py`) uses the recurrence (4) in two implementations and the direct simulation in (t, i), which agree. It checks the identities $\mathcal{S}(P_i, [b; U_i]) = ([b; U_{i+1}], \Omega_i)$ of the two cases of the proof of Lemma 5.3, with the bounds on the first kink of U_{i+1}, on 3000 random objects U_i with at most 7 kinks in each case; Lemma 5.4 for $2 \leq g \leq 8$ at four placements, 28 cases; Lemma 5.6(i) and (iii) with their firings and outgoing parts, and the outgoing parts of the pairs of (ii), for all $2 \leq a, d \leq 11$, and the diagrams of (i) and (iii) on the whole quadrant; and the identities of the two cases of the proof of Lemma 5.7 on 3000 random objects each. It runs the proof of Theorem 5.8 as an algorithm, in which every scattering is computed and compared with the form that the proof asserts, and (C0) is invoked once the prepended kink is at distance L from the rest: with $L = 10$ and (4) this handles all 816 objects that are not backgrounds with cores of length at most 11 (11,611 steps of the chains of Lemma 5.3), and with $L = 8$ and the direct simulation all 465 objects that are not backgrounds with cores of length at most 10 (4467 steps). Without the lemmas, it computes the closure of the bare points under (C0)–(C3) breadth first on the sequences with cores of length at most 12 for $L = 6$, and at most 8 for $L = 4$, using in (C2) and (C3) the probes with cores of length at most 1: the closure contains all 1434, resp. 153, of these sequences.

The second set (`rc.py`, `ta.py`, `tc.py`, `te.py`) uses only a direct simulation written from the local rule, in which two rows of the configuration that is P on the left and X on the right are evolved on a finite array. It checks Lemma 4.4 and (7), with the diagram on the quadrant, the firings and the outgoing parts, for all $-4 \leq a, b \leq 4$; Lemma 5.4, with its firings, for $2 \leq g \leq 12$ in 810 cases; the identity $X = [b; [r; K_{r+2}^-]]$ used in Corollary 5.5 at 25 placements; and Lemma 5.6(i), (ii) and (iii) in 41, 41 and 40 cases. It checks the identities and the bounds on the first kink

of the two cases of the proof of Lemma 5.3 in 10,810 and 14,320 cases, and those of the two cases of the proof of Lemma 5.7 in 11,456 and 2864 cases; these cover all 2513 objects that are not backgrounds with cores of length at most 13, with the prepended kink at distance 2, 3, 4, 5 and 8 and the prepended zero at distance 3, 4, 5, 6 and 9 from the first kink, and two positions of the probe. The same identities hold for 400 random objects with cores of lengths 14 to 40. The proof of Theorem 5.8 is run as an algorithm in which each chain of Lemma 5.3 is followed through all its $2L$ scatterings, each scattering $\mathcal{S}(P_i, [b; U_i])$ is compared with the one computed from U_i alone, and the bound $u_n \geq b + L + 2$ and the hypothesis of (C0) are checked position by position: it handles all 818 sequences with cores of length at most 11 for $L \in \{1, 2, 4, 6\}$, all 1434 with cores of length at most 12 for $L = 10$ (58,164 scatterings), all 2515 with cores of length at most 13 for $L = 12$ (122,242 scatterings), all 467 with cores of length at most 10 for $L = 16$ (29,894 scatterings), and 1500, resp. 300, random objects with cores of lengths 14 to 30 for $L = 8$, resp. $L = 20$. Without the lemmas, the scattering of every ordered pair of sequences of a set $\mathcal{U}$ at both relative parities is tabulated, and the properties (C0)–(C3) are applied to the bare points until nothing is added, points outside $\mathcal{U}$ being discarded, so that the result is a subset of C_L. It is all of $\mathcal{U}$ when $\mathcal{U}$ consists of the 153 sequences with cores of length at most 8 and $1 \leq L \leq 6$ (23,850 simulations), of the 818 sequences with cores of length at most 11 and $L \in \{4, 6, 8, 10, 12\}$ (682,120 simulations), and of the 1434 sequences with cores of length at most 12 and $L \in \{4, 6, 8, 10, 12, 13\}$ (2,096,356 simulations). When (C0) is left out, the result for the 267 sequences with cores of length at most 9 consists of the four bare points, in accordance with the paragraph after Definition 5.1.

Appendix B. Front strips of 155R

This appendix continues Section 8. Throughout, $f = 155$ and $F = F_{155}$.

Let c agree with b_β on $(-\infty, s)$, as in Proposition 8.1, put $\beta' = \beta_{(s-1) \bmod 2}$, and for $j \in \mathbb{Z}$ and $t \geq -1$ let $Z_j(t) = Y_c(t, s - t + j)$, the value at time t on the left-moving light ray $p = s + j$, which we call *line* j. For $j < 0$ and $t \geq 0$, $Z_j(t) = \beta_{(s+j) \bmod 2}$, because $Y_c(t, i)$ depends only on $c_{[i-t, i+t]}$; so $Z_{-1} = \beta'$ and $Z_{-2} = 1 \oplus \beta'$, and Z_0 is constant by Proposition 8.1. We call c an 11-*front* if $Z_0 = \beta' = 1$ (the case of Proposition 8.1) and a 00-*front* if $Z_0 = \beta' = 0$; if $Z_0 \neq \beta'$, line 0 also carries background values. The relation (3) at the centre $(t, s - t + j - 1)$ reads

$$Z_j(t+1) = Z_j(t) \oplus Z_{j-2}(t) \oplus Z_{j-2}(t-1) \oplus N\big(Z_{j-2}(t), Z_{j-1}(t), Z_j(t)\big) \qquad (t \geq 0), \tag{18}$$

so each line is driven by the two lines to its left; we call $(Z_{j-2}(t), Z_{j-1}(t))$ the *context* of line j at time t. For $f = 155$, $N(0, 0, z) = 1 \oplus z$, $N(1, 1, z) = 1$ and $N(0, 1, z) = N(1, 0, z) = 0$.

Lemma B.1 (front strips of 155R)**.** *Let $f = 155$ and c, s, Z_j be as above.*

(i) *Let $m \geq 1$, and suppose that the lines Z_j, $j < m$, are periodic on $t \geq t_0$ with a common period L. For $t \geq t_0 + 1$ put $T_t(z) = z \oplus Z_{m-2}(t) \oplus Z_{m-2}(t-1) \oplus N(Z_{m-2}(t), Z_{m-1}(t), z)$, so that $Z_m(t+1) = T_t(Z_m(t))$ and $T_{t+L} = T_t$, and let $T = T_{t_0+L} \circ \cdots \circ T_{t_0+1}$. Then exactly one of the following holds.*

- Slaved: *the context of line m is 00 at some $t \geq t_0 + 1$. Then T is constant, and Z_m is L-periodic on $t \geq t_0 + L + 1$ whatever its values at times $t \leq t_0 + L$.*
- Free: *the context is never 00 on $t \geq t_0 + 1$, and $T = \mathrm{id}$. Then Z_m is L-periodic on $t \geq t_0 + 1$, and the two values of $Z_m(t_0 + 1)$ give two L-periodic continuations.*
- Doubling: *the context is never 00 on $t \geq t_0 + 1$, and $T(z) = 1 \oplus z$. Then $Z_m(t + L) = 1 \oplus Z_m(t)$ for $t \geq t_0 + 1$, so Z_m is $2L$-periodic there and both values of $Z_m(t_0 + 1)$ lie on one cycle.*

In particular every line is eventually periodic, with a period that is a power of 2.

(ii) *Let $j \geq 1$, and suppose that Z_{j-2} and Z_{j-1} are constant on $t \geq t_0$, with values a and b. Then for $t \geq t_0 + 1$, $Z_j(t+1) = 1$ if $ab = 00$, $Z_j(t+1) = 1 \oplus Z_j(t)$ if $ab = 11$, and $Z_j(t+1) = Z_j(t)$ otherwise.*

(iii) *Suppose that the lines Z_j, $0 \leq j < K$, are periodic on $t \geq t_0$ (the front is* recurrent at width K*). Let j_1 be the least $j < K$ such that Z_j is not constant on $t \geq t_0$, with $j_1 = K$ if there is none, and let $\omega = Z_{-2}Z_{-1}\cdots Z_{j_1-1}$ be the word of the constant values. Then $j_1 \geq 1$, ω contains no* 000*, it contains* 11 *at most as its last two letters, and if $j_1 < K$ it ends in* 11*. Hence either ω contains neither* 11 *nor* 000 *and $j_1 = K$, so that on the lines $-2, \ldots, K-1$ and $t \geq t_0$ the diagram is constant on every line, like the diagram $\varphi(i+t)$ of a point of Φ^-; or ω ends in* 011*, its last letter being on line J, and the lines $j \geq J$ satisfy* (18) *with $Z_{J-1} = Z_J = 1$ constant, as the lines $j - J \geq 0$ of an* 11*-front do: the strip is a shifted* 11*-front.*

Proof. (i) For $ab = 00$ the map $z \mapsto z \oplus k \oplus N(a,b,z)$ is the constant $1 \oplus k$; for $ab \neq 00$, $N(a,b,z)$ does not depend on z and the map is $z \mapsto z \oplus k'$ with k' independent of z. So T is constant if some T_τ with $t_0 + 1 \leq \tau \leq t_0 + L$ is constant, and a translation $z \mapsto z \oplus \varepsilon$ otherwise; the three cases are T constant, $\varepsilon = 0$ and $\varepsilon = 1$. In the slaved case, if T_τ is constant, then $Z_m(\tau+1) = T_\tau(Z_m(\tau)) = T_{\tau+L}(Z_m(\tau+L)) = Z_m(\tau+L+1)$, and since $T_{t+L} = T_t$, Z_m is L-periodic from $\tau + 1 \leq t_0 + L + 1$ on. In the other cases every T_t is a translation, and $Z_m(t+L) = T_{t+L-1} \circ \cdots \circ T_t(Z_m(t))$ for $t \geq t_0 + 1$; this composition consists of the same translations as T, so it equals T. The lines $j \leq 0$ are constant on $t \geq 0$, so induction on m gives the last statement.

(ii) For $t \geq t_0 + 1$, $Z_{j-2}(t) \oplus Z_{j-2}(t-1) = 0$ in (18), and the values of $N(a,b,\cdot)$ listed above give the three cases.

(iii) The lines $j \leq 0$ are constant, so $j_1 \geq 1$, and ω begins with $Z_{-2}Z_{-1} \in \{01, 10\}$, so a factor 000 of ω occupies lines $j-2, j-1, j$ with $1 \leq j < j_1$, and a factor 11 that is not at the end of ω occupies lines $j-2, j-1$ with $1 \leq j < j_1$. In the first case (ii) would give $Z_j(t) = 1$ for $t \geq t_0 + 2$, contradicting $Z_j = 0$ on $t \geq t_0$. In the second case (ii) would make Z_j alternate on $t \geq t_0 + 1$, which is not constant. If $j_1 < K$, the lines $j_1 - 2$, $j_1 - 1$ are constant with values ab, and if $ab \neq 11$, (ii) makes Z_{j_1} constant on $t \geq t_0 + 2$, hence, being periodic, on $t \geq t_0$; so $ab = 11$. In the second alternative the letter before the final 11 exists and is 0, since ω has length at least 3 and contains no 11 before its last two letters. For $j \geq J + 1$ the relation (18) involves only the lines $j - 2 \geq J - 1$, $j - 1$ and j, and for an 11-front the lines -1 and 0 are constant 1; so the shift $j \mapsto j - J$ carries the one system to the other. □

For the 11-front the tree of periodic continuations of Lemma B.1(i) begins as follows (computer-assisted): lines 1 and 2 are doubling (lines -1 and 0 are constant 1, so line 1 alternates by (ii)), lines 3–6 are slaved, line 7 is free, lines 8–10 are slaved, and at line 11 one of the two branches doubles while the other is slaved; at line 39 there are 34 branches, with periods 8 and 16. For the 00-front there are 220 periodic continuations at width $K = 18$; in 86 of them ω contains no 11, and 134 are shifted 11-fronts, 28 of which have their final 11 on the last two lines.

Remark B.2 (the front hypothesis (Fr) for 155R). Let h be a block map of radius R that commutes with F and fixes every point of $\Phi^+ \cup \Phi^-$, let c, s and Z_j be as above, and put $\Delta = Y_{h(c)} \oplus Y_c$ and $\Delta_j(t) = \Delta(t, s - t + j)$. For $t \geq 0$ the cells of $F^t(c)$ to the left of $s - t$ are those of $F^t(b_\beta)$, and $b_\beta \in \Phi^+$, so $\Delta_j(t) = 0$ for $j < -R$ and $t \geq 0$: the configuration $h(c)$ agrees with b_β on $(-\infty, s - R)$, and its front lies at most R cells to the left of that of c. Lemma L for 155 implies the following hypothesis (Fr): for all such h and c and every j, $\Delta_j(t) = 0$ for all large t; that is, h fixes the eventually periodic front windows of Lemma B.1(i).

Lemma B.1(i) reformulates (Fr). Suppose that (Fr) fails for c, and let m be the least j with $\Delta_j(t) \neq 0$ for arbitrarily large t. Then $m \geq -R$, and there is t_1 with $\Delta_j(t) = 0$ for all $j < m$ and $t \geq t_1$. So $Y_{h(c)}$ and Y_c have the same lines $j < m$ on $t \geq t_1$, and the difference of the two instances of (18) gives, for $t \geq t_1 + 1$,

$$\Delta_m(t+1) = \Delta_m(t) \cdot \big[(Z_{m-2}(t), Z_{m-1}(t)) \neq 00\big],$$

because $N(0,0,z) \oplus N(0,0,z \oplus 1) = 1$ and $N(a,b,\cdot)$ is constant for $ab \neq 00$. So Δ_m is nonincreasing on $t \geq t_1 + 1$; since it is nonzero at arbitrarily large t, $\Delta_m(t) = 1$ for all $t \geq t_1 + 1$, and the context of line m is never 00 there. By Lemma B.1(i), line m is free or doubling, and $h(c)$ carries on it the other free continuation, or the other half of the doubled cycle, which is a time shift by L. If $m \leq 0$, the context of line m is the background pair 01 or 10, so the line is free; for $m < 0$, h moves the front by $-m \leq R$ cells to the left, and for $m = 0$ it complements the edge bit. Thus (Fr) is equivalent to: for no such h and c is there a line $m \geq -R$, free or doubling for c, such that from some time on Δ vanishes on the lines $j < m$ and is 1 on line m.

The full closure of Remark 8.2 does not use front windows. A weaker closure, which uses the relation (3) only to force the value at the point S, one step back in time, from the other four values of a stencil, forces every window for $2 \leq R \leq 4$ (computer-assisted; Appendix A.2) when the windows of the fronts at small offsets, and of their time reversals, are added to the base, and it stalls without them. A proof of (Fr), together with such a closure at every radius (call this (Cl)), would give Lemma L for 155.

Use of AI tools. Large language models (Anthropic's Claude models) were used extensively in this work: to propose and develop proofs, to design, write and run the computer searches and their independent re-implementations, to check proofs and computations, and to draft the manuscript. The models are not authors. The author has reviewed the content and takes full responsibility for it. All code, and the logs of the computations, accompany the paper; data files of more than 5 MB are regenerated by the programs (Appendix A).

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